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Polymer Chain Dynamics and NMR

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Abstract The universal features of polymer dynamics are specifically represented by laws for (anomalous) segment diffusion and chain relaxation modes. Nuclear magnetic resonance (NMR)-based techniques provide direct access to these phenomena. This in particular refers to NMR relaxation and diffusion studies. Methods suitable for this purpose are described in detail. Three basic classes of polymer dynamics models, namely the Rouse model, the tube/reptation model, and the renormalized Rouse models are outlined and discussed with respect to predictions for NMR measurands. A wealth of experimental NMR data are reviewed and compared with predictions of the model theories. It is shown that characteristic features of all three types of models can be verified in great detail provided that the model premisses are suitably mimicked in the experiments. Rouse dynamics is shown to be relevant for polymer melts with molecular weights below the critical value and for solutions of diminished entanglement effect. Features specific for the renormalized Rouse model reveal themselves in the form of high- and low-mode-number limits of the spin–lattice relaxation dispersion. These results are considered to mirror the analytical structure of the Generalized Langevin Equation. Finally, anomalous-diffusion and relaxation laws characteristic for the tube/reptation model can be perfectly reproduced in experiment if the polymer chains are confined in a nanoporous, solid matrix whereas bulk melts are not in accord with these predictions. The dynamics of chains confined in artificial tubes can be treated analytically assuming a harmonic radial potential for the polymer/wall interaction. These results derived for a real tube closely render the characteristic features of the original Doi/Edwards model predicted for a fictitious tube.

Keywords NMR · Relaxation · Diffusion · Rouse model · Reptation · Renormalized Rouse model

Abbreviations

BWR	Bloch/Wangsness/Redfield theory
CLF	contour length fluctuation
CPMG	Carr/Purcell/Meiboom/Gill
FID	free-induction decay
GLE	generalized Langevin equation
NMR	nuclear magnetic resonance
ODF	order director fluctuations
PB	poly(butadiene)
PDES	poly(diethylsiloxane)
PDMS	poly(dimethylsiloxane)
PE	poly(ethylene)
PEO	poly(ethylene oxide)
PGSE	pulsed gradient spin echo method
PHEMA	poly(hydroxyethylmethacrylate)
PIB	poly(isobutylene)
PIP	poly(isoprene)
PMMC	polymer mode-mode coupling model
PS	poly(styrene)
PTHF	poly(tetrahydrofurane)
PU	phenylurazole
RF	radio frequency
RRM	renormalized Rouse model
SGSE	steady gradient spin echo method
ThRRM	three times renormalized Rouse model
TRRM	twice renormalized Rouse model
WLF	Williams/Landel/Ferry

1 Introduction

The objective of this review is to outline the basic theories and models of linear polymer chain dynamics [1], and to demonstrate the potential of NMR (nuclear magnetic resonance) techniques for probing the characteristics of molecular motions in such systems. From the methodological point of view we will mainly focus on field-cycling NMR relaxometry and field-gradient NMR diffusometry, but will also make some reference to transverse relaxation and the residual dipolar interaction occurring in entangled polymer systems. This combination of methods turned out to be most powerful for the examination of chain modes and the translational diffusion properties of polymer segments and the coil center-of-mass. The techniques referred to are best suited to shed light on almost every aspect of polymer dynamics.

Particular emphasis will be laid on the experimental verification of features of standard theories of polymer dynamics such as the Rouse model, the tube/reptation model [1] or the renormalized Rouse models. Based on the special conditions under which predictions of these theories match experimental findings well, the application limits and the deficiencies of these models for more general scenarios become obvious and help to ameliorate the underlying model ansatz for chain dynamics.

In order to elucidate the *universal* character of polymer chain dynamics, we will attempt to consider as many different polymer species as possible for which NMR relaxometry and diffusometry data are available. On the other hand this means that we must restrict ourselves predominantly to melts of linear homopolymers with a narrow molecular weight distribution to keep the range of literature referred to at manageable levels. Only where enlightening information is expected will we refer to solutions, networks, and mesomorphic systems in addition to isotropic melts. Empirically the conclusions suggested by the huge amount of experimental NMR data available in the literature provide a complete picture rendering dynamical features in great detail.

This is in contrast to the stage of development of model theories where the "many-entangled-excluded-volume-chain" problem still has to be overcome, although promising approaches exist. In particular it will turn out that the theoretical concepts available for this sort of system are unable to describe many experimental details quantitatively. On the other hand, predictions of all model theories referred to in this context will be shown to be compatible with certain experimental findings provided that the sample and measuring conditions are adapted to the scenarios that were anticipated in the theoretical derivations.

Apart from the introductory section, the article is subdivided into four major sections: *NMR Methods; Modeling of Chain Dynamics and Predictions for NMR Measurands; Experimental Studies of Bulk Melts, Networks, and Concentrated Solutions;* and *Chain Dynamics in Pores*. First, the NMR techniques of interest in this context will be described. Second, the three fundamental polymer dynamics theories, namely the *Rouse model*, the *tube/reptation model*, and the *renormalized Rouse theories* are considered. The immense experimental NMR data available in the literature will be classified and described in the next section, where reference will be made to the model theories wherever possible. Finally, recent experiments, analytical treatments, and Monte Carlo simulations of polymer chains confined in pores mimicking the basic premiss of the tube/reptation model are discussed.

2
NMR Methods

NMR provides a rich variety of methods for polymer studies. This in particular refers to the chemical structure of such compounds. Here we deliberately restrict ourselves to investigations on length scales, where chemical bonds and interactions merely matter indirectly via parameters such as segment lengths or friction coefficients. That is, we are dealing with dynamic properties of Kuhn segment chains for times or frequencies beyond those relevant for fluctuations within Kuhn segments. Any chemical shift-based spectroscopy is beyond the scope of this review. Figure 1 shows a schematic representation of the fluctuation time or angular frequency ranges covered by the methods of interest in this context. We will be referring mainly to field-cycling NMR relaxometry combined with conventional high-field methods, to field-gradient NMR diffusometry, and, to a minor degree, to dipolar broadening-based techniques such as the dipolar correlation effect. For an introduction to these methods the reader is referred to monographs like Ref. [2].

Diffusion in polymers can manifest itself in "normal" or "anomalous" form depending on the time or displacement length scale probed in the experiment. The criterion for this distinction is the time dependence of the mean squared displacement

$$\left\langle R^2 \right\rangle \propto t^{\kappa} \tag{1}$$

valid in the form of a power law in random unrestricted media. An exponent κ=1 indicates normal diffusion as is expected for center-of-mass diffusion, i.e., root mean square displacements much larger than the Flory radius. On the other hand, κ<1 is the signature of anomalous (subdiffusive) displacements. This is of particular interest for segment motions within the length scale of the random polymer coil, where all time dependences are governed by chain modes. That is, characteristic laws are expected for this so-called segment diffusion behavior that may be decisive for the validity of chain dynamics theories. It is therefore of paramount importance to have experimen-

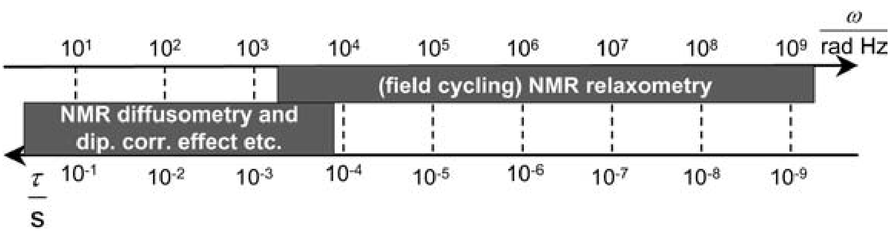


Fig. 1. Schematic representation of the ranges of fluctuation time constants and angular frequencies accessible by the two main NMR techniques under consideration in this article

tal tools in hand probing displacements and chain dynamics on the right time and length scales. The NMR techniques to be described turned out to be particularly powerful in this respect.

2.1 NMR Diffusometry

The standard *pulsed gradient spin echo* (PGSE) and *steady gradient spin echo* (SGSE) methods are based on spatially constant field gradients $\vec{G} = \vec{\nabla} B_0 = const$, where B_0 is the magnitude of the main magnetic field. In the following, all main field gradients are assumed along the z direction so that the total magnetic flux density at a position z is given by $B(z)=B_0+Gz$. The spin echoes to be considered here are the result of ordinary coherence refocusing of the Hahn type [2].

The attenuation factor of the echo amplitude due to translational diffusion is given by

$$A_{diff}(T_E) = \langle e^{i\varphi(T_E)} \rangle. \tag{2}$$

T_E is the echo time and φ is the (precession) phase a spin possesses at time $t=T_E$. The brackets represent an ensemble average over all spins in the sample. In the following, this expression will be analyzed for different pulse sequences.

Any echo attenuation by relaxation occurs independently of diffusive attenuation and is of marginal interest in this context. It can readily be separated by either keeping all pulse intervals constant (and varying the gradient strength) or by compensating for relaxation losses in experimental protocols especially designed for this purpose.

2.1.1 Hahn Spin Echo in the Short Gradient Pulse Limit ($\delta \ll \Delta$)

We consider the coherence evolution of uncoupled spins I=1/2 during the pulsed gradient Hahn spin echo sequence schematically shown in Fig. 2a. A suitable basis for the treatment is the spherical product operator formalism. Explanations, definitions, and rules of the spin operator formalism needed in this context can be found in Ref. [2]. Times just before and immediately after RF (radio frequency) and field gradient pulses will be indicated by minus and plus signs, respectively.

The initial reduced density operator is given by the equilibrium density operator,

$$\sigma(0-) = \sigma_0 = I^{(0)}. \tag{3}$$

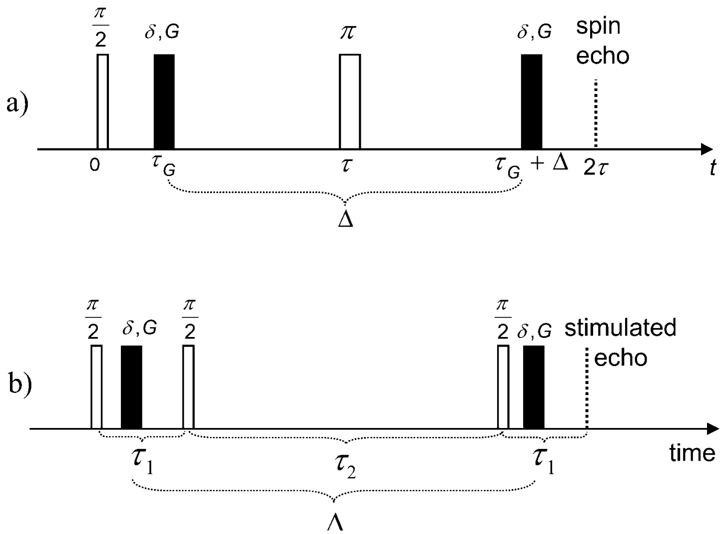


Fig. 2a, b. Radio frequency pulse sequences for the Hahn spin echo (**a**) and the stimulated spin echo (**b**) combined with magnetic field gradient pulses (*black*). Echoes other than the stimulated echo are not indicated in sequence (b). The stimulated echo variant of the pulsed gradient spin echo (PGSE) method is of interest in viscous systems with small diffusion coefficients and, as a consequence, transverse relaxation times much shorter than the spin–lattice relaxation time. Note that in order to obtain distinct echoes at the indicated positions a small but finite background gradient is necessary

The sequence begins with a 90° pulse of width τ_{90} and a rotating-frame phase direction assumed along the x axis. The rotating-frame Hamiltonian during this RF pulse is defined by $H_{90}=-\hbar\gamma B_1 I_x$, where B_1 is the amplitude of the rotating flux density component of the RF field. The reduced density operator immediately after this pulse reads

$$\sigma(0+) = e^{-\frac{i}{\hbar}H_{90}\tau_{90}}\sigma(0-)e^{+\frac{i}{\hbar}H_{90}\tau_{90}} = \frac{i}{\sqrt{2}}\left(I^{(+1)} + I^{(-1)}\right) = \sigma(\tau_G-). \qquad (4)$$

During the field gradient pulses, the coherences evolve under the action of the rotating-frame Hamiltonian $H_G=-\hbar\gamma G z_j I_z$, where z_j is the position coordinate of the spin-bearing particle on the gradient axis during the j-th gradient pulse. This produces

$$\sigma(\tau_G+) = e^{-\frac{i}{\hbar}H_G\delta}\sigma(\tau_G-)e^{+\frac{i}{\hbar}H_G\delta} = \frac{i}{\sqrt{2}}\left(I^{(+1)}e^{i\varphi_1} + I^{(-1)}e^{-i\varphi_1}\right) = \sigma(\tau-), \qquad (5)$$

where $\varphi_1=\gamma G z_1\delta$. The subsequent 180° RF pulse of length τ_{180} and a rotating-frame phase direction assumed along the y direction produces

$$\sigma(\tau+) = e^{-\frac{i}{\hbar}H_{180}\tau_{180}}\sigma(\tau-)e^{+\frac{i}{\hbar}H_{180}\tau_{180}} = \frac{i}{\sqrt{2}}\left(I^{(-1)}e^{i\varphi_1} + I^{(+1)}e^{-i\varphi_1}\right)$$
$$= \sigma(\tau_G + \Delta-) \tag{6}$$

with $H_{180}=-\hbar\gamma B_1 I_y$. The second gradient pulse causes further phase shifts according to

$$\sigma(\tau_G + \Delta+) = e^{-\frac{i}{\hbar}H_G\delta}\sigma(\tau_G + \Delta-)e^{+\frac{i}{\hbar}H_G\delta}$$
$$= \frac{i}{\sqrt{2}}\left(I^{(-1)}e^{i(\varphi_1-\varphi_2)} + I^{(+1)}e^{-i(\varphi_1-\varphi_2)}\right) = \sigma(2\tau), \tag{7}$$

where $\varphi_2=\gamma G z_2 \delta$.

Coherence evolution in the presence of a field gradient winds the transverse magnetization up along the gradient direction in the form of a helix of pitch $2\pi/(\gamma G\delta)$. The periodicity implied in this helix may be characterized by a "wave number"

$$k = \gamma G \delta. \tag{8}$$

The spin echo amplitude of a spin ensemble being at z_1 during the first gradient pulse and at z_2 during the second is thus represented by the reduced density operator

$$\sigma(2\tau) = \frac{i}{\sqrt{2}}\left(I^{(-1)}e^{ik(z_1-z_2)} + I^{(+1)}e^{-ik(z_1-z_2)}\right). \tag{9}$$

The phase factors depend on the particle displacement $Z=z_1-z_2$ which is distributed after the gradient pulse interval Δ according to the propagator $P(Z,\Delta)$, i.e., the probability density that a particle has been displaced a distance Z along the gradient direction in a time Δ. The average over all particles is given by

$$\left\langle e^{\pm ikZ}\right\rangle = \int_{-\infty}^{+\infty} P(Z,\Delta)\exp\{\pm ikZ\}dZ = \int_{-\infty}^{+\infty} P(Z,\Delta)\cos\{kZ\}dZ, \tag{10}$$

where we have anticipated that $P(Z,\Delta)$ is an even function of Z and that k is constant within the sample.

2.1.1.1
Gaussian Propagators

The propagator for normal displacements, Z, along the gradient direction with the diffusion coefficient D is given by the Gaussian function

$$P(Z,\Delta) = \frac{1}{(4\pi D\Delta)^{1/2}}\exp\left\{-\frac{Z^2}{4D\Delta}\right\}. \tag{11}$$

Displacements along other space directions are not directly probed in this experiment and are therefore irrelevant. Evaluating Eq. 10 based on this propagator results in $\langle e^{\pm ikz}\rangle = \exp\{-k^2 D\Delta\}$ so that Eq. 9 becomes

$$\sigma(2\tau) = I_y e^{-k^2 D\Delta}. \tag{12}$$

Equation 2 takes the form

$$A_{diff}(T_E) = \left\langle e^{i\varphi(T_E)} \right\rangle \equiv A_{diff}(k, \Delta) = \exp\left\{-k^2 D\Delta\right\}. \tag{13}$$

Inserting $\langle Z^2\rangle = 2D\Delta$ in Eq. 13 provides a direct relation between echo attenuation and the mean squared displacement in the diffusion time Δ:

$$A_{diff}(T_E) = \exp\left\{-\frac{1}{2}k^2\langle Z^2\rangle\right\} \underset{\text{system}}{\overset{\text{isotropic}}{=}} \exp\left\{-\frac{1}{6}k^2\langle R^2\rangle\right\} \tag{14}$$

which refers to both normal and anomalous displacements provided the propagator is Gaussian.

2.1.1.2
Non-Gaussian Propagators

In contrast to center-of-mass diffusion, segment diffusion is characterized by non-Gaussian propagators. The attenuation of spin echoes is then no longer governed by unspecific equations like Eqs. 13 or 14. However, in special cases such as the reptation model, a formalism for the evaluation of the echo attenuation can be set up [3].

If such an analytical formalism is not available, the mean squared displacement, i.e., the second moment of the propagator, can be evaluated approximately from the experimental echo attenuation function by restricting oneself to the low wave number limit. The phase factor given in Eq. 2 can be expanded according to

$$A_{diff} = \left\langle e^{\pm ikZ}\right\rangle = \left\langle 1 \pm \frac{ikZ}{1!} - \frac{k^2Z^2}{2!} \mp \frac{ik^3Z^3}{3!} + \cdots \right\rangle. \tag{15}$$

Keeping in mind that in the absence of flow the propagator is an even function of Z, one finds in the lowest nontrivial approximation for small wave numbers

$$A_{diff} = \left\langle e^{\pm ikZ}\right\rangle \overbrace{\approx}^{k^2\langle Z^2\rangle \ll 1} 1 - \frac{1}{2}k^2\langle Z^2\rangle \overset{\text{isotropy}}{=} 1 - \frac{1}{6}k^2\langle R^2\rangle \approx \exp\left\{-\frac{1}{6}k^2\langle R^2\rangle\right\}, \tag{16}$$

where $\langle R^2\rangle$ is the three-dimensional mean squared displacement in the gradient pulse interval Δ. Plotting the echo amplitude as a function of k^2 and evaluating the initial slope of the decay curve thus provides the diffusion time dependence of the mean squared displacement, $\langle R^2\rangle=\langle R^2(\Delta)\rangle=6D(\Delta)\Delta$, where $D(\Delta)$ is the time-dependent diffusion coefficient.

2.1.2 Hahn Spin Echo for Long Gradient Pulses

We are again referring to the pulse scheme shown in Fig. 2a. The gradient pulse width, δ, is now assumed to be comparable to the interval Δ. Displacements during the gradient pulses are therefore no longer negligible. That is, the phase shift accumulated by a spin depends on the trajectory of the spin-bearing particle during the gradient pulse. For constant gradients along the z axis, this can be expressed by

$$\varphi_j(\delta) = \gamma G \int_0^\delta z_j(t')dt', \tag{17}$$

where the subscript j=1,2 indicates the first or the second gradient pulse. The attenuation factor, Eq. 2, thus becomes

$$A_{diff}(T_E) = \langle \exp\{i\varphi(T_E)\}\rangle = \left\langle \exp\left\{ i\gamma G\left[\int_0^\delta z_1(t')dt' - \int_0^\delta z_2(t')dt'\right]\right\}\right\rangle \tag{18}$$

in analogy to the treatment in the previous section. The analytical form of this attenuation function is best obtained by solving the Bloch/Torrey equation [4], that is, Bloch's equations supplemented by a diffusion term according to

$$\frac{dM_+(z,t)}{dt} = -i\gamma G z M_+(z,t) - \frac{M_+(z,t)}{T_2} + D\frac{d^2}{dz^2}M_+(z,t), \tag{19}$$

where $M_+=M_x+iM_y$ is the local complex transverse magnetization. The resulting attenuation factor by ordinary diffusion (Gaussian propagator) is [5]

$$A_{diff}(k,\Delta) = \exp\left\{-k^2 D(\Delta-\delta/3)\right\}, \tag{20}$$

which takes the form of the short pulse limit given in Eq. 13 for $\delta<<\Delta$.

2.1.3 Stimulated Echo

The longest diffusion time (which is essentially given by Δ) that can be probed with the Hahn echo methods is limited by the transverse relaxation time T_2. As a consequence, the echo attenuation achievable in this time limit may not be sufficient for very slow diffusion even for the strongest field gradients technically feasible. Also, in the case of anomalous diffusion it may be desirable to probe a diffusion time range as wide as possible in order to acquire the time dependence of the mean square displacement. Since the spin–lattice relaxation time T_1 tends to be much longer than T_2 in viscous media, it is of interest to employ echo signals the relaxation decay of which is partly governed by T_1. The stimulated echo is a favorable example of this sort, although a factor of two in the signal strength is sacrificed relative to the Hahn echo [6].

Figure 2b shows a typical RF pulse sequence combined with field gradient pulses for probing diffusion with the aid of the stimulated echo. In the first RF pulse interval the transverse magnetization is attenuated by transverse relaxation, and the spin coherences evolve under the action of the magnetic field gradient. The second RF pulse "stores" half of the magnetization in the z direction in spatially modulated form (often called "magnetization grid" or "grating") in complete analogy with the optical "forced Rayleigh scattering" technique [7]. The modulation degree of the grating is leveled off by diffusion and spin–lattice relaxation. The third RF pulse converts the z magnetization back to spin coherences which then evolve into the stimulated echo. A detailed description of the spin coherence evolution can be found in Ref. [2].

The total attenuation of the stimulated echo in liquids after two free-evolution intervals τ_1 and the grating storage interval τ_2 is given by

$$A_{stim.echo} = A_{relax} A_{diff}, \tag{21}$$

where

$$A_{relax} = \exp\{-2\tau_1/T_2\}\exp\{-\tau_2/T_1\}. \tag{22}$$

A_{diff} represents any of the formulas given in Eqs. 13–16 and Eq. 20 for the respective limits. If $\tau_1 << T_2 << T_1 > \tau_2$, the maximum diffusion time can be adjusted much longer than in the Hahn echo case.

2.1.4
Spin Echoes and Steady B_0 Gradients

The efficiency of spatial encoding by field gradients depends on the wave number given in Eq. 8, that is the "area" $G\delta$ of the gradient pulses in the free-evolution intervals of the pulse sequences shown in Fig. 2. To generate short and intense gradient pulses with extremely short rise and decay times is technically difficult and, even worse, is susceptible to motional artifacts due to pulsed magnetic forces on the gradient coil system. For very short transverse relaxation times (which are usually accompanying slow translational diffusion) it may therefore be advisable to use steady gradients instead of pulsed ones. Any gradient switching is then avoided and the gradient encoding efficiency in the coherence free-evolution intervals is optimal.

Figure 3a shows the steady-gradient RF pulse scheme for the Hahn echo. The pulsed-gradient parameters δ and Δ defined in Fig. 2a now take identical values and are equal to the RF pulse spacing,

$$\delta = \Delta = \tau. \tag{23}$$

Equation 20 thus becomes

$$A_{diff}(\tau) = \exp\left\{-\frac{2}{3}k^2 D\tau\right\} = \exp\left\{-\frac{2}{3}\gamma^2 G^2 D\tau^3\right\}, \tag{24}$$

where we have anticipated Gaussian propagators as above. If the propagator does not have a Gaussian form and if no special analytical formalism is available for the evaluation of experimental attenuation data, the low-wave-number limit given in Eq. 16 can again be employed as an approach.

Figure 3b shows the analogous variant for the stimulated echo. This is the preferential method for extremely short transverse relaxation times. In this case the pulsed-gradient parameters turn into $\delta=\tau_1$ and $\Delta=\tau_1+\tau_2$, so that Eq. 20 for Gaussian propagators becomes

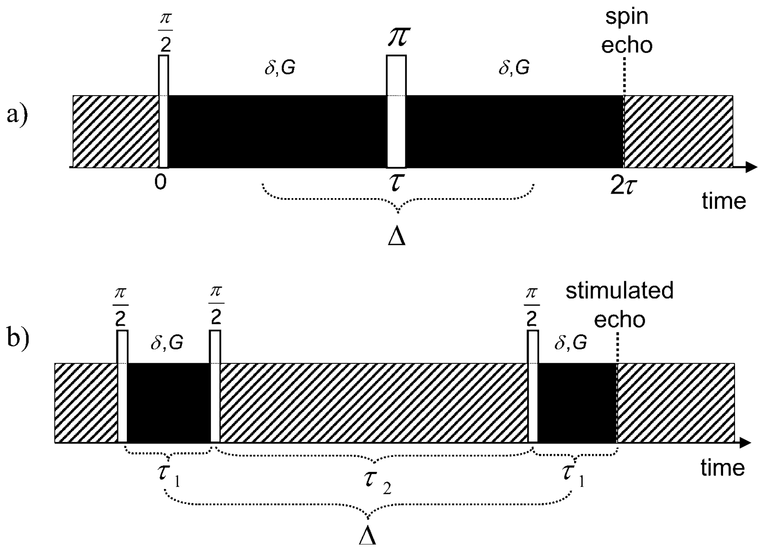


Fig. 3a, b. Radio frequency (RF) pulse sequences for the Hahn spin echo (**a**) and the stimulated spin echo (**b**) combined with steady magnetic field gradients. The *hatched sections* of the gradients are irrelevant for diffusion measurements. Echo attenuation by diffusive displacements is solely caused by the *black sections* of the gradient cut out by the RF pulses and the echo time. The fringe field of superconducting magnets turned out to provide particularly stable and strong gradients

$$A_{diff}(\tau_1,\tau_2)=\exp\left\{-\gamma^2G^2D\tau_1^3\left(\frac{2}{3}+\frac{\tau_2}{\tau_1}\right)\right\}. \tag{25}$$

In the limit $\tau_1<<\tau_2$, this expression takes a form analogous to Eq. 14,

$$A_{diff}(\tau_1<<\tau_2)\approx\exp\left\{-k^2D\tau_2\right\}=\exp\left\{-\frac{1}{6}k^2\left\langle R^2(\tau_2)\right\rangle\right\}, \tag{26}$$

where the wave number is now defined by $k=\gamma G\tau_1$. For non-Gaussian propagators, one may again employ the low-wave-number approximation

$$\begin{aligned}A_{diff}&=\left\langle e^{\pm ikZ}\right\rangle\overbrace{\approx}^{k^2\langle Z^2\rangle\ll1}1-\frac{1}{2}k^2\left\langle Z^2\right\rangle\overset{\text{isotropy}}{=}1-\frac{1}{6}k^2\left\langle R^2\right\rangle\approx\exp\left\{-\frac{1}{6}k^2\left\langle R^2\right\rangle\right\}\\&=\exp\left\{-k^2D(\tau_2)\tau_2\right\}\end{aligned} \tag{27}$$

with the time-dependent diffusion coefficient $D=D(t)$.

Steady gradients of considerable strength (e.g., up to 60 T/m with a conventional 9.4 T wide-bore magnet) are readily available in the fringe field of superconducting magnets [8]. Applications to diffusion studies in polymers have been reported [9–12]. Even stronger steady gradients can be produced with a special Maxwell coil design for superconducting magnets [13].

Since the fringe field gradient has a fixed strength at the position of the probe head, the only experimental parameters of the pulse schemes shown in Fig. 3 that can be varied are the time intervals between the RF pulses. That is, the echo attenuation by relaxation must be considered and compensated for. This can conveniently be done with the aid of self-compensating pulse schemes [14].

Some care must be taken with fringe field methods if motional averaging of dipolar interactions is incomplete. Echoes then tend to be modulated by the so-called dipolar correlation effect [15, 16]. One can account for this phenomenon by dividing the (normalized) echo amplitudes recorded with and without gradient at the same Larmor frequency and at the same pulse intervals [11, 12]. Any influence by relaxation and the dipolar correlation effect can be eliminated in this way.

2.2 NMR Spin–Lattice Relaxometry

The term "relaxometry" is normally used in context with techniques for the measurement of spin–lattice relaxation times in general. Transverse relaxation and effects due to residual dipolar couplings will be considered in the next section.

As indicated in the scheme shown in Fig. 1, diffusometry and relaxometry are complementary to each other with respect to the time or frequency scales they are probing. This is mainly due to the potential of the field-cycling NMR relaxometry technique that permits us to measure relaxation times down to the kHz regime, corresponding to a time scale on which field-gradient diffusometry becomes applicable. Above 10 MHz, the experimental frequency scale can be supplemented by spin–lattice relaxation measurements with conventional NMR spectrometers operating at several hundred MHz by present-day standards. Apart from spin–lattice relaxation in the laboratory frame, we will also present some experimental rotating-frame spin–lattice relaxation data later for comparison.

2.2.1 General Theoretical Background

The studies to be reviewed below mainly refer to proton or deuteron magnetic resonance in melts or concentrated solutions of polymers of molecular masses close to or above the critical value M_c. Under such circumstances the predominant spin–lattice relaxation mechanism of protons (spin 1/2) is based on fluctuating dipole–dipole couplings of like spins, that is, we are dealing with the homonuclear case. Deuterons (spin 1), in contrast, possess a finite electric quadrupole moment which is subject to quadrupole coupling to local molecular electric field gradients which is much more efficient than dipolar interactions.

The relaxation formalisms of dipolar-coupled two-spin 1/2 systems (protons) on the one hand, and of quadrupolar-coupled spin 1 nuclei (deuterons) on the other, have very much in common and lead to largely equivalent analytical expressions [2, 17]. This very much facilitates comparisons of experimental results obtained with either technique.

The reason for this analogy is the fact that the spatial part of the dipolar as well as of the quadrupolar coupling Hamiltonian can be commonly expressed by second-order spherical harmonics $Y_{2,m}(\vartheta,\varphi)$ with m=0, ±1, ±2. For a definition of these functions see Ref. [2], p. 403. The only terms of the Hamiltonian relevant for spin–lattice relaxation in the laboratory frame are those for m=±1 and m=±2 in connection with spin operator terms inducing single-quantum and double-quantum transitions, respectively, in the two-spin 1/2 (dipolar coupling) or single-spin 1 (quadrupolar coupling) systems [2]. These transitions do not conserve Zeeman spin energy. They are rather connected with energy exchange between Zeeman spin energy and lattice energy as required for spin–lattice relaxation. The term "lattice" comprises all motional degrees of freedom of the spin-bearing molecules.

The spherical harmonics $Y_{2,m}(\vartheta,\varphi)$ are expressed in polar coordinates, that is the polar angle ϑ and the azimuthal angle φ. These coordinates define the orientation of the internuclear vector relative to the external magnetic flux density $\vec{B}_0$ in the case of dipolar coupling, or the orientation of the principal electric field gradient (i.e., of a molecular axis) again relative to $\vec{B}_0$ in the case of quadrupolar interactions, where one usually anticipates rotationally symmetric electric field gradients.

Molecular motions in the sense of reorientations of molecules or chemical groups lead to fluctuating polar coordinates, $\vartheta=\vartheta(t)$, $\varphi=\varphi(t)$. As a consequence, the dipolar or quadrupolar Hamiltonians become time dependent and, hence, tend to induce spin transitions as predicted by time-dependent perturbation theory. With dipolar coupling, there is a third variable fluctuating as a result of molecular motion, namely the internuclear distance $r=r(t)$ of the two-spin 1/2 system. This, however, matters only with intermolecular or intersegment interactions while all intrasegment couplings are associated with constant r values. Below we will see that the comparison of proton relaxation with deuteron relaxation of the same polymer species permits one to distinguish intra- from intermolecular interactions and to elucidate the dynamical implications connected with these interactions on this basis.

In the frame of the Bloch/Wangsness/Redfield (BWR) relaxation theory [2, 17], the fluctuations of the spin Hamiltonians are described with the aid of (preferably normalized) autocorrelation functions of the type

$$G_m(t)=\left\langle \frac{Y_{2,m}(\vartheta_0,\varphi_0)\,Y_{2,-m}(\vartheta_t,\varphi_t)}{r_0^3 r_t^3}\right\rangle \Big/ \left\langle \frac{\left|Y_{2,m}(\vartheta_0,\varphi_0)\right|^2}{r_0^6}\right\rangle \text{(dipolar coupling)},$$

$$G_m(t)=\left\langle Y_{2,m}(\vartheta_0,\varphi_0)\,Y_{2,-m}(\vartheta_t,\varphi_t)\right\rangle \text{(quadrupolar coupling)}. \tag{28}$$

The subscripts 0 and t of the spatial variables indicate the time at which they are to be taken. Actually the expressions given in Eq. 28 are statistically stationary functions, so that only the time interval matters rather than the absolute time. The brackets stand for an ensemble average over all spin systems in the sample.

According to time-dependent perturbation theory, the transition probability per time unit is proportional to the spectral density (or intensity function) of the fluctuating coupling inducing the transition. The spectral density is given as the Fourier transform of the autocorrelation functions,

$$I(\omega)=\int_{-\infty}^{\infty} G_m(t)\,e^{-i\omega t}\,dt. \tag{29}$$

Note that the spectral density defined in this way is independent of the subscript m for isotropic systems since it is based on the normalized autocorrelation functions given in Eq. 28.

The spin–lattice relaxation rate directly reflects the spin transition probabilities per time unit for single- and double-quantum transitions, and hence is proportional to a linear combination of spectral densities in the form

$$\frac{1}{T_1} = C_{coupl}[I(\omega) + 4I(2\omega)], \tag{30}$$

where $\omega=\gamma B_0$ is the resonance angular frequency depending on the gyromagnetic ratio γ of the nuclei and the external magnetic flux density B_0. The analytical form of Eq. 30 is valid for systems of two like, dipolar-coupled spins 1/2 as well as for spins 1 quadrupolar coupled to local electric field gradients. The prefactor C_{coupl} is a constant specific for the type of the dominating spin coupling. The first spectral density term in the brackets on the right-hand side of Eq. 30 refers to single-quantum, the second to double-quantum transitions. The latter consequently is a function of twice the single-quantum resonance frequency.

Later in the review we will distinguish intrasegment from intersegment dipolar couplings giving rise to two-proton spin–lattice relaxation contributions according to

$$\frac{1}{T_1} = \frac{1}{T_1^{\text{intra}}} + \frac{1}{T_1^{\text{inter}}}. \tag{31}$$

The analytical form of Eq. 31 anticipates stochastic independence of the two types of fluctuating couplings. This appears to be a reasonable assumption since intersegment couplings are modulated by translational segment diffusion, whereas the variations of intrasegment interactions originate from chain modes in the time/frequency range of interest. As a consequence, intrasegment couplings fluctuate much faster than intersegment interactions. The latter therefore matter only at low frequencies. Note that the relaxation of quadrupole nuclei such as deuterons is dominated at any frequency solely by intrasegment interactions to local electric field gradients. Deuteron relaxation is therefore a favorable means for the discrimination of intra- from intersegment relaxation mechanisms.

The overall correlation time of the fluctuating spin couplings is defined by

$$\tau_c = \int_0^\infty G_m(t)dt. \tag{32}$$

The BWR theory resulting in Eq. 30 is valid only in the limit

$$T_1 >> \tau_c. \tag{33}$$

Below we will discuss numerous NMR relaxometry applications to polymers. The information on the type of segment–internal fluctuations, chain modes or center-of-mass displacements is contained in the autocorrelation functions in the time domain or, equivalently, in the spectral density in the frequency domain according to Eq. 29. In order to probe characteristic features of polymer dynamics it is therefore of interest to measure the frequency dependence of the spin–lattice relaxation time or rate, which then directly reflects the spectral density related to the type of segment or chain motion. In the next section we describe methods suitable for the recording of such spin–lattice relaxation "dispersion" curves.

2.2.2 The Field-Cycling NMR Relaxometry Technique

Spin–lattice relaxation dispersions according to Eq. 30 can be recorded over several decades of the frequency with the aid of field-cycling NMR relaxometry [18–21]. Combined with ordinary high-field NMR relaxometry, the accessible ranges for protons and deuterons are

$$10^3\,\mathrm{Hz} < \frac{\omega}{2\pi} < 10^9\,\mathrm{Hz} \text{ and } 10^2\,\mathrm{Hz} < \frac{\omega}{2\pi} < 10^8\,\mathrm{Hz}, \tag{34}$$

respectively. The high-frequency limit is given by the available high-field magnets. At low frequencies several factors may restrict the applicability of the field-cycling technique: (1) the "local field" representing the secular part of the spin couplings may exceed the external flux density in the relaxation interval of the field cycle; (2) the compensation of the earth field (or other magnetic stray fields in the lab) at the sample position may be imperfect; (3) the low-field spin–lattice relaxation times may become too short to permit field switching fast enough for reliable measurements; and (4) the low-field spin–lattice relaxation time may become shorter than the correlation time of the longest correlation function component so that the validity condition of the BWR formula given in Eq. 33 is violated.

Figure 4 schematically shows a typical field cycle. The external magnetic flux density is cycled in a sequence of three different values: the polarization field, the relaxation field, and the detection field. The sample is polarized in the polarization field, B_p, which is chosen as high as compatible with the cooling device of the resistive magnet coil [21]. This takes a couple of spin–lattice relaxation times taken at this particular field value. After that the magnetization is equal to the Curie equilibrium magnetization $M_0(B_p)$.

The magnetic flux density is then electronically switched down to the preselected relaxation field, B_r, at which spin–lattice relaxation is to be examined. On the one hand, the field-switching rate must be large enough to avoid excessive relaxation losses of the magnetization during the switching process. On the other hand, it should be slow enough to permit adiabatic field changes in case the relaxation field is perceptibly superimposed by local fields (of arbitrary directions other than that of $\vec{B}_r$).

In the relaxation field the magnetization is aligned along the external magnetic field direction and is initially equal to the Curie equilibrium magnetization in the polarization field, $M(0)=M_z(0)=M_0(B_p)$, where we have ignored potential relaxation losses during the switching interval. It then relax-

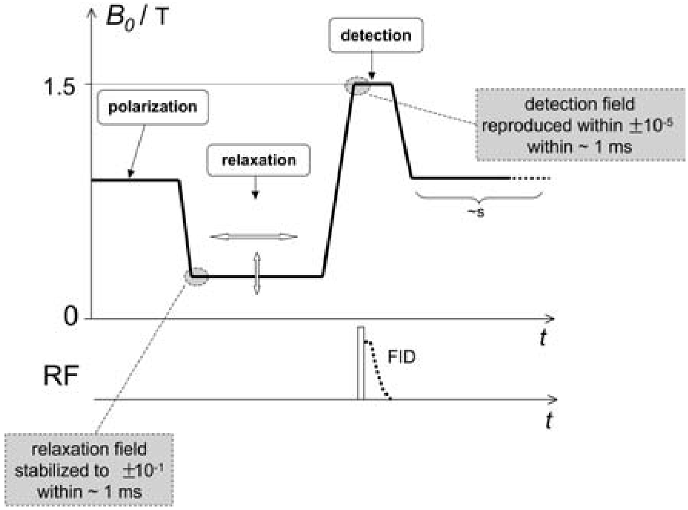


Fig. 4. Schematic representation of a typical cycle of the main magnetic field B_0 employed with field-cycling NMR relaxometry. The free induction decay (FID) is recorded after a 90° radio frequency (RF) pulse in the detection field. The repetition time amounts several times the spin–lattice relaxation time in the polarization field. The most critical sections of the cycle are brought out by *gray boxes*

es toward the new Curie equilibrium magnetization of the relaxation field, $M_0(B_r)$, so that the magnetization evolves according to the following solution of Bloch's equation:

$$M_z(\tau) = M_0(B_r) + \left[M_0\left(B_p\right) - M_0(B_r)\right] \exp\{-\tau/T_1(B_r)\}. \tag{35}$$

The magnetization remaining after the relaxation interval τ is finally detected with the aid of a 90° RF pulse or a suitable spin echo sequence in the form of a NMR signal after switching the magnetic flux density to a fixed detection field. Since the acquisition of a free-induction signal takes only a few milliseconds at most, the detection field period can be kept extremely short and the detection flux density, hence, correspondingly strong without overloading the magnet coil. After having recorded the signal, the flux density is switched back to the polarization field value, and after a period of a couple of spin–lattice relaxation times the whole cycle begins again.

Incrementing the relaxation interval τ thus permits one to record the relaxation curve for a given relaxation flux density B_r. The spin–lattice relaxation dispersion is then scanned point by point by stepping B_r through a series of discrete values spread over the desired flux density (or frequency) range.

It can be shown that relaxation losses in the finite switching intervals diminish the dynamic range of the variation of the relaxation decay and hence the experimental accuracy, but that they do not cause any systematic experimental error provided that the passages between the different field levels are reproducible when incrementing the relaxation interval τ for a given relaxation flux density B_r [2]. However, some care must be taken to avoid experimental artifacts that may arise at the critical field crossover points marked in Fig. 4.

At low fields, proton and especially deuteron spin–lattice relaxation times of viscous polymer systems may easily fall short of a millisecond. That is, coming from the large polarization field of typically 0.5 T, relaxation fields that can be as low as 10^{-5} T must be reached *and* settled and stabilized with a desired accuracy of better than 10% within a total passage time in the order of one millisecond. The short settling time is a stringent condition which is not easy to fulfill practically. Likewise the passage from the relaxation field to the detection field should occur in a transition time of the same order. In particular, the detection flux density needed for magnetic resonance must be hit and reproduced with an accuracy of about 10^{-5} in subsequent transients.

It is therefore of paramount importance to check and calibrate the field cycle with the aid of a test device providing the accuracy, the time resolution and, in particular, the dynamic range required. A fast field probe placed at the sample position connected to a 12 to 16 bit transient recorder with a MHz bandwidth turned out to be a safe way for avoiding any experimental artifacts by imperfections of the field cycle.

Field-cycling NMR relaxometry experiments can be favorably supplemented by ordinary high-field relaxation measurements employing the inversion–recovery or saturation–recovery variants. Comparative spin–lattice relaxation experiments "in the rotating frame" ($T_{1\rho}$) are also of interest particularly in the presence of molecular order [22]. A detailed description of the diverse NMR methods referred to can be found in Ref. [2].

The experimental frequency/temperature "window" opened by NMR relaxometry is schematically shown in Fig. 5. Considering temperatures above the glass transition and below thermal degradation suggests a typical range between 300 and 400 K. The proton/deuteron frequency range accessible by the field-cycling technique in combination with high-field relaxometry is within 10^2 and 10^9 Hz. This window almost completely matches the dynamic range of chain modes of polymers with molecular masses up to 10^5. Only at low temperatures and high frequencies does some influence of local segment–internal fluctuations tend to show up. On the other hand, at high temperatures and low frequencies, center-of-mass motions of polymers with molecular masses below 10^5 may manifest themselves in the form of low-frequency cutoffs of the dispersion curves. However, for the most part, spin–lattice relaxation in the experimental window is dominated by chain modes irrespective of the molecular mass.

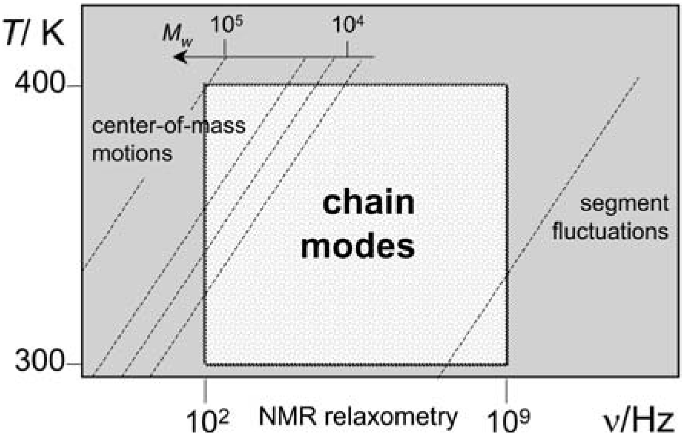


Fig. 5. Schematic representation of the experimental temperature/frequency "window" conveniently accessible by field-cycling NMR relaxometry in combination with conventional high-field techniques. It specifically addresses the chain-mode regime of typical polymers, whereas local segment fluctuations and center-of-mass motions can only be probed at low temperatures/high frequencies and high temperatures/low frequencies/low molecular weights combinations, respectively

2.3
Residual Dipolar Broadening and Transverse Relaxation

The standard BWR theory for transverse relaxation anticipates complete motional averaging of all local fields due to dipolar interaction (or by any other spin coupling) on a time scale corresponding to the linewidth in the absence of molecular motions. This time scale is defined by the reciprocal rigid-lattice linewidth, $(\Delta\omega)_{rl}^{-1}$. The so-called *motional narrowing* condition reads [2]

$$(\Delta\omega)_{rl}\tau_c \ll 1, \tag{36}$$

where τ_c is the correlation time defined in Eq. 32 for m=0.

In the motional narrowing limit, transverse relaxation is governed by spin transitions induced by fluctuating (both secular and nonsecular) spin interactions rather than by spin evolution in secular local fields of the same origin, but without being connected with spin transitions. The distinction between transverse relaxation by spin transitions on the one hand, and by evolution in secular local fields on the other, is best done by considering the spin operator terms in the Hamiltonians of the spin interactions (see the tables given in Chap. 46 of Ref. [2]). Spin operators inducing non-spin-energy conserving transitions are nonsecular by definition. Operators that do not induce spin transitions and operator terms that induce only spin-energy-conserving transitions are secular. Secular spin interactions cause a distribution of the eigenenergies and, hence, a distribution of the local Larmor frequencies. The latter scenario becomes relevant if the motional narrowing condition given by Eq. 36 is violated. With all macromolecules, this will most likely be the case.

The most prominent spin interaction in polymers is dipolar coupling. If motional narrowing according to Eq. 36 is absent or incomplete, one speaks of *dipolar broadening* or *residual dipolar broadening*, respectively. Residual dipolar broadening contains some useful information on chain dynamics. Numerous techniques in principle suitable for retrieving this information have been suggested.

The free-induction decay of transverse magnetization has been analyzed in terms of polymer dynamics [23–26]. A solid echo technique was employed for the same purpose [27, 28]. The so-called dipolar correlation effect on the stimulated echo turned out to be a particularly simple and robust tool in this context too [15, 29, 30]. Finally, double-quantum NMR spectroscopy was suggested [31, 32] as a means of probing features of chain dynamics.

At temperatures far above the glass transition residual dipolar couplings in flexible polymers are getting negligible. Also, heteronuclear species such as ^{13}C are subject to secular dipolar coupling only to a minor degree. Under such circumstances, the (average) transverse relaxation can be studied using standard techniques. Corresponding data reveal characteristic molecular weights and temperatures of entangled polymer systems in a quite spectacular way [33–35]. The crossover from unentangled to entangled chain dynamics was visualized clearly this way. Finally it should be mentioned that transverse relaxation curves are suitable to distinguish the special dynamics of chain-end blocks compared to that of segments in the middle of the polymer [36] by analyzing relaxation curve components representing different segment mobilities.

2.3.1 The Anderson/Weiss Approach

Transverse relaxation under incomplete motional narrowing conditions in multispin systems can be treated with the aid of the Anderson/Weiss approximation [2]. Precession phase shifts caused by spin interactions with many coupling partners is assumed to be subject to the central limit theorem. That is, the distribution of the local fields B_{loc}, i.e., the distribution of the corresponding angular frequency offsets $\Omega=\gamma B_{loc}$, and hence the distribution of the phase shifts accumulated in a given time interval t, is described by a Gaussian function. With this approximation one finds for the normalized transverse relaxation decay function

$$S(t) = \exp\left\{ -\left\langle \Omega^2 \right\rangle_{rl} \int_0^t (t-\tau) g(\tau) d\tau \right\}, \tag{37}$$

where $\langle \Omega^2 \rangle_{rl}$ is the mean squared angular frequency offset (or the *second moment* or the *variance* of the distribution of angular frequency offsets) in the absence of motions (i.e., for the "rigid lattice" limit). The normalized autocorrelation function of the frequency offsets is given by

$$g(\tau) = \frac{\langle \Omega(0)\Omega(\tau) \rangle}{\left\langle \Omega^2 \right\rangle_{rl}}. \tag{38}$$

The brackets indicate ensemble averages. An analogous formalism for the dipolar correlation effect can be found in Refs. [2, 15, 16].

Expressing the autocorrelation function given in Eq. 38 by its Fourier transform, the spectral density $I(\omega)$,

$$g(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} I(\Omega) e^{i\Omega\tau} d\Omega, \tag{39}$$

leads to [2]

$$S(t) = \exp\left\{ -\frac{1}{\pi} \int_0^{\infty} \frac{\left\langle \Omega^2 \right\rangle_{rl}}{\Omega^2} I(\Omega) [1 - \cos(\Omega t)] d\Omega \right\}. \tag{40}$$

With incomplete motional narrowing, the transverse relaxation decay obviously reflects fluctuations with rates $\Omega l t; \sqrt{\left\langle \Omega^2 \right\rangle_{rl}}$, that is a range complementary and partly overlapping with that of field-cycling NMR relaxometry. The representation given in Eq. 40 is particularly useful for the distinction of different dynamic components (see also Fig. 18).

3
Modeling of Chain Dynamics and Predictions for NMR Measurands

Polymers are molecules subject to complex intra- and intermolecular interactions combined with many intramolecular degrees of motional freedom. The degree of polymerization ϵ typically ranges from 10^2 to 10^6. Assuming only three rotamers per monomer, the number of possible conformations already takes the astronomical number value of the order 3^{ϵ}. The time constants characterizing the molecular motions are spread over up to six orders of magnitude. In view of these facts, it is a rather demanding task to model polymers and their molecular dynamics.

At room temperatures $T \approx 300$ K, conformational transitions typically take $\tau_s = 10^{-11}$–10^{-9} s. All elementary conformational changes displace atoms of the polymer on a distance length scale $l \leq 1$ nm. These characteristic quantities permit the distinction of "local" and "global" properties.

A detailed description of statically or dynamically local properties must be based on the stereochemical composition of the monomers and their potential energy landscape [1, 37–44]. On the other hand, the global, i.e., large-scale, properties of macromolecules tend to be insensitive to the stereochemical details of the monomers. Therefore, a coarse-grained description becomes adequate by considering model chains. All chemical specificity can then be represented by a few characteristic parameters to be specified in the following.

A number of more or less equivalent chain models have been suggested [1, 42]. We will discuss only the simplest and most popular one, that is the "random walk" or "freely jointed" chain. Macromolecules are represented (i.e., approximated) by a linear sequence of Kuhn segments of length b (see Fig. 6). The number of monomers per Kuhn segment is taken just large enough to permit the neglect of any stereochemical restriction of the orientation of the Kuhn segments relative to each other. There is no mutual orientation correlation. Merely the linear order of segments, that is the "linear memory", is retained. The conformational statistics of such a freely jointed chain of N Kuhn segments $N >> 1$ is mathematically equivalent to random walks in three-dimensional space.

In the Rouse model to be discussed below, each Kuhn segment is subdivided into a "bead" (where the segment mass is concentrated) and a massless "spring". The elasticity of these springs is based on the conformational entropy of the segments which depends on how far they are stretched. This bead/spring model is characterized by the following basic parameters partially also illustrated in Fig. 6.

The *Kuhn segment length* is defined by the root mean squared value

$$b \equiv \sqrt{\langle x^2 \rangle} \tag{41}$$

of the lengths x of the segments in a chain at a given instant. Likewise the *Flory radius* represents the root mean squared value

$$R_F \equiv \sqrt{\left\langle \vec{R}_{ee}^2 \right\rangle} = \sqrt{Nb^2} \tag{42}$$

of the chain end-to-end vector $\vec{R}_{ee}$ in an ensemble of independent polymer chains with N segments each. Note that $\left\langle \vec{R}_{ee} \right\rangle = 0$ in an isotropic distribution of Kuhn segments, i.e., in the absence of an external force field or any mesomorphic order. Equations 41 and 42 anticipate that the length distribution is Gaussian.

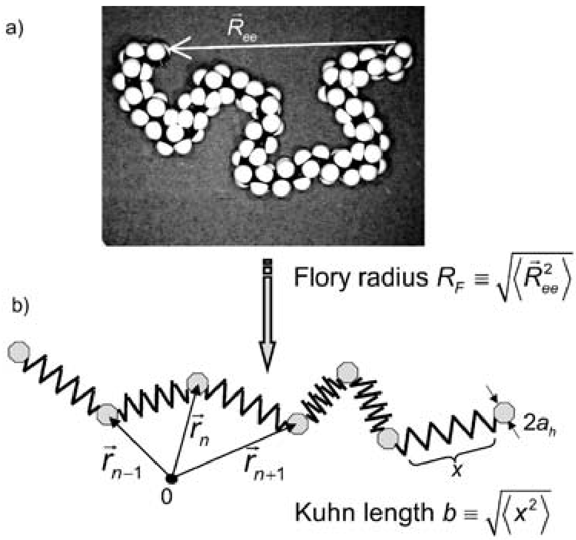


Fig. 6. Freely jointed chain of Kuhn segments and bead/spring chain model. For coarse-grain treatments for long time and length scales, the polymer may be modeled by a chain of beads and massless entropic springs. The beads are assumed to have a hydrodynamic radius a_h. The Kuhn segment length, b, is defined as the root mean squared length of the springs. The root mean squared chain end-to-end distance is called the Flory radius R_F

The freely jointed segment chain model provides a satisfactory description of large-scale conformational properties of real chains in polymer melts and concentrated solutions, where effects of excluded-volume distortions of the (Gaussian) conformational statistics can be neglected [1, 37–44]. Typical orders of magnitude for flexible polymers such as polyethylene (PE), polyethylene oxide (PEO), polystyrene (PS) etc. are $b \simeq 0.5 \div 2$ nm, $N \simeq 10 \div 10^4$, $R_F \simeq 2 \div 10^2$ nm. A Kuhn segment roughly represents 2–5 monomers. Numbers like these can be estimated based on the conformational statistics expected under the restrictions of the conformational degrees of freedom, that is, for the angles between the valence bonds in the real chain. They can also be evaluated from experimental data as empirical characteristic parameters of the macromolecule [37–39].

3.1 The Rouse Model

The dynamics of an isolated Kuhn segment chain in its bead-and-spring form is considered in a viscous medium without hydrodynamic backflow or excluded-volume effects. The treatment is based on the Langevin equation generalized for Brownian particles with internal degrees of freedom. A first, crude formalism of this sort was reported by Kargin and Slonimskii [45]. Independently of this work, Rouse published later a more detailed version [46] which is nowadays referred to as the "Rouse model".

The effective intramolecular interactions between the segments are approximated by entropic harmonic interactions, reflecting the Gaussian character of the large-scale chain conformation. Intermolecular interactions (with the surrounding viscous medium) are taken into account by friction and stochastic forces acting on the segments.

The *entropic spring constant* and the *friction coefficient* of a Kuhn segment are given by

$$K = \frac{3k_B T}{b^2} \tag{43}$$

and

$$\zeta = 6\pi\eta a_h, \tag{44}$$

respectively, where k_B is Boltzmann's constant, T is the absolute temperature, η is the viscosity of the medium surrounding the segment, and a_h is the hydrodynamic radius of the segments.

The equation of motion of the n^{th} segment (see Fig. 6b) reads

$$\frac{\partial}{\partial t}\vec{p}_n(t) = K(2\vec{r}_n - \vec{r}_{n+1} - \vec{r}_{n-1}) - \zeta\frac{\partial\vec{r}_n}{\partial t} + \vec{f}_n^L(t), \tag{45}$$

where $\vec{p}_n(t)$ is the momentum, and $\vec{f}_n^L(t)$ is the Langevin stochastic force acting on this segment. In the continuum limit, the segment number n can be treated as a continuous variable ranging from 0 to N, so that Eq. 45 can be rewritten in approximate form as

$$\frac{\partial}{\partial t}\vec{p}_n(t) \approx K\frac{\partial^2\vec{r}_n}{\partial n^2} - \zeta\frac{\partial\vec{r}_n}{\partial t} + \vec{f}_n^L(t). \tag{46}$$

The boundary conditions reflecting the existence of only one neighbor segment at the chain ends are

$$\left.\frac{\partial}{\partial n}\vec{r}_n(t)\right|_{n=0,N} = 0. \tag{47}$$

The first term on the right-hand side of Eq. 46 represents an intramolecular force whereas the second and third terms are forces of an intermolecular nature.

The inertial force term $\frac{\partial}{\partial t}\vec{p}_n(t)$ on the left-hand side of Eq. 46 is important only for times $t << m/\zeta$, where m is the mass of a Kuhn segment. For typical experimental situations with polymer melts one estimates $m/\zeta \div 10^{-13}$–10^{-12} s. That is, the inertial force term can be neglected on the time scale relevant for chain dynamics studies. Equation 46 thus reads

$$\zeta \frac{\partial}{\partial t} \vec{r}_n(t) \approx \frac{3k_B T}{b^2} \frac{\partial^2 \vec{r}_n(t)}{\partial n^2} + \vec{f}_n^L(t). \tag{48}$$

The differential operator $\frac{3k_BT}{b^2}\frac{\partial^2}{\partial n^2}$ represents entropically elastic forces and has a discrete set of eigenfunctions. In this context, these eigenfunctions are called the *Rouse normal coordinates* defined by

$$\vec{X}_p(t) = \frac{1}{N} \int_0^N dn \cos\left(\frac{\pi p}{N} n\right) \vec{r}_n(t), \tag{49}$$

where p=0, 1, 2, ... is the normal-mode number. Strictly speaking, this series of eigenfunctions is infinite. However, normal coordinates with mode numbers p=N, N+1 ... must be considered as a mathematical artifact owing to the replacement of a discrete set of ordinary differential equations (see Eq. 45) by partial differential equations in the continuum limit, Eq. 48. Fortunately for most physical quantities of interest here, it does not matter whether the set of normal coordinates is limited or not provided that $N >> 1$.

The equation of motion given by Eq. 48 can be solved by combining it with the normal coordinate expression defined in Eq. 49. Forming the correlation function $\langle \vec{X}_p(t) \cdot \vec{X}_{p'}(0) \rangle$ of the normal coordinates leads then to

$$\langle \vec{X}_p(t) \cdot \vec{X}_{p'}(0) \rangle = \delta_{pp'} \frac{Nb^2}{2\pi^2 p^2} \exp\left\{-\frac{t}{\tau_p}\right\}, \tag{50}$$

where we have employed the fact that the Cartesian components α and β (where $\alpha,\beta = x,y,z$) of the Langevin stochastic forces acting on the n^{th} and k^{th} segments at times t_2 and t_1, respectively, are correlated as

$$\left\langle f_{\alpha n}^L(t_2) f_{\beta n}^L(t_1) \right\rangle = 2\zeta k_B T \delta_{\alpha\beta} \delta(t_2 - t_1) \delta(n - k). \tag{51}$$

Equation 50 represents the Rouse modes in the proper sense with the relaxation time of the p^{th} mode

$$\tau_p = \tau_s \left(\frac{N}{p}\right)^2. \tag{52}$$

The relaxation time of the N^{th} Rouse mode,

$$\tau_s = \frac{b^2 \zeta}{3\pi^2 k_B T}, \tag{53}$$

is called the *segmental relaxation time*, referring to the mode with the shortest relaxation time in the frame of the Rouse model. On the other hand, the longest relaxation time associated with the mode number p=1 is the so-called *terminal chain relaxation time for the Rouse model* or simply *Rouse relaxation time*. It is given by

$$\tau_1 \equiv \tau_R = \tau_s N^2. \tag{54}$$

The Kronecker symbol $\delta_{pp'}$ in Eq. 50 indicates that the normal modes are mutually orthogonal to each other. That is, the Rouse modes are in reality referring to the autocorrelation functions

$$C_p(t) = \left\langle \vec{X}_p(t) \cdot \vec{X}_p(0) \right\rangle \tag{55}$$

of the normal coordinates. This is the function we will refer to in the treatments of experimental NMR parameters of interest in this context.

3.1.1 Translational Segment Diffusion of a Rouse Chain

The mean squared displacement of the n^{th} Kuhn segment can be expressed in the following way:

$$\begin{aligned} \left\langle (\vec{r}_n(t) - \vec{r}_n(0))^2 \right\rangle &= \left\langle \left(\vec{X}_0(t) - \vec{X}_0(0)\right)^2 \right\rangle + 8\sum_{p=1}^{N-1} \left[C_p(t) - C_p(0)\right] \cos^2\left(\frac{\pi}{N} pn\right) \\ &\equiv g_{cm}(t) + g_{rel}^n(t). \end{aligned} \tag{56}$$

The first term on the right-hand side represents the mean squared displacement of the chain center-of-mass, $g_{cm}(t)$, the second term refers to displacements of segment n relative to the center-of-mass, $g_{rel}^n(t)$. Averaging Eq. 56 over all segment positions n gives

$$\begin{aligned} \left\langle \vec{R}^2(t) \right\rangle \equiv \left\langle (\vec{r}(t) - \vec{r}(0))^2 \right\rangle &= \left\langle \left(\vec{X}_0(t) - \vec{X}_0(0)\right)^2 \right\rangle + 4\sum_{p=1}^{N-1} \left[C_p(t) - C_p(0)\right] \\ &\equiv g_{cm}(t) + g_{rel}(t), \end{aligned} \tag{57}$$

where $g_{rel}(t) = \frac{1}{N} \sum_{n=1}^{N-1} g_{rel}^n(t)$. Combining Eqs. 57 and 50 leads to the following limits for the mean squared segment displacement in the frame of the Rouse model:

$$\left\langle \vec{R}^2(t) \right\rangle = \begin{cases} \frac{2}{\sqrt{\pi^3}} b^2 \sqrt{\frac{t}{\tau_s}} & \text{for} \quad \tau_s \ll t \ll \tau_R \\ 6D_R t & \text{for} \quad t \gg \tau_R \end{cases}, \tag{58}$$

where

$$D_R = \frac{1}{3\pi^2} \frac{b^2}{\tau_s N} \tag{59}$$

is the Rouse chain diffusion coefficient. On a time scale shorter than the Rouse relaxation time, segment diffusion is subdiffusive, and the mean squared displacement follows a power law $\langle R^2\rangle \propto N^{0} t^{1/2}$. This is in contrast to the behavior at times longer than τ_R where the relation $\langle R^2\rangle \propto N^{-1} t^{1}$ holds.

3.1.2
Spin–Lattice Relaxation of a Rouse Chain

For the treatment of spin–lattice relaxation, we need an expression for the autocorrelation function of the Kuhn segment tangent vector, which in the continuum limit is given by

$$\vec{b}_n(t) = \frac{\partial}{\partial n}\vec{r}_n(t) = -\frac{2\pi}{N}\sum_{p=1}^{N-1}\vec{X}_p p \sin\left(\frac{\pi}{N}pn\right). \tag{60}$$

The autocorrelation function of this vector is readily found as an expression based on the autocorrelation function of the normal coordinates, Eq. 55,

$$\left\langle \vec{b}(t)\cdot\vec{b}(0)\right\rangle = \frac{2\pi^2}{N^2}\sum_{p=1}^{N-1} p^2 C_p(t), \tag{61}$$

where we have averaged over all segments n, and where the orthogonality of the normal coordinates was taken into account again. In analogy to Eq. 58 the following limits can be derived for this tangent vector autocorrelation function:

$$\left\langle \vec{b}(t)\cdot\vec{b}(0)\right\rangle = \begin{cases} \frac{\sqrt{\pi}}{2} b^2 \left(\frac{\tau_s}{t}\right)^{1/2} & \text{for } \tau_s \ll t \ll \tau_R \\ \frac{b^2}{N}\exp\left\{-\frac{t}{\tau_R}\right\} & \text{for } t \gg \tau_R \end{cases}. \tag{62}$$

According to Refs. [47–49], the spin–lattice relaxation rate is related to the tangent vector autocorrelation function by

$$\frac{1}{T_1} \propto \int_{-\infty}^{\infty} \left\langle \vec{b}(t)\cdot\vec{b}(0)\right\rangle^2 e^{-i\omega t} dt. \tag{63}$$

This expression is based on a variant of Eq. 28 especially suited for the description of chain modes [47]. Inserting Eq. 62 we find the limits

$$\frac{1}{T_1} \propto \begin{cases} -\tau_s \ln(\omega\tau_s) & \text{for } \tau_R^{-1} \ll \omega \ll \tau_s^{-1} \\ \omega^0 \tau_s \ln N & \text{for } \omega \ll \tau_R^{-1} \end{cases}. \tag{64}$$

The logarithmic frequency dependence [50, 51] in the limit $\tau_R^{-1} << \omega << \tau_s^{-1}$ will be of particular interest for experimental tests of the model.

Equations 63 and 64 are valid for intrasegment spin interactions. Since the Rouse model is experimentally relevant mainly for modes of relatively short chains or short chain sections, any contribution from intersegment dipolar interactions are negligible on this time/frequency scale, and need not be treated explicitly.

3.2
The Tube/Reptation Model

The Rouse model describes the dynamical properties of melts of macromolecules of a relatively small number of Kuhn segments, $N \leq N_c$, reasonably well. The critical number N_c is the number of Kuhn segments for the critical molecular mass M_c. Flexible polymers have critical Kuhn segment numbers typically in the range $N_c=40 \div 60$ [1, 42–44, 52]. On the other hand, chain dynamics in concentrated systems of polymers with $N>>N_c$ is much slower than expected on the basis of the Rouse model. Alluding to chain entanglements that are considered to become relevant in this case, one speaks of "entangled" dynamics. For example, experimental terminal relaxation times and center-of-mass self-diffusion coefficients scale as $\tau_t \propto N^{3 \div 3.4}$ and $D \propto N^{-2 \div 2.5}$, whereas the Rouse model predicts much weaker dependences, $\tau_R \propto N^2$ and $D \propto N^{-1}$, respectively.

The term "entanglement" is understood as the global representation of interchain interactions, which have not been considered in the Rouse equation of motion, Eq. 46. The Rouse equation of motion is the simplest way to describe effectively single-chain dynamics. In the frame of this model, intermolecular interactions are merely accounted for by way of a molecular mass independent local friction coefficient ζ. The fact that the Rouse theory predicts a chain length-dependent relaxation time, $\tau_R=\tau_s N^2$, already indicates some internal inconsistency, because this molecular weight dependence in combination with the uncrossability property of polymer chains (the "connectivity restriction") suggests a molecular weight dependent friction. That is, the applicability of the Rouse model is rather limited (see also Ref. [58] and the literature cited therein). In the following, two of the most prominent attempts to take entanglement interactions into account will be discussed in more detail. The first model of interest is the tube/reptation model.

The tube concept and the reptation model were originally introduced by Edwards [53] and de Gennes [54], respectively, and then investigated in more detail by Doi and Edwards [1, 55] and many other authors [56–66]. Entanglement interactions between neighboring chains are modeled by a fictitious tube around the chain referred to. This chain is also termed the "tagged chain" in an ensemble of otherwise identical chains. All other macromolecules form the so-called matrix.

Let us assume for the moment that the matrix is frozen while the tagged chain is still mobile. The tagged chain cannot cross contours of matrix chains due to intermolecular *excluded volume interactions* and the *chain connectivity*. It appears intuitively self-evident that the motions of the tagged chain will be restricted to a tube-like region of space around the initial chain conformation. After some time, the only degree of translational freedom, namely displacements of the tagged chain in this tube back and forth, will thread the tagged chain from the "initial tube" into a "new tube". The diameter of this sort of tube would be equal to typical nearest neighbor distances between two monomers from different chains, d=0.3–0.5 nm.

In real polymer melts the matrix is as mobile as the tagged chain. In the tube/reptation model this is attempted to be taken into account by assuming a fictitious tube of a much larger effective diameter (typically 5–7 nm). It is postulated that segment motions become highly anisotropic for times $t >> \tau_e=\tau_s N_e^2$, where $N_e \approx N_c/2$. Intermolecular entanglement forces effectively localize the macromolecule in a space region in the form of a curved tube of diameter $d=bN_e^{1/2}$ around the *contour line* or *primitive path* of the chain (see Fig. 7). The tube conformation corresponds to that of a freely jointed chain, that is, the tube conformation has a Gaussian end-to-end length distribution. The characteristic time

$$\tau_e \equiv \tau_s N_e^2 \approx \frac{1}{4}\tau_s N_c^2 \tag{65}$$

is called the *entanglement time*, and N_e is the number of Kuhn segments effectively corresponding to an "entanglement".

Segments moving inside the tube are considered to move in a viscous medium as it is anticipated in the Rouse model. In the interior of the tube, entanglement interactions are assumed to be absent. However, as soon as a segment approaches the tube "walls", entanglement forces become infinitely large so that the polymer is effectively confined by the topological constraints imposed by the tube. The chain-end segments are excepted from this restraint: they are not subject to entanglement forces and can therefore move isotropically. The consequence is that the chain can migrate in a reptile manner (*reptation*) into a new conformation completely uncorrelated to that of the initial chain coil.

3.2.1 The Doi/Edwards Limits

There are four dynamic processes characterized by the four characteristic time constants of the tube/reptation model (see Fig. 7): the segment fluctuation time τ_s (given in Eq. 53), the entanglement time τ_e (given in Eq. 65), the (longest) Rouse relaxation time τ_R (given in Eq. 54), and the (tube) disengagement time τ_d (given below in Eq. 71). With these four time constants, four dynamic limits can be defined, which will be termed the Doi/Edwards limits $(\mathrm{I})_{\mathrm{DE}}$ to $(\mathrm{IV})_{\mathrm{DE}}$. Table 1 shows the limiting laws for the mean squared

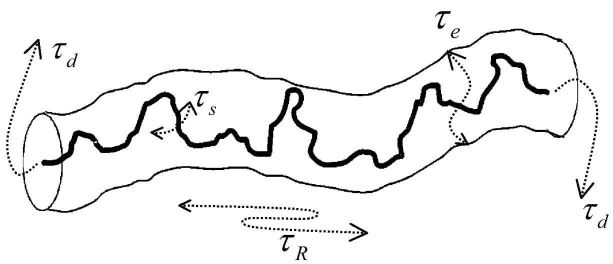


Fig. 7. Illustration of a polymer chain (the tagged chain) confined in the fictitious tube of diameter d formed by the matrix. The contour line of the tube is called the primitive path having a random-walk conformation with a step length a=d. The four characteristic types of dynamic processes (*dotted arrow lines*) and their time constants τ_s, τ_e, τ_R, and τ_d defined in the frame of the Doi/Edwards tube/reptation model are indicated

segment displacement and the spin–lattice relaxation time commonly expected on this basis. Figure 7 illustrates the chain processes dominating in the four limits.

Limit (I)$_{\mathrm{DE}}$ ($\tau_s << t << \tau_e$) represents Rouse dynamics. Entanglement interactions are not yet effective. Segment displacements are isotropic. On this time scale the segments of the tagged chain can be considered to be surrounded by an isotropic viscous medium just as assumed in the Rouse model. The length scale of the Rouse modes is however restricted by the diameter of the fictitious tube. If a regime like this exists, then it should reveal itself by the mean squared segment displacement and the (intrasegment) spin–lattice relaxation time laws

$$\left\langle R^2 \right\rangle \propto M^0 t^{1/2} \tag{66}$$

and

$$T_1 \propto -M^0 / \ln(\omega \tau_s), \tag{67}$$

respectively (Table 1).

On a longer time scale, in **limit (II)$_{\mathrm{DE}}$** ($\tau_e << t << \tau_R$), the tube constraints become effective, and chain modes are consequently connected with elongations along the tube contour (the "primitive path"). The distribution of relaxation times extends up to the value known as the (longest) Rouse relaxation time given in 54. That is, segment motions are anisotropic on the time scale of this limit. Since segments distant enough from each other are fluctuating still independently, the mean squared segment displacement remains molecular mass independent. Two particularly characteristic power laws for limit (II)$_{\mathrm{DE}}$ refer to the mean squared segment displacement,

$$\left\langle R^2 \right\rangle \propto M^0 t^{1/4}, \tag{68}$$

Table 1. Theoretical dependences on time (t), angular frequency (ω), and molecular mass (M) predicted by the tube/reptation model for the mean squared segment displacement and the intrasegment spin–lattice relaxation time in the four Doi/Edwards limits. The factors C_I, C_{II}, C_{III}, and C_N are frequency and molecular mass independent constants

	Limit	Mean squared segment displacement $\langle R^2\rangle=$	Refs	Intrasegment spin–lattice relaxation time $T_1^{\text{intra}}=$	Refs
$(\mathrm{I})_{\mathrm{DE}}$	$\tau_s << (t, 1/\omega) << \tau_e$	$\frac{2}{\pi^{3/2}} b^2 \left(\frac{t}{\tau_s}\right)^{1/2}$ $\propto M^0 t^{1/2}$	[1]	$-C_I \frac{M^0}{\tau_s \ln(\omega\tau_s)}$	[47, 50]
$(\mathrm{II})_{\mathrm{DE}}$	$\tau_e << (t, 1/\omega) << \tau_R$	$b^2 N_e^{1/2} \left(\frac{t}{\tau_s}\right)^{1/4}$ $\propto M^0 t^{1/4}$	[1, 54]	$C_{II} M^0 \omega^{3/4}$	[54, 70]
$(\mathrm{III})_{\mathrm{DE}}$	$\tau_R << (t, 1/\omega) << \tau_d$	$\frac{2}{\pi} b^2 \left(\frac{N_e}{3N}\frac{t}{\tau_s}\right)^{1/2}$ $\propto M^{-1/2} t^{1/2}$	[1, 54]	$C_{III} M^{-1/2} \omega^{1/2}$	[54]
$(\mathrm{IV})_{\mathrm{DE}}$	$\tau_d << (t, 1/\omega)$	$2 \frac{k_B T}{\zeta} \frac{N_e}{N^2} t$ $\propto M^{-2} t^1$	[1,54]	$C_{IV} M^{-(1.5 \ldots 2.0)} \omega^0$	[123]

and to the (intrasegment) spin–lattice relaxation time,

$$T_1 \propto M^0 \omega^{3/4} \tag{69}$$

(Table 1). Equations 68 and 69 are totally equivalent as can be seen from a simple probability consideration. In the frame of this and the following time limit, the correlation function given in Eq. 28 can essentially be considered to represent the probability that the spin-bearing segment is still or again in the tube section where it was initially [54, 70, 71]. That is,

$$G_m(t) \propto \frac{a^2}{\langle R^2(t)\rangle}, \tag{70}$$

where $a = b\sqrt{N_e}$ is the step length of the primitive path. Inserting Eq. 68 and carrying out the Fourier transform required by Eqs. 29 and 30 directly leads to the proportionality given in Eq. 69.

Reptation in the proper sense begins to show up only beyond the Rouse time scale, namely in **limit (III)$_{\mathbf{DE}}$** defined by $\tau_R << t << \tau_d$. The disengagement time is defined by

$$\tau_d = 3\tau_s \frac{N^3}{N_e}, \tag{71}$$

that is, $\tau_d >> \tau_R$ for $N >> N_e$. Segment motions are still anisotropic in the sense of correlated fluctuations along the tube contour. The chain moves coherently as a whole and effectively like a single random walker back and forth. The mean squared segment displacement in curvilinear coordinates along the tube contour is a linear function of time. However, if considered in Euclidean space it is proportional to the square root of time as a consequence of the Gaussian tube conformation. Since the chain moves as a whole and, hence, is subject to friction proportional to the chain length, the mean squared curvilinear displacement is now inversely proportional to the chain length, whereas mean squared displacements measured in the Euclidean space are inversely proportional to the square root of the chain length. The latter is again a consequence of the random coil character anticipated for the tube. The power laws characteristic for this limit are

$$\left\langle R^2 \right\rangle \propto M^{-1/2} t^{1/2} \tag{72}$$

and

$$T_1 \propto M^{-1/2} \omega^{1/2}, \tag{73}$$

where the latter refers to intrasegment spin interactions (Table 1). The complete equivalence of Eqs. 72 and 73 can be shown the same way as above with the aid of Eq. 70.

Finally, for $t >> \tau_d$ **limit (IV)$_{\mathrm{DE}}$** comes into play. This is the regime of normal diffusion, when all correlations with the initial conformation are lost, and the mean squared displacement obeys the Einstein relation

$$\left\langle R^2 \right\rangle = 6 D_c t, \tag{74}$$

where

$$D_c = \frac{k_B T}{3\zeta} \frac{N_e}{N^2} \tag{75}$$

is the isotropic center-of-mass self-diffusion coefficient. The dispersion of the spin–lattice relaxation time becomes independent of the frequency in the form of a low-frequency plateau, as is expected for isotropic rotational tumbling of the spin-bearing object in the so-called extreme narrowing limit reading $\omega\tau_d << 1$ in the present case.

Note that all limits of the time and molecular weight dependences of the mean squared segment displacement discussed so far are reflected by corresponding limits of the frequency and molecular weight dependences of the spin–lattice relaxation time (see Table 1). Experimental tests based on both phenomena therefore appear to be particularly favorable.

3.2.2
Echo Attenuation by Reptation

The evaluation formulas given in Eq. 20 or 25 for attenuation of the stimulated echo by diffusion anticipates Gaussian propagators. As illustrated in Fig. 8, waiting time delays matter in NMR diffusion measurements with polymers subject to reptation [72, 73]. This follows from the fact that segments diffusing along the curvilinear path of the randomly coiled tube contour are intermittently "trapped" in loops that contribute little to displacements along the field gradient direction. Effectively non-Gaussian propagators are the consequence for the Doi/Edwards limits $(\mathrm{II})_{\mathrm{DE}}$ and $(\mathrm{III})_{\mathrm{DE}}$. One may try an approximate evaluation according to Eq. 16. However, it turned out to be more indicative for the characteristic reptation features to consider a special formalism specifically developed for the reptation problem [3].

In the short gradient "pulse" limit of the fringe-field NMR diffusometry technique, which is of particular interest in context with slowly diffusing polymers (see Fig. 3), the effective diffusion time is given by $t \approx \Delta = \tau_2$. Under this prerequisite the stimulated-echo attenuation factor can be analyzed according to

$$A_{diff}(k,t) = \left\langle e^{i\vec{k}\cdot\vec{R}(t)} \right\rangle_R = \underbrace{\left\langle e^{i\vec{k}\cdot\vec{R}_s(t)} \right\rangle_{R_s}}_{\text{segments; } A_s(k,t)} \underbrace{\left\langle e^{i\vec{k}\cdot\vec{R}_c(t)} \right\rangle_{R_c}}_{\text{center-of-mass; } A_c(k,t)} , \tag{76}$$

where we have anticipated that segment and center-of-mass displacements are independent of each other because of the different time scales on which these processes take place.

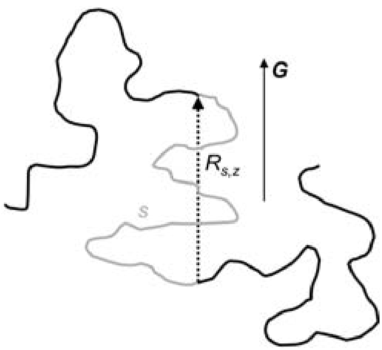


Fig. 8. Polymer segments in a "tube" are displaced along the randomly coiled contour line resulting in the "curvilinear displacement" s in the experimental diffusion time (drawn in *gray*). Field-gradient NMR diffusometry measures displacements $R_{s,z}$ along the gradient direction (**G**). Contour loops act as "traps" for segments the displacements of which are monitored along **G**

The ensemble averages indicated by the brackets refer to displacements along the field gradient direction. The subscripts of the ensemble averages indicate displacements R in general, displacements R_s relative to the center-of-mass by segment diffusion (limits $(\mathrm{I})_{\mathrm{DE}}$, $(\mathrm{II})_{\mathrm{DE}}$, and $(\mathrm{III})_{\mathrm{DE}}$), and displacements of the center-of-mass R_c (limit $(\mathrm{IV})_{\mathrm{DE}}$), respectively.

Attenuation by center-of-mass diffusion (limit $(\mathrm{IV})_{\mathrm{DE}}$) is described by the standard expression given in Eqs. 13 or 26

$$A_c(k,t) = \exp\left\{-k^2 D_c t\right\}, \tag{77}$$

where D_c is the center-of-mass diffusion coefficient of the whole chain (see Eq. 75). In limits $(\mathrm{II})_{\mathrm{DE}}$ and $(\mathrm{III})_{\mathrm{DE}}$, displacements achieved in the interval of limit $(\mathrm{I})_{\mathrm{DE}}$ due to free Rouse modes are relatively small and can safely be neglected. The spin-echo attenuation factor is then dominated [3] by

$$A_s(k,t) = \underbrace{\left\langle \underbrace{\int \left(\frac{4\pi}{3} a|s|\right)^{-3/2} e^{R_s^2/a|s|} e^{i\vec{k}\cdot\vec{R}_s(t)} d^3\vec{R}_s}_{\text{average over all } R_s \text{ for a given } s} \right\rangle}_{\text{average over all } s}$$

$$= \exp\left\{\frac{k^4 a^2 \langle s^2(t)\rangle}{72}\right\} \operatorname{erfc}\left\{\frac{k^2 a\sqrt{\langle s^2(t)\rangle}}{6\sqrt{2}}\right\} \tag{78}$$

where the mean squared *curvilinear* segment displacement,

$$\left\langle s^2(t)\right\rangle = \frac{2D_0 t}{N + \frac{12a^2 D_0 t}{N^2 b^4}} + \frac{2b\sqrt{D_0 t}}{\sqrt{3\pi} + 18\frac{\sqrt{D_0 t}}{Nb}} \tag{79}$$

combines the expressions given in [1] for limits $(\mathrm{II})_{\mathrm{DE}}$ and $(\mathrm{III})_{\mathrm{DE}}$. The step length of the so-called primitive path is denoted by $a = b\sqrt{N_e}$. The quantity $D_0 = k_B T/\zeta$ is the (fictitious) diffusivity of a Kuhn segment in the absence of chain connectivity.

3.2.3 General Remarks on the Tube/Reptation Model

The tube diameter, $d = a = bN_e^{1/2}$, is the main phenomenological parameter of the model. Actually this is a fitting parameter for the description of the viscoelastic plateau on the basis of the tube/reptation formalism [1, 42]. Typical values for melts of flexible polymers are in the mesoscopic range of d = 5–7 nm. This is much larger than the intermolecular distance of typically σ=0.3–0.5 nm. The inequality $d >> \sigma$ partly spoils the uniformity and the logic of the picture inherent to the tube/reptation model: that large tubes contain many chains and not just a single "tagged chain". Nevertheless, Rouse displacement dynamics is anticipated on a length scale up to the tube diameter. That is, the chain connectivity constraint and excluded-volume interactions are ignored for this limit. In the frame of this model, these restraints are taken into account only outside the tube but not inside.

A further inconsistency refers to the treatment of shear stress relaxation. The Doi/Edwards tube cannot exist without long-living intermolecular correlations pertaining to a time scale of the order $\tau_d = \frac{1}{3}\tau_s N^3/N_e$. Nevertheless, in the calculation of the stress tensor on the basis of the tube/reptation model all intermolecular correlations are considered to be fast decaying so that their neglect appears to be justified. That is, the stress tensor is treated on the basis of intramolecular interactions only by ignoring all intermolecular contributions [66–69, 74–77].

A number of modified versions of the tube/reptation model have been suggested in the literature [56, 57, 59–65] in order to improve the description of experimental data. Terms like "contour length fluctuation", "reptons", "double reptation", "constraint release" etc. have been introduced. The main goal of these modifications was to account for finite-length effects in the sense that segments near the chain ends behave in a different way compared to segments in the middle, and that a tube formed of chains of finite lengths cannot be really static. Nevertheless, all of these modifications adopt the analytical features of the original tube/reptation model in the limit $N \rightarrow \infty$. The changes are expected to become effective for the Doi/Edwards limits $(\mathrm{III})_{\mathrm{DE}}$ and $(\mathrm{IV})_{\mathrm{DE}}$, but not for $(\mathrm{I})_{\mathrm{DE}}$ and $(\mathrm{II})_{\mathrm{DE}}$. These corrections unavoidably require the introduction of new phenomenological elements or parameters lacking a clear, mathematically well-defined, microscopic basis. The improved coincidence between theory and experiment is thus paid for by complications and ambiguities in the formalism.

Numerous computer simulations have been carried out in order to examine the transition from Rouse to reptation dynamics [70, 78–88]. Entanglement effects on chain dynamics clearly showed up. However, the discussion as to what extent the characteristic features of the reptation model for concentrated polymer liquids are verified by these simulations still remains controversial. It also should be mentioned that a series of phenomenological "nonreptative" models were published recently [89–94]. They mainly focus on viscoelastic properties of entangled polymer systems.

Below we will come back to the reptation model in context with the dynamics of polymers confined in tube-like pores formed by a solid matrix. For a system of this sort the predictions for limits $(\mathrm{II})_{\mathrm{DE}}$ and $(\mathrm{III})_{\mathrm{DE}}$ (see Table 1) could be verified with the aid of NMR experiments [11, 95] as well as with an analytical formalism for a harmonic radial potential and a Monte Carlo simulation for hard-pore walls [70]. The latter also revealed the crossover from Rouse to reptation dynamics when the pore diameter is decreased from infinity to values below the Flory radius.

3.3 Memory Function Formalisms

The illustrative and intuitive nature of the tube/reptation model is juxtaposed by its partly phenomenological character that makes it impossible to trace the formalism back to elementary principles on a molecular level. The strategic idea to project the whole multiparticle complexity of an entangled polymer system to the assumption of a fictitious tube, so to speak, is the beginning and the end of our thinking in the frame of this model.

The memory function approaches to be outlined in the following are of a more formal nature but nevertheless are relying on some assumptions that are not yet founded on basic principles. At the present state of the art some phenomenological premiss is unavoidable. However, memory function approaches are analytically more flexible and can be adapted to experimental data by scrutinizing the general laws anticipated for short-time segmental dynamics. On the other hand, they certainly do not inspire our imagination in the same way as the reptation model does.

In this section we will first refer to the formalism originally introduced by Zwanzig and Bixon [96, 97], which was then applied to polymer dynamics by Schweizer [98, 99] and others [100–108]. This sort of theory is based on the Zwanzig/Mori projection operator technique in connection with treatments of the *Generalized Langevin Equation* [109–115]. It should be noted that this equation can be considered as the "microscopic" basis of phenomenological approaches based on the memory function formalism [116–122].

3.3.1 The Generalized Langevin Equation (GLE)

Polymers can be described adequately in terms of classical formalisms. The set of all position vectors and momentums of all segments in the system to be treated is denoted by γ, that is a point in phase space. In principle any physical property of the system can be described by an appropriate function $A(\gamma)$ of phase variables. Consider now two arbitrary physical quantities corresponding to the functions $A(\gamma)$ and $B(\gamma)$. The scalar product of these functions is given by

$$\langle A|B\rangle \equiv \int d\gamma A^*(\gamma)B(\gamma)\rho_{eq}(\gamma) \equiv \langle A^*(\gamma)B(\gamma)\rangle_{eq}, \tag{80}$$

where $A^*(\gamma)$ is the conjugate complex of $A(\gamma)$, and $\rho_{eq}(\gamma)$ is the equilibrium Gibbs canonical distribution function of the whole system. The set of all functions of this sort forms the so-called Liouville space L.

One is now interested in the kinetics of a certain set of physical quantities A_1,A_2, ... A_n, the so-called quantities of interest. The set of all linear combinations of these quantities forms the linear subspace $L^n\{A_i\}\in L$. The projection operator with respect to $L^n\{A_i\}$ is defined by

$$\hat{P} \equiv \sum_{k,m} |A_k\rangle \| \langle A|A\rangle \|^{-1}_{km} \langle A_m|, \tag{81}$$

where $\| \langle A|A\rangle \|^{-1}_{\mathrm{km}}$ is an element of the matrix inverse to the static correlation matrix $\| \langle A_k|A_m\rangle \|$, the ket $|A_m\rangle$ is the vector in the Liouville space L corresponding to the quantity A_m (in fact $|A_m\rangle = A_m(\gamma)$), and the bra $\langle A_k|$ is the vector in the Liouville space L corresponding to the quantity A_k (in fact $\langle A_k| = A_k^*(\gamma)$).

Any physical quantity obeys the equation of motion:

$$\frac{d}{dt} A_n(t) = \{A_n; H\} \equiv i\hat{L} A_n\,, \tag{82}$$

where $\{f;g\}$ is the Poisson bracket of quantities f and g, $\hat{L} \equiv i\{H; \ldots\}$ is the Liouville operator, and H is the Hamiltonian of the system. Note that $A_n(t)$ is a short notation for $A_n(\gamma)$ in the Heisenberg representation. It is defined by

$$A_n(t) \equiv \exp\left\{i\hat{L}t\right\} A_n(\gamma). \tag{83}$$

The Mori transformation can be used to formally obtain an exact system of equations for "quantities of interest":

$$\frac{d}{dt} A_n(t) = \sum_k i\omega_{nk} A_k(t) - \sum_k \int_0^t d\tau K_{nk} A_k(t-\tau) + F_n^{\mathrm{Q}}(t), \tag{84}$$

where $\omega_{nk} = \sum\limits_m \| \langle A|A\rangle \|^{-1}_{km} \left\langle A_m \middle| \hat{L} \middle| A_n \right\rangle$ is the frequency matrix. The quantity

$$F_n^{\mathrm{Q}}(t) = \exp\left\{i\hat{Q}\hat{L}\hat{Q}\right\} i\hat{Q}\hat{L} |A_n\rangle \tag{85}$$

is a generalized stochastic Langevin force associated with the physical quantity A_n. The operator given by $\hat{Q} = 1 - \hat{P}$ projects on the subspace orthogonal to $L^n\{A_r\}$. Finally, the memory matrix is defined by

$$K_{nk}(\tau) \equiv \sum_m \| \langle A|A\rangle \|^{-1}_{km} \left\langle F_m^{\mathrm{Q}}(0) \middle| F_n^{\mathrm{Q}}(\tau) \right\rangle. \tag{86}$$

Equation 84 is often called the generalized Langevin equation (GLE) [109, 115].

Note that the time evolution of experimental observables $A_n(t)$ is governed by "real dynamics" which is determined according to Eq. 83 by the real propagator $\exp\left\{i\hat{L}t\right\}$. The situation with the time evolution of the stochastic force $F_n{}^{\mathrm{Q}}(t)$ and the memory matrix $K_{nk}(t)$ is much more complicated. According to Eq. 85, their evolution is governed by "projected dynamics" the propagator of which is given by $\exp\left\{i\hat{Q}\hat{L}\hat{Q}t\right\}$.

3.3.2 The GLE for the Tagged Macromolecule

The position and momentum vectors of the segments of the tagged chain are denoted by $\vec{r}_1, \vec{r}_2, \ldots \vec{r}_N, \vec{p}_1, \vec{p}_2, \ldots \vec{p}_N$, where N is the number of segments. The dynamical state of the tagged chain is given by its phase space point

$$\gamma_N \equiv \{\vec{r}_1, \vec{r}_2, \ldots \vec{r}_N, \vec{p}_1, \vec{p}_2, \ldots \vec{p}_N\}. \tag{87}$$

The "quantities of interest" (analogous to $A_k(\gamma)$) are the single-chain *phase densities* determined by the following relation:

$$f(\Gamma_N; \gamma_N) \equiv \delta(\Gamma_N - \gamma_N) \equiv \prod_{i=1}^{N} \delta\left(\vec{R}_i - \vec{r}_i\right) \delta\left(\vec{P}_i - \vec{p}_i\right), \tag{88}$$

where $\Gamma_N \equiv \left\{\vec{R}_1, \vec{R}_2, \ldots, \vec{R}_N; \vec{P}_1, \vec{P}_2 \ldots, \vec{P}_N\right\}$ is a general phase space point, and δ is Dirac's delta function. The variables forming Γ_N are called the *field variables*. In the following we will use the short notation

$$f(\Gamma_N; \gamma_N) \equiv f(\Gamma_N). \tag{89}$$

The physical meaning and importance of the phase density f(Γ_N) is elucidated by its mean equilibrium value

$$\langle f(\Gamma_N) \rangle = \rho_N^*(\Gamma_N) \tag{90}$$

which is the single-chain distribution function over all segment coordinates and momentums.

The effective single-chain Hamiltonian, or effective single-chain free energy of the tagged macromolecule, can be determined by the relation

$$H_N^* \equiv H_N^*(\Gamma_N) = -kT \ln \rho_N^*(\Gamma_N). \tag{91}$$

The effective Hamiltonian $\delta H^*{}_N$ can be represented as

$$H_N^* = \sum_{i=1}^{N} \frac{P_i^2}{2m} + W^*\left(\{\vec{R}_i\}\right), \tag{92}$$

where $W^*\left(\{\vec{R}_i\}\right)$ is the so-called effective intramolecular potential energy, or the potential of the mean force. The projection operator on the Liouville space of the tagged macromolecule can be represented by

$$\begin{aligned} \hat{P}A(\gamma) &= \frac{1}{\rho_N^*(\gamma_N)} \int d\gamma' \delta\left(\gamma_N - \gamma_N'\right) A(\gamma') \rho_{eq}(\gamma') \\ &= \int d\gamma_M A(\gamma) \exp\{-\beta \delta H(\gamma)\} \equiv \langle A(\gamma) \rangle_{\gamma_N}^*, \end{aligned} \tag{93}$$

where γ_M is the short notation for all position vectors and momentums of the matrix segments. That is $\gamma_N+\gamma_M=\gamma$ and $\delta H(\gamma)=H(\gamma)-H_N^*(\gamma_N)$. $A(\gamma)$ is an arbitrary property of the system. $\rho_{eq}(\gamma)$ is the equilibrium distribution function of the total system.

The quantity $\delta H(\gamma)$ does not depend on the momentums of the tagged-chain segements in accordance with Eq. 92, and can be therefore considered as the Hamiltonian of the matrix, which is moving under the condition that the conformation of the tagged chain is fixed. The quantity $\langle A(\gamma)\rangle_{\gamma N}^*$ can be regarded as a conditionally averaged value of $A(\gamma)$ under the condition that γ_N has a fixed value.

The generalized Langevin equation for phase densities has the following structure:

$$\frac{\partial}{\partial t}f(\Gamma_N;t)=\int d\Gamma_N' i\Omega\left(\Gamma_N|\Gamma_N'\right)f(\Gamma_N';t)-$$
$$-\int_0^t d\tau\int d\Gamma_N' K\left(\Gamma_N|\Gamma_N';\tau\right)f\left(\Gamma_N';t-\tau\right)+F^Q(\Gamma_N;t), \qquad (94)$$

where $\Omega(\Gamma_N|\Gamma'_N)$ is the frequency matrix, $K(\Gamma_N|\Gamma'_N;\tau)$ is the memory matrix, and $F^Q(\Gamma_N;t)\equiv\exp\left\{i\hat{Q}\hat{L}_\gamma\hat{Q}t\right\}i\hat{Q}\hat{L}_\gamma f(\Gamma_N)$ is the generalized Langevin force associated with the phase density $f(\Gamma_N)=\delta(\Gamma_N-\gamma_N)$. The subscript γ indicates that the Liouville operator acts on the variables of the phase space point γ. $\hat{Q}=1-\hat{P}$ is the projection operator on the subspace orthogonal to L_N.

The frequency matrix can be expressed as

$$\Omega\left(\Gamma_N|\Gamma_N'\right)=i\left\{H_N^*\left(\Gamma_N'\right);\delta\left(\Gamma_N-\Gamma_N'\right)\right\}_{\Gamma_N'}, \qquad (95)$$

where the subscript Γ_N' means that the differentiation involved in the Poisson bracket {;} is to be done over variables forming Γ_N' only. The generalized stochastic Langevin force $F^Q(\Gamma_N;t)$ is written in the form

$$F^Q(\Gamma_N;t)\equiv\exp\left\{i\hat{Q}\hat{L}_\gamma\hat{Q}t\right\}\{\delta(\Gamma_N-\gamma_N);\delta H(\gamma)\}_\gamma. \qquad (96)$$

The memory matrix reads

$$K\left(\Gamma_N|\Gamma_N';t\right)=\frac{1}{\rho_N^*\left(\Gamma_N'\right)}\left\langle\left\{\delta\left(\Gamma_N'-\gamma\right);\delta H(\gamma)\right\}_\gamma\exp\left\{i\hat{Q}\hat{L}_\gamma\hat{Q}t\right\}\cdot\right.$$
$$\left.\cdot\{\delta(\Gamma_N-\gamma);\delta H(\gamma)\}_\gamma\right\rangle_{eq}. \qquad (97)$$

Note that the time evolution of the generalized stochastic Langevin force $F^Q(\Gamma_N;t)$ and the memory matrix are governed by so-called projected dynamics determined by the projected propagator $\exp\left\{i\hat{Q}\hat{L}\,\hat{Q}t\right\}$, which must be distinguished from the real dynamics determined by the real propagator.

Let the function $a(\gamma_N)$ be an arbitrary characteristic quantity of the tagged chain for the moment. The generalized Langevin equation for $a(\gamma_N)$ is then obtained from Eq. 94. To do this one needs to multiply both sides of Eq. 94 by $a(\Gamma_N)$ and integrate over Γ_N. After some calculations the following result is found:

$$
\begin{aligned}
\frac{d}{dt}a(t) = & \left\{a(\gamma(t)); H_N^*(\gamma_N(t))\right\} + \exp\left\{i\hat{Q}\hat{L}_\gamma\,\hat{Q}t\right\}\{a;\delta H\} + \\
& + \int_0^t \frac{d\tau}{\rho_N^*(\gamma_N(t-\tau))} \sum_k \frac{\partial}{\partial \vec{p}_k(t-\tau)} \cdot \\
& \cdot \left\langle \delta\vec{F}_k^{\text{inter}}\delta(\gamma_N(t-\tau)-\gamma_N)\exp\left\{i\hat{Q}\hat{L}_\gamma\,\hat{Q}\tau\right\}\{a;\delta H\}_\gamma \right\rangle_{eq},
\end{aligned} \tag{98}
$$

where $\delta\vec{F}_k^{\text{inter}} = \hat{Q}\vec{F}_k = -\frac{\partial}{\partial \vec{r}_n}H(\gamma) + \frac{\partial}{\partial \vec{r}_n}W^*\{\vec{r}_i\}$ is the fluctuating part of the intermolecular force acting on the n$^{\text{th}}$ segment of the tagged macromolecule.

Equation 98 is now considered for $a(\gamma_N) = \vec{r}_n$ and $a(\gamma_N) = \vec{p}_n$ leading to

$$
\frac{d}{dt}\vec{r}_n(t) \equiv \vec{v}_n(t) = \frac{1}{m}\vec{p}_n(t) \tag{99}
$$

$$
\frac{d}{dt}\vec{p}_n = -\frac{\partial}{\partial \vec{r}_n}W^*(\{\vec{r}_i\}) - \sum_k \int_0^t d\tau\Gamma_{nk}^{\alpha\beta}(\tau;t-\tau)v_k^\beta(t-\tau)\vec{e}_\alpha + \vec{F}_n^Q(t), \tag{100}
$$

where $\vec{e}_\alpha$ is the unit vector aligned along the axis labeled α. The memory matrix is given by

$$
\Gamma_{nk}^{\alpha\beta}(\tau;t-\tau) = \frac{1}{kT\rho_N^*(t-\tau)}\left\langle F_k^{Q\beta}(0)\delta(\gamma_N(t-\tau)-\gamma_N)F_n^{Q\alpha}(\tau)\right\rangle_{eq}. \tag{101}
$$

The quantity $V^\beta{}_k(t-\tau)$ is the β component of the k^{th} segment's velocity at a time moment $t-\tau$. The function $F_n^{Q\alpha}(\tau) \equiv \exp\left\{i\hat{Q}\hat{L}\,\hat{Q}\tau\right\}\hat{Q}F_n^\alpha(\gamma)$ is the α component of the generalized stochastic Langevin force acting on the segment number n at time τ. The Dirac delta function in Eq. 101 allows us to rewrite this expression in a form of conditional averaging over the matrix variables under the condition that phase variables of the tagged chain have constant values equal to their real values $\gamma_N(t-\tau)$:

$$
\Gamma_{nk}^{\alpha\beta}(\tau;t-\tau) = \frac{1}{kT}\left\langle F_k^{Q\beta}(0)F_n^{Q\alpha}(\tau)\right\rangle^*_{\gamma_N(t-\tau)}. \tag{102}
$$

The system of Eqs. 99 and 100 is an exact representation of segment dynamics in a system of many entangled Kuhn chains. However, this system

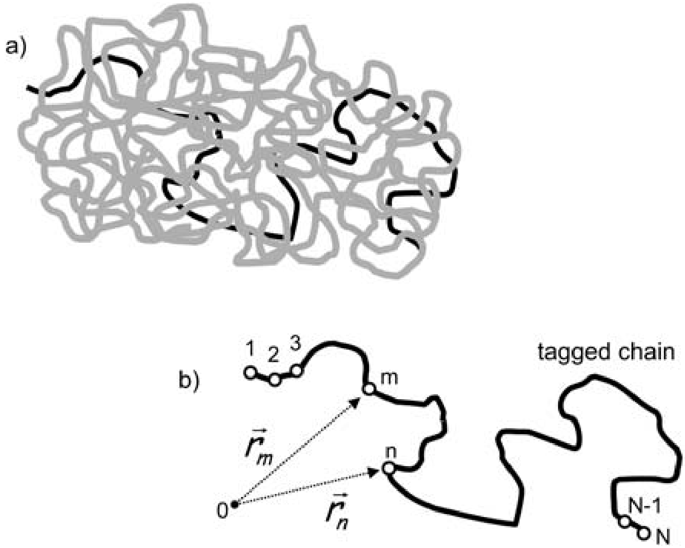


Fig. 9a, b. The global effect of the "matrix chains" (*gray*) on the dynamics of the "tagged chain" (*black*) is represented by the memory function (**a**). The Kuhn segments of the tagged chain are numbered consecutively (**b**)

cannot be solved rigorously, even not in principle, because it is not closed. That is, it contains two new, unknown objects, namely the potential of the mean force $W^*\{\vec{r}_i\}$ defined by Eq. 92 and the memory matrix $\Gamma_{nk}^{\alpha\beta}(\tau; t-\tau)$ given in Eq. 102. All one can do is to replace them by an intuitive ansatz or some physically plausible approximation. These two functions represent the global "entanglement" effect of the matrix on the tagged chain (for an illustration see Fig. 9).

3.3.2.1
The Potential of the Mean Force and the Memory Matrix

The potential of the mean force $W^*\{\vec{r}_i\}$ in principle is determined by equilibrium statistical mechanics. The coarse grain configuration of polymer chains in melts obeys Gaussian statistics with respect to the end-to-end distance, for instance. This suggests the following approximation of the potential of the mean force:

$$W^*\{\vec{r}_i\} = \frac{3k_B T}{2b^2} \sum_{n=1}^{N} (\vec{r}_n - \vec{r}_{n-1})^2. \tag{103}$$

Substituting this expression in Eq. 100 reproduces the effectively intramolecular entropic force acting on the n^{th} segment, which is exactly equal to the first term on the right-hand side of the Rouse equation of motion given by Eq. 45.

The analytical form of the memory matrix $\Gamma_{nk}^{\alpha\beta}(\tau; t-\tau)$ given in Eq. 101 is much less clear. The memory matrix $\Gamma_{nk}^{\alpha\beta}(\tau; t-\tau)$ contains all the (unknown) information about intermolecular interactions, i.e., all consequences of entanglement effects. The fluctuating part of the intermolecular force $\vec{F}_n(\tau)$ acting on the n^{th} segment of the tagged chain can be expressed by matrix density fluctuations around this segment:

$$\vec{F}_n^Q(\tau) = \int d^3\vec{r}\vec{f}(\vec{r})\delta\rho_n^Q(\vec{r},\tau,\gamma_n(t-\tau)), \tag{104}$$

where $\vec{f}(\vec{r}) = -\frac{\partial}{\partial\vec{r}}U(\vec{r})$ is the intermolecular force which a matrix segment exerts on the segment of the tagged chain at a distance $\vec{r}$. The function $\delta\rho_n^Q(\vec{r},\tau,\gamma_n(t-\tau))$ is given by

$$\delta\rho_n^Q(\vec{r},\tau,\gamma_n(t-\tau)) = \exp\left\{i\hat{Q}\hat{L}\,\hat{Q}\tau\right\}\left(\rho_n(\vec{r},\gamma_n(t-\tau)) - \rho_n^*(\vec{r},\gamma_n(t-\tau))\right), \tag{105}$$

where $\rho_n(\vec{r},\gamma_n(t-\tau))$ is the density of matrix segments at a distance $\vec{r}$ from the n^{th} segment of the tagged chain with a conformation given by $\gamma_n(t{-}\tau)$. The function $\rho_n^*(\vec{r},\gamma_n(t-\tau))$ is a conditional equilibrium density of the matrix segments at a distance $\vec{r}$ from the n^{th} segment of the tagged chain under the condition that the conformation of the tagged chain is fixed and given by $\gamma_n(t{-}\tau)$.

On this basis the memory matrix can be expressed as an integral over the time-dependent autocorrelation function of matrix density fluctuations for projected dynamics:

$$\Gamma_{nk}^{\alpha\beta}(\tau,t-\tau) = \frac{1}{k_B T}\int d^3\vec{r}\int d^3\,\vec{r}\,'f_n^{\alpha}(\vec{r})f_k^{\beta}(\vec{r}\,')\left\langle\delta\rho_k^Q(0;\vec{r}\,')\delta\rho_n^Q(\tau;\vec{r})\right\rangle^*_{\gamma_n(t-\tau)}, \tag{106}$$

where the dependence of the density fluctuations $\delta\rho_k^Q(0;\vec{r}\,')$ and $\delta\rho_n^Q(\tau;\vec{r})$ around segments k and n on the tagged chain variables at time $t{-}\tau$ via $\gamma_n(t{-}\tau)$ was omitted for simplicity.

3.3.2.2
The Rouse Equation of Motion as a Special Case of the GLE

The Rouse equation of motion can be treated as a special case of Eq. 100. To show this, the memory function given in Eq. 106 is subjected to the following approximations:

(i) The time-dependent correlation function $\left\langle \delta\rho_k^Q(0;\vec{r}\,')\delta\rho_n^Q(\tau;\vec{r})\right\rangle^*_{\gamma_n(t-\tau)}$ for projected dynamics is a function of the conformation of the tagged chain and its Kuhn segment momentums at time $t-\tau$. We make use of the approximation

$$\left\langle \delta\rho_k^Q(0;\vec{r}\,')\delta\rho_n^Q(\tau;\vec{r})\right\rangle^*_{\gamma_n(t-\tau)} \approx \left\langle \delta\rho_k^Q(0;\vec{r}\,')\delta\rho_n^Q(\tau;\vec{r})\right\rangle_{eq}, \tag{107}$$

that is, we replace the conditional average $\langle\ldots\rangle^*_{\gamma_n(t-\tau)}$by an equilibrium average $\langle\ldots\rangle_{eq}$. The correlation function then does not depend on the chain conformation and its segment momentums at time $t-\tau$. Since flexible polymer melts are isotropic, the memory function becomes a scalar with respect to space rotations, i.e., $\Gamma_{nk}^{\alpha\beta}(\tau,\, t-\tau) \sim \delta_{\alpha\beta}$, where $\delta_{\alpha\beta}$ is the Kronecker symbol.

(ii) The memory matrix given in Eq. 106 has a nonlocal character with respect to the Kuhn segment number. There can be nonnegligible correlations in the density fluctuations between distant tagged-chain segments n and k with $|n-k|>>1$. The simplest approximation is to neglect this nonlocality, which is mathematically equivalent to setting $\Gamma_{nk}^{\alpha\beta} \propto \delta_{nk}$.

(iii) The most crucial approximation is the assumption that the memory function decays fast enough to become negligibly small on a molecular mass independent time scale. In accordance with Eq. 106 this means that the local matrix density distribution around the n^{th} segment of the tagged chain equilibrates on a microscopic time scale.

Taking all approximations together, the memory matrix can be expressed as

$$\Gamma_{nk}^{\alpha\beta}(\tau; t-\tau) = 2\zeta\delta_{\alpha\beta}\delta_{nk}\delta(\tau). \tag{108}$$

Accounting for this and Eq. 103, Eq. 100) can be shown to coincide with the Rouse equation of motion given by Eq. 45.

Fast equilibration of the local matrix density fluctuations around the segments of the tagged chain contradicts chain connectivity in a sense. Only if chains are allowed to cross each other, can one expect that a local equilibrium distribution of the matrix segments around a certain tagged chain configuration can be reached on a molecular mass independent microscopic time scale. Anyway, very fast local motions do indeed exist, leading to an approximate local equilibrium distribution of the matrix segments around the segments of the tagged chain with a molecular mass independent accuracy.

From an empirical point of view it is well known that the Rouse model describes satisfactorily the dynamical properties of polymer melts with $N<N_C$. It is instructive to extract the fast decaying Rouse contribution from the total memory matrix given by Eq. 102 and rewrite Eq. 100 as

$$\frac{d}{dt}\vec{p}_n = -\frac{3k_BT}{b^2}\frac{\partial^2}{\partial n^2}\vec{r}_n(t) - \zeta\vec{v}_n(t) - \sum_k \int_{\tau_0}^{t} d\tau \Gamma_{nk}^{\alpha\beta}(\tau; t-\tau) v_k^\beta(t-\tau)\vec{e}_\alpha + \vec{F}_n^Q(t), \tag{109}$$

where ζ is the "bare" local friction coefficient connected with the fast local motions and $\tau_0 << t$ is the characteristic time associated with these fast local motions. Here the approximation given in Eq. 103 was taken into account.

From Eqs. 100, 102, and 109, we obtain the self-diffusion coefficient

$$D = \frac{k_BT}{\zeta_{eff}^*}, \tag{110}$$

where

$$\zeta_{eff}^* = \frac{1}{3k_BT}\sum_{n,k}\int_0^\infty \left\langle \vec{F}_k^Q(0)\cdot\vec{F}_n^Q(\tau)\right\rangle d\tau = N\zeta + \frac{1}{3k_BT}\sum_{n,k}\int_{\tau_0}^\infty \left\langle \vec{F}_k^Q(0)\cdot\vec{F}_n^Q(\tau)\right\rangle d\tau \tag{111}$$

is the effective friction coefficient of the macromolecule.

3.4 Renormalized Rouse Models

"Renormalization" in this context means an attempt to find a physically plausible ansatz for the unknown memory matrix given by Eq. 106. In principle one could postulate a power law with an exponent being a fitting parameter to experimental data after having derived expressions of observables on this basis. However, in the frame of the renormalized Rouse models (RRM) a somewhat less formal and less phenomenological approach is possible.

The preaveraging approximation given in Eq. 107 is again used. The memory matrix then becomes a scalar relative to space rotations, and does not depend on the momentary configuration of the tagged chain:

$$\Gamma_{nk}^{\alpha\beta}(\tau) = \Gamma_{nk}^{\alpha\beta} = \frac{1}{3k_BT}\int d^3\vec{r}\int d^3\vec{r}\,'\vec{f}(\vec{r})\vec{f}(\vec{r}\,')\left\langle\delta\rho_k^Q(0,\vec{r}\,')\delta\rho_n^Q(\tau,\vec{r})\right\rangle_{eq}. \tag{112}$$

The general equation of motion (Eq. 109) becomes a linear function of the tagged chain variables, and the system is considered to be isotropic.

Intermolecular interactions determining the force $\vec{f}(\vec{r})$ with which a segment from a chain acts on a segment from another one are approximated by a hard sphere interaction with a certain diameter d comparable with the Kuhn segment length b. The "entanglement effects" or the "topological interactions" are the consequence of chain connectivity and excluded-volume intermolecular interactions.

The matrix density correlation function $\left\langle \delta\rho_k^Q(0,\vec{r}\,')\delta\rho_n^Q(\tau,\vec{r})\right\rangle_{eq}$ for projected dynamics is approximated as a k-space integral (for a detailed argument see Ref. [98]),

$$\Gamma_{nm}(t) = \frac{8}{27}\frac{\rho_m d^6 g^2(d)}{k_B T}\int\limits_0^{b^{-1}} dk k^4 \omega_{nk}^Q(k,t)\hat{S}^Q(k,t), \tag{113}$$

where a mistake in the numerical factor published in Ref. [98] is corrected. The function $g(r)$ is the radial intermolecular distribution function; $\omega_{nk}^Q(k,t)$ and $\hat{S}^Q(k,t)$ are the Fourier transforms of $\omega_{nk}^Q\left(\vec{R};t\right)$ and $\hat{S}^Q(\vec{r};t)$, respectively.

The intrachain dynamical structure factor for projected dynamics is defined by

$$\hat{\omega}_{nm}^Q(k,t) = \hat{\omega}_{nm}(k)\exp\left\{-\frac{k^2}{6}\left\langle \vec{R}^2(t)\right\rangle_Q\right\}, \tag{114}$$

where $\hat{\omega}_{nm}(k)$ is the static intrachain structure factor. Likewise, the collective dynamical structure factor for projected dynamics of the matrix surrounding the tagged chain is given by

$$\hat{S}^Q(k,t) = \hat{S}(k)\exp\left\{-\frac{k^2}{6}\left\langle \vec{R}^2(t)\right\rangle_Q\right\}, \tag{115}$$

where $\hat{S}(k)$ is the static collective structure factor.

Single-chain $\hat{\omega}_{nm}^Q(k,t)$ and collective $\hat{S}^Q(k,t)$ dynamical structure factors for projected dynamics in Eq. 113 reflect the fact that local density fluctuations around the tagged chain are relaxed by the projected motions both of the tagged chain and the matrix chains. The projected nature of the dynamics is hidden in Eqs. 114 and 115 in the projected mean squared displacement, which qualitatively describes typical distances over which elementary density fluctuations become dispersed around the tagged chain during the time t.

The essence of the first and second renormalization ansatzes to be described in the following refers to heuristic replacements of the mean squared segment displacement for projected dynamics, $\left\langle \vec{R}^2(t)\right\rangle_Q$. In the (once) renormalized Rouse model this unknown function is replaced by the result of the ordinary Rouse model, $\left\langle \vec{R}^2(t)\right\rangle_R$, given in Eq. 58 [98]. In the twice renormalized Rouse model, it is substituted by the result of the first renormalization.

3.4.1 The First Renormalization

Equation 109 becomes mathematically closed by equating

$$\left\langle \vec{R}^2(t) \right\rangle_Q = \left\langle \vec{R}^2(t) \right\rangle_R. \tag{116}$$

From Eq. 109 we find the integrodifferential equation for the autocorrelation function of the normal modes (for a definition see Eq. 49)

$$\frac{\partial}{\partial t} C_p(t) + \int_0^t \Gamma_p(t-\tau) \frac{\partial}{\partial \tau} C_p(\tau) d\tau = -\frac{p^2}{\tau_R} C_p(t), \tag{117}$$

where the contribution of the inertia term was neglected. The memory function with respect to the correlation function of the p^{th} normal mode is defined by

$$\Gamma_p(t) = \frac{1}{\zeta} \int_0^N \Gamma_m(t) \cos\left(\frac{\pi}{N} pm\right) dm, \tag{118}$$

where chain end effects are neglected, so that

$$\Gamma_{nk}^{\alpha\beta}(t) = \delta_{\alpha\beta} \Gamma_{|n-m|}(t). \tag{119}$$

On this basis the following integral representation is obtained for the memory function $\Gamma_p(t)$ [49]

$$\Gamma_p(t) = \frac{16}{3\sqrt{\pi}\pi^2} \frac{\psi b^3}{\left\langle \vec{R}^2(t) \right\rangle_Q^{3/2}} \frac{1}{\tau_s} \int_0^{\sqrt{\frac{\langle \vec{R}^2(t) \rangle_Q}{3b^2}}} \frac{q^6 \exp\{-q^2\}}{q^4 + \left(\frac{2\pi p \langle \vec{R}^2(t) \rangle_Q}{N} \right)^2} dq, \tag{120}$$

where $\psi \equiv \rho_m d^3 \left(\frac{d}{b}\right)^3 g^2(d) \hat{S}(0)$ is a dimensionless parameter characterizing the influence of the entanglement effects on the chain dynamics. The quantity $\hat{S}(0)$ is given by $\hat{S}(0) = \rho_m k_B T \kappa_T$, where κ_T is the isothermal compressibility.

The exact solution of Eq. 117 is nonexponential in general. The relaxation time of the p^{th} normal mode can be defined in the following way:

$$\tau_p = \frac{1}{C_p(0)} \int_0^\infty C_p(t) dt = \tau_s \left(\frac{N}{p}\right)^2 \left[1 + \hat{\Gamma}_p(0)\right], \tag{121}$$

where $\hat{\Gamma}_p(0) = \int\limits_0^\infty \Gamma_p(t)dt$. For the self-diffusion coefficient the following relation can be obtained

$$D = \frac{k_B T}{N\zeta\left[1 + \hat{\Gamma}_0(0)\right]}. \tag{122}$$

This treatment of the RRM differs from the original version by Schweizer [98], where the mode-number-dependent term $\left(2\pi p\left\langle \vec{R}^2(t)\right\rangle_Q \Big/ N\right)^2$ in expressions corresponding to Eq. 120 was neglected. However, this term is quite essential [49, 58, 105] for the normal-mode relaxation of entangled polymers.

In the short time limit, $t \ll \tau_e' \propto \tau_s/\psi$, the integral term in Eq. 109 is negligible, so that chain dynamics tends to approach "unentangled" Rouse behavior. At longer times the integral term starts to dominate over the local friction term, $\zeta\vec{v}_n(t)$. That is, chain dynamics becomes "entangled". The details of the transition unentangled–entangled motion depend on the parameter ψ.

From Eqs. 121 and 120 we find for $\left\langle \vec{R}^2(t)\right\rangle_Q = \left\langle \vec{R}^2(t)\right\rangle_R$ in the *high-mode-number limit*, $p>N/6\pi$,

$$\tau_p^{RR} = \tau_s \left(\frac{N}{p}\right)^2 \left[1 + 0.05\psi\left(\frac{N}{p}\right)^2\right]. \tag{123}$$

If the entanglement parameter is big enough, $\psi \gg \frac{1}{\pi^2}$, and for normal modes

$$\frac{N}{6\pi} < p < 0.22\sqrt{\psi}N \tag{124}$$

the entanglement effect becomes essential, and Eq. 123 approaches

$$\tau_p^{RR} \simeq 0.05\psi\tau_s \left(\frac{N}{p}\right)^4. \tag{125}$$

For $t \gg \tau_e' \propto \tau_s/\psi$, chains are subject to entangled behavior. The mean squared displacement can be approximated as

$$\left\langle \vec{R}_n^2(t)\right\rangle \simeq 0.5 b^2 \left(\frac{t}{\psi\tau_s}\right)^{1/4} \tag{126}$$

and the autocorrelation function of the segment tangent vector becomes

$$\left\langle \vec{b}_n(t) \cdot \vec{b}_n(0)\right\rangle \simeq 0.4 b^2 \left(\frac{\psi\tau_s}{t}\right)^{1/4}. \tag{127}$$

At longer times, $t >> 0.05(6\pi)^4\psi\tau_s$, the influence of normal modes with numbers $p << N/6\pi$ becomes essential. This limit is called the *low-mode-number limit*. In this case both contributions in the denominator of the integrand are important and the following result for the normal mode relaxation time can be obtained

$$\tau_p^{RR} = \tau_s \left(\frac{N}{p}\right)^2 \left[1 + 2.14\psi\sqrt{\frac{N}{p}}\right] \simeq 2.14\psi\tau_s \left(\frac{N}{p}\right)^{2.5}. \tag{128}$$

The terminal relaxation time of the RRM scales as

$$\tau_1^{RR} \simeq 2.14\psi\tau_s N^{2.5}. \tag{129}$$

In the limit $N\rightarrow\infty$ and $p>>N$, the time dependence of the normal mode relaxation functions can be approximated by an exponential decay as

$$C_p(t) = C_p(0)\exp\left\{-\frac{t}{\tau_p}\right\}. \tag{130}$$

Otherwise Eq. 117 can be solved only numerically. It was found, that the solution of the equation can be approximated by the stretched exponential (Fig. 10)

$$C_p(t) \approx C_p(0)\exp\left\{-\left(\frac{t}{\tilde{\tau}_p}\right)^{\beta_p}\right\}, \tag{131}$$

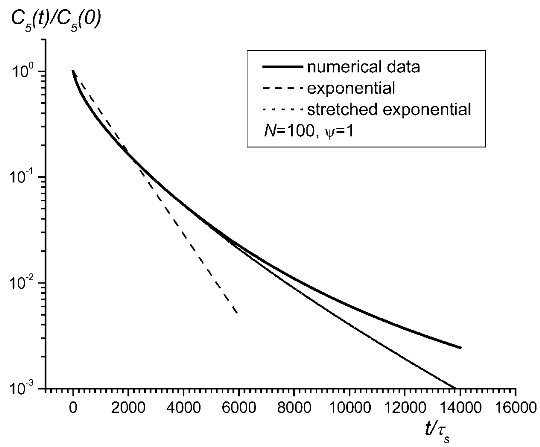


Fig. 10. Numerically evaluated, normalized autocorrelation function for the RRM for p=5 together with a fitted curve for a stretched exponential, Eq. 131, and an exponential curve. The stretched-exponential parameter is β_5=0.66

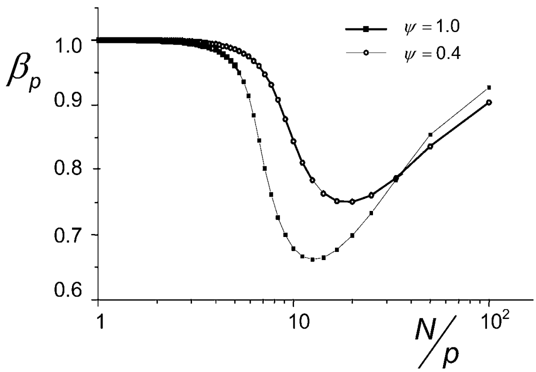


Fig. 11. Mode number dependence of the stretching-parameter β_p for the RRM (see Eq. 131) for different entanglement parameters ψ. The number of Kuhn segments was assumed to be N=100

where $\tilde{\tau}_p = \frac{\beta_p}{\Gamma(\beta_p)} \tau_p$ and β_p is a mode-number-dependent stretching parameter (see Fig. 11). $\Gamma(\beta_p)$ is the gamma function. In the same limit, the mean squared segment displacement obeys

$$\left\langle \vec{R}_n^2(t) \right\rangle = \begin{cases} 0.23 b^2 \left(\frac{t}{\psi \tau_s} \right)^{2/5} & \text{for} \quad t \ll \tau_1^{RR} \\ 6 D_{RR} t & \text{for} \quad t \gg \tau_1^{RR} \end{cases}, \tag{132}$$

where

$$D_{RR} = \frac{0.21}{3\pi^2} \frac{b^2}{\psi \tau_s N^{3/2}} \tag{133}$$

is the center-of-mass diffusion coefficient in the RRM. The correlation function of the segment tangent vector calculated in the Markovian approximation is in the low-mode-number limit given by

$$\left\langle \vec{b}_n(t) \cdot \vec{b}_n(0) \right\rangle \simeq \begin{cases} b^2 \left(\frac{\psi \tau_s}{t} \right)^{2/5} & \text{for} \quad t \ll \tau_1^{RR} \\ \frac{b^2}{N} \exp\left\{ -\frac{t}{\tau_1^{RR}} \right\} & \text{for} \quad t \gg \tau_1^{RR} \end{cases}. \tag{134}$$

These results should be asymptotically correct in the limit $N \to \infty$ and $t >> 0.05(6\pi)^4 \psi \tau_s$. For finite N, the exponents predicted in Eq. 132 may change slightly due to the stretched exponential decay of $C_p(t)$ given in Eq. 131. The limiting laws for the observables of interest in the frame of this review, the mean squared displacement and the (intrasegment) spin–lattice relaxation rate, are summarized in Table 2 (compare [49]).

Table 2. Theoretical dependences on time (t), angular frequency (ω), molecular mass (M), or number of Kuhn segments (N) predicted by the (once) renormalized Rouse model for the mean squared segment displacement and the intrasegment spin–lattice relaxation time. The factors C_I, C_I^{RR}, C_{II}^{RR}, and C_{III}^{RR} are frequency and molecular mass independent constants

	Limit	Mean squared segment displacement $\langle R^2\rangle=$	intrasegment spin–lattice relaxation time $T_1^{\text{intra}}=$	Refs
Rouse	$\tau_s<<(t,1/\omega)<<\tau_s/\psi$	$\frac{2}{\pi^{3/2}}b^2\left(\frac{t}{\tau_s}\right)^{1/2}$ $\propto M^0t^{1/2}$	$-C_I\frac{M^0}{\tau_s\ln(\omega\tau_s)}$	[47, 48]
$(\mathrm{I})_{\mathrm{RR}}$	"high – mode – number limit" $\frac{N}{6\pi}<p<0.22N\sqrt{\psi}$ $\frac{\tau_s}{1.9\psi}\ll(t,1/\omega)\ll(6\pi)^4 1.9\psi\tau_s$	$0.5b^2\left(\frac{t}{\tau_s\psi}\right)^{1/4}$ $\propto M^0t^{1/4}$	$C_I^{RR}M^0\omega^{1/2}$	[49]
$(\mathrm{II})_{\mathrm{RR}}$	"low – mode – number, short – time limit" $p<\frac{N}{6\pi}$ $\frac{(6\pi)^4}{20}\psi\tau_s\ll(t,1/\omega)\ll\tau_1^{RR}$ $\approx 2.14\psi\tau_sN^{2.5}$	$0.23b^2\left(\frac{t}{\psi\tau_s}\right)^{2/5}$ $\propto M^0t^{2/5}$	$C_{II}^{RR}M^0\omega^{1/5}$	[47, 49]
$(\mathrm{III})_{\mathrm{RR}}$	"low – mode – number, long – time limit" $p<\frac{N}{6\pi}$ $\tau_1^{RR}\approx 2.14\psi\tau_sN^{2.5}\ll(t,1/\omega)$	$6D_{RR}t$ $\approx\frac{0.42}{\pi^2}\frac{b^2}{\psi\tau_s}N^{-3/2}t$	$\left[\frac{C_{III}^{RR}b^4N^{-2}\tau_1^{RR}}{1+\left(\omega\tau_1^{RR}/2\right)^2}\right]^{-1}$	present work

3.4.2 The Second Renormalization

The RRM predictions for the chain length dependences of the self-diffusion coefficient, $D^{RR}\propto N^{-1.5}$, and of the terminal relaxation time, $\tau_1^{RR}\propto N^{2.5}$, are weaker than suggested by experiments: $D^{\exp}\propto N^{-(2\div 2.5)}$, $\tau^{\exp}\propto N^{3.5}$. This means that, at least at long times, the memory matrix $\Gamma_{nm}(t)$ decays more slowly than predicted by the renormalized Rouse model. That is, the entanglement effect is underestimated. On the other hand, RRM predictions are more realistic for entangled polymers than those of the ordinary Rouse model. This suggests one should attempt another ansatz for projected dynamics on the basis of the RRM results. Substituting $\left\langle\vec{R}_n^2(t)\right\rangle_Q=\left\langle\vec{R}_n^2(t)\right\rangle_{RR}$ in the formulas given in Eqs. 114 and 115 leads to the twice renormalized Rouse model (TRRM).

The *high-mode-number limit*, $p>N/6\pi$, for the normal mode relaxation time suggests:

$$\tau_p^{TRR}\simeq\tau_s\left(\frac{N}{p}\right)^2\left[1+0.148\psi^2\left(\frac{N}{p}\right)^2\right] \tag{135}$$

similar to Eq. 123 found for the RRM. In the limit $t \gg \tau'_e = \tau_s/\psi$ and $\psi^2 >> 1/\pi^2$, the TRRM predicts

$$\left\langle \vec{R}_n^2(t) \right\rangle \simeq 0.4 b^2 \left(\frac{t}{\psi \tau_s} \right)^{1/4} \tag{136}$$

and

$$\left\langle \vec{b}_n(t) \cdot \vec{b}_n(0) \right\rangle \simeq 0.5 b^2 \left(\frac{\psi \tau_s}{t} \right)^{1/4} \tag{137}$$

analogous to the relations given in Eqs. 126 and 127.

The difference between the RRM and the TRRM becomes more pronounced in the *low-mode-number limit*, $p<N/6\pi$. The normal-mode relaxation time of order p reads

$$\tau_p^{TRR} = \tau_s \left(\frac{N}{p} \right)^2 \left[1 + 5.52 \psi^2 \frac{N}{p} \right] \simeq 5.52 \psi^2 \tau_s \left(\frac{N}{p} \right)^3 \tag{138}$$

compared to the expression given in Eq. 128. The terminal relaxation time scales with molecular mass the same way as in the case of the reptation/tube model, namely

$$\tau_1^{TRR} \simeq 5.52 \psi^2 \tau_s N^3. \tag{139}$$

However, the dependence on the mode number, $\tau_p \propto (N/p)^3$, differs from that suggested by the reptation model, $\tau_p \propto N^3/p^2$ [1]. Note that the TRRM result in this respect was confirmed by computer simulations [84, 87] in a wide range.

As in the RRM case, the memory function $\Gamma_p(t)$ given by Eq. 120 becomes fast decaying:

$$\Gamma_p(t) \propto \frac{1}{\left\langle \vec{R}_n^2(t) \right\rangle^{7/2}} \propto \frac{1}{t^{7/5}}. \tag{140}$$

The Markovian approximation for Eq. 117 therefore becomes correct in the limit $N \to \infty$ and $p<<N$. This means that the solution of Eq. 117 approaches an exponential decay according to

$$C_p(t) = C_p(0) \exp \left\{ -\frac{t}{\tau_p^{TRR}} \right\}. \tag{141}$$

More generally the solutions of Eq. 117 are nonexponential for the TRRM and, as it was found in numerical investigations, can be approximated over two orders of magnitude of $C_p(t)$ by the stretched exponential given in Eq. 131. Only the fitting parameters β_p and $\tilde{\tau}_p$ differ from those found for the RRM. Illustrations are shown in Figs. 12 and 13. The stretched exponen-

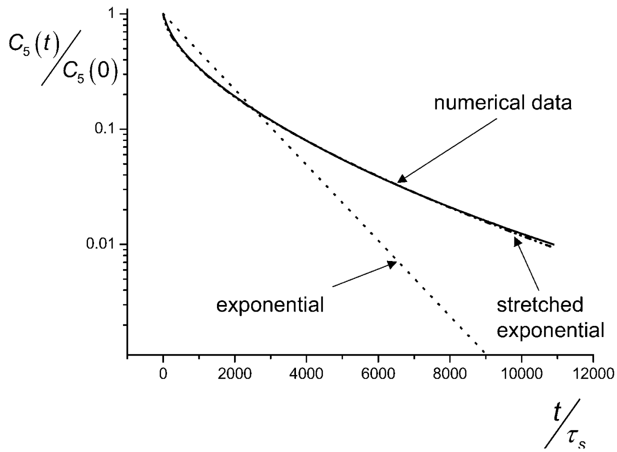


Fig. 12. Numerically evalutated, normalized autocorrelation function for the TRRM for p=5 together with a fitted curve for a stretched exponential, Eq. 131, and an exponential function. The parameters are β_5=0.61, ψ=1

tial decay reproduces the numerical data very well for $C_p(t)/C_p(0) \geq 0.01$ (Fig. 12). The stretching parameter β_p for the TRRM as a function of the normal mode number p shows a minimum and approaches unity for decreasing p (Fig. 13). Qualitatively the same behavior was found in computer

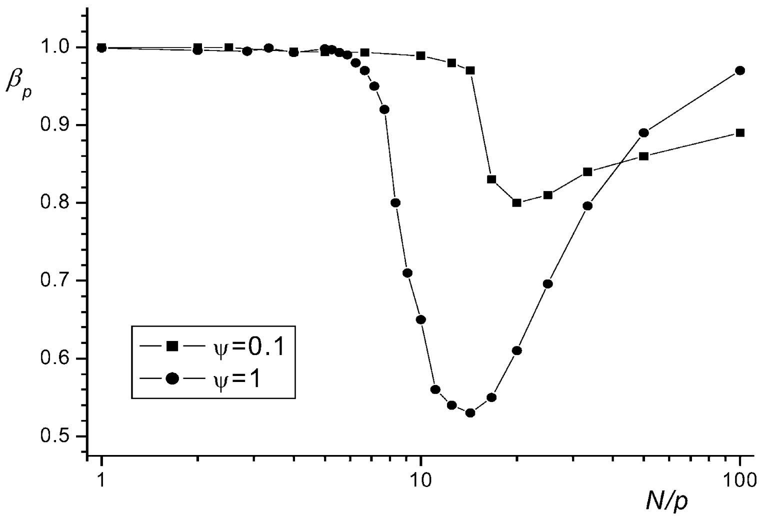


Fig. 13. Mode number dependence of the stretching-parameter β_p (see Eq. 131) for the TRRM for different entanglement parameters ψ=0.1, 1. The number of Kuhn segments was assumed to be N=100

simulations [84, 87]. Comparing Figs. 10 and 11 with Figs. 12 and 13 suggests that the RRM and the TRRM are similar in this respect.

In the low-mode-number limit, the TRRM predicts for the mean squared segment displacement

$$\left\langle \vec{R}_n^2(t) \right\rangle_{TRR} = \begin{cases} 0.16b^2 \left(\frac{t}{\psi^2 \tau_s} \right)^{1/3} & \text{for} \quad \tau_e \ll t \ll \tau_1^{TRR} \\ 6D_{TRR}t & \text{for} \quad t \gg \tau_1^{TRR} \end{cases}, \tag{142}$$

where

$$D_{TRR} \simeq \frac{0.052}{3\pi^2} \frac{b^2}{\psi^2 N^2 \tau_s}. \tag{143}$$

Remarkably the TRRM gives the same molecular mass dependence for the self-diffusion coefficient as the tube/reptation model.

In the same limit the autocorrelation function of the tangent vector turns out to be

$$\left\langle \vec{b}_n(t) \cdot \vec{b}_n(0) \right\rangle_{TRR} \simeq \begin{cases} 1.58b^2 \left(\frac{\psi^2 \tau_s}{t} \right)^{1/3} & \text{for} \quad \tau_e \ll t \ll \tau_1^{TRR} \\ \frac{b^2}{N} \exp\left\{ -\frac{t}{\tau_1^{TRR}} \right\} & \text{for} \quad t \gg \tau_1^{TRR} \end{cases}. \tag{144}$$

Table 3. Theoretical dependences on time (t), angular frequency (ω), molecular mass (M), or number of Kuhn segments (N) predicted by the twice renormalized Rouse model for the mean squared segment displacement and the intrasegment spin–lattice relaxation time. The factors C_I, $C_I{}^{TRR}$, $C_{II}{}^{TRR}$, and $C_{III}{}^{TRR}$ are frequency and molecular mass independent constants

	Limit	Mean squared segment displacement $\langle R^2 \rangle =$	Intrasegment spin–lattice relaxation time $T_1{}^{\text{intra}} =$
Rouse	$\tau_s << (t, 1/\omega) << \tau_s/\psi$	$\frac{2}{\pi^{3/2}} b^2 \left(\frac{t}{\tau_s} \right)^{1/2}$ $\propto M^0 t^{1/2}$	$-C_I \frac{M^0}{\tau_s \ln(\omega \tau_s)}$
$(\mathrm{I})_{\mathrm{TRR}}$	"high – mode – number limit" $\frac{N}{6\pi} < p < 0.22N\sqrt{\psi}$ $\frac{\tau_s}{\psi} \ll (t, 1/\omega) \ll (6\pi)^4 \psi^2 \tau_s$ $\psi \gg \frac{1}{\pi^2}$	$0.4b^2 \left(\frac{t}{\tau_s \psi} \right)^{1/4}$ $\propto M^0 t^{1/4}$	$C_I{}^{TRR} M^0 \omega^{1/2}$
$(\mathrm{II})_{\mathrm{TRR}}$	"low – mode – number, short – time limit" $p < \frac{N}{6\pi}$ $\frac{(6\pi)^4}{20} \psi \tau_s \ll (t, 1/\omega)$ $\ll \tau_1^{TRR} \approx 5.52\psi^2 \tau_s N^3$	$0.16b^2 \left(\frac{t}{\psi^2 \tau_s} \right)^{1/3}$ $\propto M^0 t^{1/3}$	$C_{II}{}^{TRR} M^0 \omega^{1/3}$
$(\mathrm{III})_{\mathrm{TRR}}$	"low – mode – number, long – time limit" $p < \frac{N}{6\pi}$ $\tau_1^{TRR} \approx 5.52\psi^2 \tau_s N^3 \ll (t, 1/\omega)$	$6D_{TRR}t$ $\approx \frac{0.104}{\pi^2} \frac{b^2}{\psi^2 \tau_s} N^{-2} t$	$\left[\frac{C_{III}^{TRR} b^4 N^{-2} \tau_1^{TRR}}{1 + \left(\omega \tau_1^{TRR}/2 \right)^2} \right]^{-1}$

If the entanglement parameter obeys $\psi \leq 1/\pi^2$, only the low-mode-number regime of the entangled motion exists for times $t >> \tau_e = \psi^{-4}\tau_s$ in the TRRM as well as in the RRM. Table 3 summarizes the limiting laws for the observables of interest in the frame of this review.

3.4.3 General Remarks on Renormalized Rouse Models

In principle one could continue the renormalization procedure by constructing an infinite series of *n* times renormalized Rouse models by equating the mean squared segment displacement for projected dynamics iteratively. However, the renormalization procedure is nothing more than a heuristic way of closure of the equations of motion. It must be distinguished from ordinary iteration procedures serving the solution of *well-defined* equations. The renormalization procedure by way of contrast is an attempt to *define* the equation. This means that an infinite repetition of renormalization steps does not necessarily lead to solutions more appropriate for the problem to be solved. The renormalization steps must rather be considered as trial and error attempts to find good solutions just in a phenomenological sense. However, there is some limitation how far one can go in principle with this iteration strategy.

The three times renormalized Rouse model, ThRRM, means that we set $\langle \vec{R}_n^2(t) \rangle_Q = \langle \vec{R}_n^2(t) \rangle_{TRR}$ for the memory function $\Gamma_p(t)$ in Eq. 120. High-mode-number and low-mode-number limits can be distinguished again for the time scale of entangled dynamics. In the high-mode-number limit, the predictions of the ThRRM coincide with those of the TRRM and the RRM apart from numerical factors (see Eqs. 135–137 and 123, 126, 127, respectively). An essential difference arises only in the low-mode-number limit, where the following scaling predictions are found:

$$\tau_p^{ThRR} \propto \psi^3 \tau_s \left(\frac{N}{p} \right)^{3.5}, \tag{145}$$

$$\langle \vec{R}_n^2(t) \rangle_{ThRR} \propto b^2 \left(\frac{t}{\psi^3 \tau_s} \right)^{2/7} \quad \text{for} \quad \tau_e \ll t \ll \tau_1^{ThRR}, \tag{146}$$

$$\left\langle \vec{b}_n(t) \cdot \vec{b}_n(0) \right\rangle_{ThRR} \propto b^2 \left(\frac{\psi^3 \tau_s}{t} \right)^{2/7} \quad \text{for} \quad \tau_e \ll t \ll \tau_1^{ThRR}, \tag{147}$$

$$D_{ThRR} \propto \frac{b^2}{\psi^3 N^{2.5} \tau_s}, \tag{148}$$

where $\tau_1^{ThRR} \propto \tau_s \psi^3 N^{3.5}$ is the terminal relaxation time for the ThRRM.

If the renormalization procedure is repeated n times, the terminal relaxation time and the self-diffusion coefficient are expected to follow the scaling predictions

$$\tau_1^{nRR} \propto \tau_s \psi^n N^{\frac{n+4}{2}} \tag{149}$$

and

$$D_{nRR} \propto \frac{b^2}{\tau_s \psi^n} N^{-\left(\frac{n+2}{2}\right)}, \tag{150}$$

respectively. The mean squared displacement scales in the low-mode-number limit as

$$\left\langle \vec{R}_n^2(t) \right\rangle_{nRR} \propto b^2 \left(\frac{t}{\psi^n \tau_s} \right)^{-\frac{2}{n+4}}, \tag{151}$$

and the renormalization ansatz is

$$\left\langle \vec{R}_n^2(t) \right\rangle_Q = \left\langle \vec{R}_n^2(t) \right\rangle_{(n-1)RR} \propto b^2 \left(\frac{t}{\psi^{(n-1)} \tau_s} \right)^{-\frac{2}{n+3}}. \tag{152}$$

Then the expression for the memory function $\Gamma_p(t)$ given in Eq. 120 can be approximated as a power law for the limit $N \rightarrow \infty$ and $t \rightarrow \infty$ while p/N=const,

$$\Gamma_p(t) \propto \frac{1}{\left\langle \vec{R}_n^2(t) \right\rangle} \propto \frac{1}{t^{\left(\frac{7}{n+3}\right)}}. \tag{153}$$

Obviously, if $n \geq 4$ the memory function is slowly decaying, and there is no limit in which a Markovian approximation would be applicable. Therefore the renormalization order n=3 is the boundary between fast and slowly decaying memory functions. That is, renormalization attempts to describe entangled polymer dynamics are realistically restricted to the RRM, TRRM or the ThRRM without any heuristic preference.

It remains to future developments to find a physical, more elementary and less heuristic basis for the renormalization ansatz, or at least some physical arguments and principles making this sort of ansatz more and more plausible. At present the justification of this sort of modeling can only be based on a phenomenological argument, namely the success of describing experimental findings.

Below it will be shown that field-cycling NMR relaxometry studies unambiguously reveal a crossover between high-frequency and low-frequency dispersion regimes that can be identified with the high-mode-number and low-mode-number limits of the renormalized Rouse models. Moreover, the variation of the power law exponents closely corresponds to that predicted by the renormalized Rouse models. These dynamic regimes cannot be explained by the tube/reptation model, for instance. The conclusion is, that the renormalized Rouse models correctly reflect the structure of the generalized Langevin equation with respect to the predicted dynamic limits, while some minor ambiguity is left with respect to the exponent values.

Finally it should be mentioned that Schweizer also suggested the Polymer Mode–Mode Coupling model (PMMC) [99, 100] as a further heuristic ansatz to account for the so-called viscoelastic plateau, a phenomenon not predicted by renormalized Rouse models, at least if based on an effectively intramolecular entropic stress tensor. In a recent review [104] all advantages and deficiencies of the PMMC ansatz are described in detail. The main problem with this sort of model arising in the present context is that a molecular mass dependence is predicted on a time scale shorter than the (longest) Rouse relaxation time. Such a behavior is inconceivable because the polymer segments do not yet "know" the chain length on that time scale. On the other hand, the predictions of the PMMC model for long times, $t \gg \tau_R$, are more realistic than those of the renormalized Rouse models. The future development of memory matrix approaches of any sort will be closely connected with investigations of the nature of "projected dynamics" as a key problem.

4 Experimental Studies of Bulk Melts, Networks and Concentrated Solutions

In this section a series of experimental NMR studies based on the techniques described above will be compared with predictions of the model theories. Of course, any model is based on idealizations approaching reality only under certain conditions. The objective of the first few paragraphs will therefore be to demonstrate the rich variety of phenomena that can influence polymer dynamics. It will be elucidated under what circumstances the essence of the model theories and their predicted limits comes to light. We will first consider three basic features of polymer dynamics, namely the *three dynamic components* governing molecular motions of polymer chains, the *dynamics of chain-end blocks* in contrast to the central segment block, and how *free volume and voids* influence dynamics and the appearance of NMR measurands.

4.1 The Three Components of Polymer Dynamics as Relevant for NMR Relaxometry

The chemical composition of a polymer described by a Kuhn segment chain (see Fig. 6) is exclusively represented by specific parameters such as the segment friction coefficient and the Kuhn segment length. That is, no information referring to a length scale shorter than the Kuhn segment and, hence, to the specific chemistry of the compound is considered in the frame of the polymer theories discussed in this context. In terms of time scales, only molecular motions taking place in the limit $t >> \tau_s$ are regarded, where τ_s is a time constant characteristic for local motions occurring inside a Kuhn segment. In terms of the Rouse model, the segment fluctuation time τ_s is given by Eq. 53. The question to be examined in this section is to what extent NMR relaxation experiments are affected by local motions ($t \leq \tau_s$) on the one hand and by chain modes ($t >> \tau_s$) on the other.

Generally one can distinguish three dynamic components contributing to motions of a chain [35, 48, 123, 124]. *Component A* represents fluctuations occurring within Kuhn segments, that is on a time scale up to $t \approx \tau_s$ and a length scale $R \leq b$. These motions may be supplemented and superimposed by monomer side-group rotational diffusion if such mobile groups exist. *Component B* refers to the hydrodynamic chain-mode regime which is of particular interest in context with chain dynamics models. *Component C*, the cutoff process of component B, finally corresponds to the terminal chain relaxation time τ_t after which all memory of the initial conformation gets lost. In terms of the reptation model, the terminal chain relaxation time is given by the tube disengagement time, $\tau_t = \tau_d$ (see Eq. 71).

4.1.1 Component A

Component A implies anisotropic main-chain-group and, if applicable, side-group reorientations covering a restricted solid angle range of the inter-dipole vector (proton resonance) or the electric field gradient principal axis (deuteron resonance). As concerns the main-chain groups, it is mainly due to rotameric isomerism [126, 127] and may be interpreted even in terms of defect diffusion models [128].

The consequence of the restricted nature of the reorientations by component A is that the dipolar or quadrupolar correlation functions do not decay to zero by these local and molecular weight independent motions. Rather a residual correlation remains at long times that decays further only by chain modes of a hierarchically higher order. The correlation functions given in Eqs. 28 and 38 can therefore be specified for component A as

$$G_A(t) = g_A(t) + G_A(\infty), \tag{154}$$

where $g_A(t >> \tau_s)=0$ and $G_A(\infty)$=const. Experiments suggest that in condensed polymer systems $G_A(\infty)/G_A(0)=10^{-3}...10^{-2}$ only [33, 125,129]. That is, most of the orientation correlation function decays already due to component A. Components B and C can therefore refer only to this small residual correlation.

The consequence of the strong correlation decay due to component A is that the T_1 minimum is predominantly determined by this component and

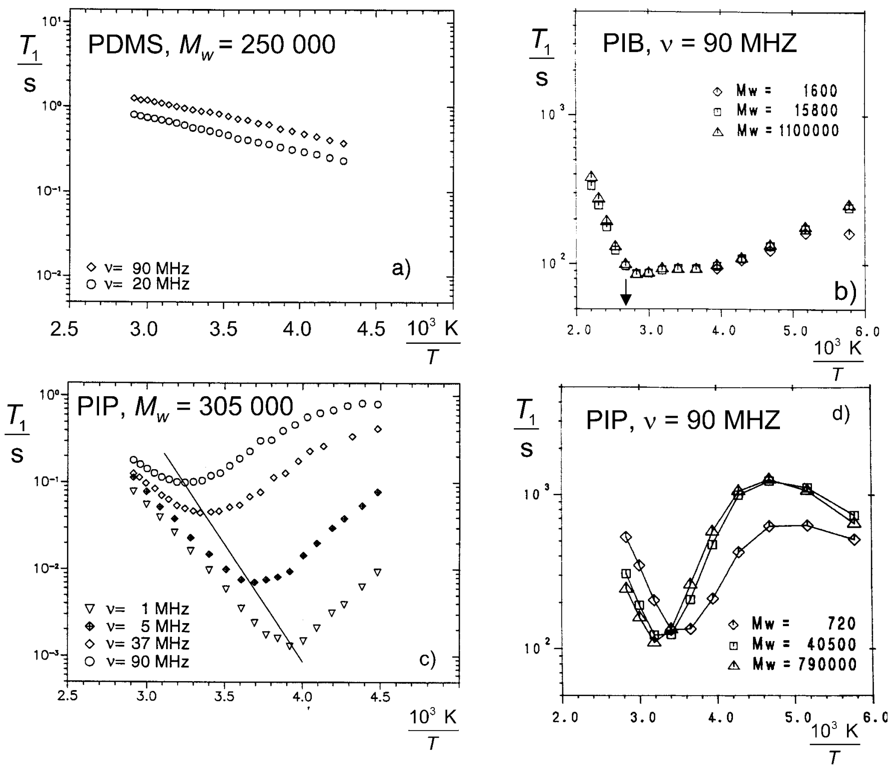


Fig. 14a–d. Proton spin-lattice relaxation time of various polymer melts as a function of the inverse temperature for different molecular masses and frequencies. The conclusion is that field-cycling NMR relaxometry of these polymers at the experimental temperatures is largely dominated by $\omega\tau_s<<1$ as a condition for the observation of components B and potentially C. **a** Polydimethylsiloxane (PDMS) for different frequencies. τ_s is too short in the temperature range indicated to fulfill the minimum condition $\omega\tau_s\approx 1$ even at 90 MHz [125]. **b** Polyisobutylene (PIB) for different molecular masses. The independence of the molecular weight above or below the critical value is obvious. The minimum is broadened by the superposition of rotational diffusion of methyl side groups. The *arrow* indicates the temperature of the field-cycling data shown in Fig. 28 [33]. **c** Polyisoprene (PIP) for different frequencies. The minimum positions (*solid line*) suggest an Arrhenius law $\tau_s=2\times10^{-19}$ s exp(58 kJ/mol/*RT*) [125]. **d** PIP for different molecular masses

indicates the value of τ_s via the "minimum condition" $\omega\tau_s \approx 1$. A series of experimental data sets are shown in Figs. 14 and 15. The slight shift of the T_1 minimum with the molecular weight observed with polyisoprene (Fig. 14d) can be straightforwardly described in a semiempirical three-component formalism [33].

The values deduced from the T_1 minima corroborate that the frequency range of field-cycling NMR relaxometry largely corresponds to the time limit $t >> \tau_s$ and, hence, addresses the chain-mode regime beyond local seg-

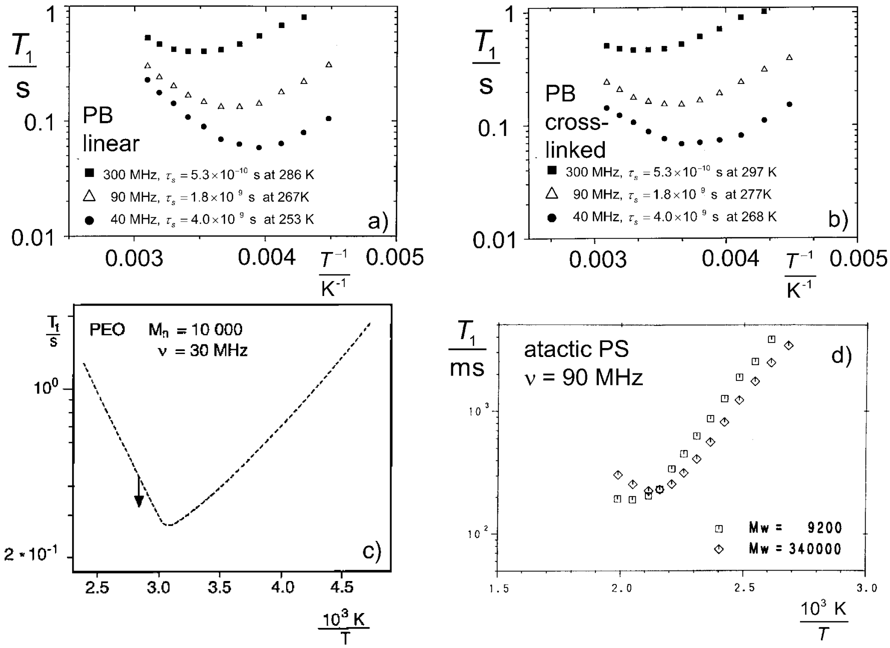


Fig. 15a–d. Temperature dependence of the proton spin–lattice relaxation time of various polymer melts at different frequencies and for different molecular masses. The segment fluctuation times τ_s can be evaluated from the T_1 minimum condition $\omega\tau_s$<<1. The temperatures at which the T_1 minimum occurs are specified for linear and cross-linked PB. It follows again that field-cycling NMR relaxometry at the experimental temperatures is largely dominated by $\omega\tau_s$<<1 as a condition for the observation of components B and potentially C. **a** Linear polybutadiene (PB; M_w=65,000) for different frequencies [131]. **b** Thermoreversibly cross-linked sample prepared from linear PB (M_w=51,000) by adding 37 phenylurazole groups per chain [131]. **c** Polyethyleneoxide (PEO). The *arrow* indicates the temperature of the field-cycling data shown in Fig. 29. M_n is the number average of the molecular mass [48, 130]. **d** Atactic polystyrene (PS) for different molecular masses. In this case, phenyl ring flips are slow enough to dominate the minimum so that segment fluctuations are concealed [33]

ment fluctuations (compare Fig. 5). In terms of the Rouse model, the segment fluctuation time τ_s is given by Eq. 53.

4.1.2 Component B

Component B represents motions in the chain-mode regime between τ_s and the terminal chain relaxation time. On a time scale up to the longest chain-mode relaxation time, which in terms of the Rouse model is given by τ_R (see Eq. 54), it is independent of the molecular mass. A relatively weak molecular weight dependence can only arise beyond τ_R, that is in terms of the reptation model in the time range $\tau_R < t < \tau_d$ (see Table 1).

4.1.3 Component C

Component C finally terminates component B and may become visible in the experimentally accessible frequency window of field-cycling NMR relaxometry in the form of a crossover to an "extreme-narrowing" plateau [2]. Since this crossover is connected with the terminal chain relaxation time τ_t, it will be strongly dependent on the molecular mass. This effect, however, will show up in the experimentally accessible frequency window of NMR relaxometry only for relatively small molecular masses (in contrast to the NMR diffusometry experiments).

4.1.4 The Different Time-scale Approach for the NMR Correlation Function

Representing the correlation function decays by components A, B, and C by the normalized partial correlation functions $G_A(t)$, $G_B(t)$ and $G_C(t)$, and interpreting these functions as probabilities that the respective fluctuations of the spin interactions have not yet taken place at time t, permits one to combine the three partial correlation functions in the total expression

$$G(t) = G_A(t)G_B(t)G_C(t). \tag{155}$$

Combining this function with Eq. 154 and making use of the *different time scale approximation*, that is $g_A(t{>}\tau_s) \approx 0$, $G_B(t{\leq}\tau_s) \approx G_B(0){=}1$, and $G_C(t{\leq}\tau_t) \approx G_C(0){=}1$ leads to

$$G(t) \approx g_A(t) + G_A(\infty)G_B(t)G_C(t), \tag{156}$$

where τ_t in this case generally represents the terminal chain relaxation time and $G_A(\infty)$ is a constant typically representing less than a few percent of the initial correlation function value.

Figure 16 shows typical proton spin–lattice relaxation dispersion data for polyethylene melts as an illustration of the three-component behavior of polymer melts. For comparison with model theories the chain-mode regime represented by component B is suited best and will be discussed in detail. It will be shown that the NMR relaxometry frequency window of typically 10^3 Hz$< \nu <10^8$ Hz (for proton resonance) almost exclusively probes the influence of chain modes represented by component B (compare Fig. 5). That is, the correlation function experimentally relevant for spin–lattice relaxation dispersion may be identified with component B according to

$$G(t) \approx G_A(\infty)G_C(0)G_B(t) \propto G_B(t) \tag{157}$$

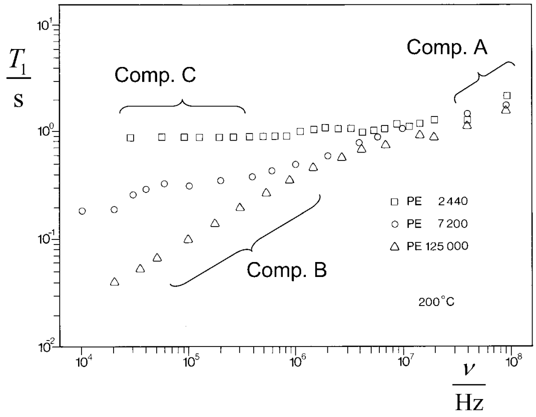


Fig. 16. Visualization of the three component regimes by way of example of proton spin-lattice relaxation dispersion in polyethylene (PE) melts of different molecular masses [123, 132]. The onset of Component A is indicated at the highest frequencies. It represents local fluctuations within a Kuhn segment and is characterized by a time constant τ_s=6.4×10^{-10} s independent of the molecular mass. It causes more than 90% of the decay of the orientation correlation function. At the temperature of the experimental data, 200 °C, the condition $\omega\tau_s \approx 1$ would be fulfilled at ν=2.5×10^8 Hz outside the frequency window shown. Hydrodynamic chain modes reveal themselves in distinct form as component B at low frequencies provided that the molecular mass (i.e., the terminal chain relaxation time) is large enough not to conceal this regime. Component C finally is a manifestation of the terminal relaxation time leading to a crossover to a low-frequency plateau for polymers short enough (compare Fig. 5). Other examples where component C showed up in field-cycling NMR relaxometry are given in Figs. 31b (PDMS, ^{1}H resonance) and 45b (deuterated PEO, ^{2}H resonance)

for polymer melts with $M >> M_c$, T >300 K, and τ_s<10^{-9} s. NMR relaxometry thus offers a unique chance to directly probe predictions by chain-mode model theories.

Before deepening the discussion of the experimental features of component B referring to the chain-mode regime, we will first consider a series of semiempirical studies corroborating the existence of the three dynamic components and other polymer characteristics one has to keep in mind when interpreting experimental polymer data. Such general characteristics can favorably be elucidated with the aid of transverse relaxation.

4.1.5 Average Transverse Relaxation

As a local mechanism, component A is completely independent of the molecular mass. Component B is also expected to be molecular weight independent for $t<\tau_R$, but adopts a weak molecular weight dependence for $\tau_R < t < \tau_t$ under entangled-dynamics conditions. This is in contrast to component C which, as a terminal process, is subject to a strong dependence on the chain length for $t \geq \tau_t$. The crossover between regimes where the more or less molecular weight dependent components dominate can also be visualized by the temperature and molecular weight dependences of the average transverse relaxation rate [33]. Figures 17, 19, and 20 show typical data of this sort plotted as the reciprocal average transverse relaxation rate.

The average transverse relaxation rate is obtained from the initial slope of the (normalized) transverse relaxation curve. Describing the (typically non-exponential) transverse relaxation curve by a linear combination of exponentials,

$$S(t) = \sum_j a_j \exp\left\{-{}^{t}/_{T_{2,i}}\right\}, \tag{158}$$

with coefficients a_j and time constants $T_{2,j}$, one obtains for the time derivative

$$\frac{dS(t)}{dt} = -\sum_j a_j \frac{1}{T_{2,j}} \exp\left\{-{}^{t}/_{T_{2,i}}\right\}. \tag{159}$$

In this expression we have neglected any initial Gauss-like decay contribution as it occurs in extreme cases such as networks or linear polymers of high molecular weight close to the glass temperature [133–137] (compare the Anderson/Weiss formalism). With the transverse relaxation data shown in Figs. 17, 19, and 20, this initial Gauss-like decay becomes only perceptible or even prevailing at the lowest temperatures investigated. In that case, a gradual crossover to solid-like behavior occurs leading to average transverse relaxation times as short as 10 μs. Typical transverse relaxation curves of linear polymer melts can be found elsewhere [36, 123, 133, 134, 137]. These data demonstrate that far above the glass temperature, and in particular with polymers having mobile side groups (such as polydimethylsiloxane, PDMS, for example), a data analysis based on the exponential function expression given in Eq. 159 is justified in good approximation. Note also, that part of the data discussed in the following refer to ^{13}C resonance of quaternary carbons (see Fig. 19c), so that a Gauss-like component is excluded in this case anyway.

Under such conditions, the initial slope of the transverse relaxation curve (often evaluated from the decay to $1/e$ of the initial value both in experiment and in theoretical Anderson/Weiss evaluations) approaches the average transverse relaxation rate according to

$$\left.\frac{dS(t)}{dt}\right|_{t\to 0} = -\sum_j a_j \frac{1}{T_{2,j}} \equiv \left\langle \frac{1}{T_{2,j}} \right\rangle. \tag{160}$$

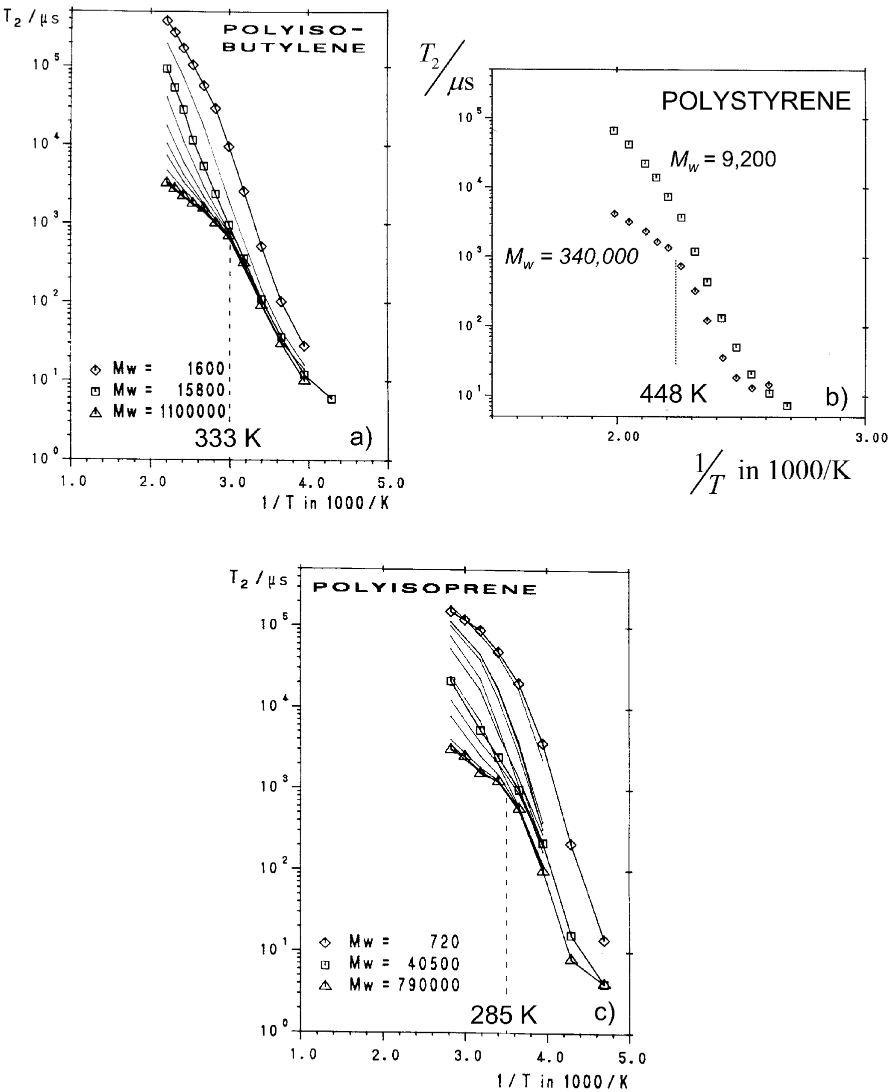


Fig. 17a–c. Effective proton transverse relaxation time $T_2 = \langle 1/T_{2,j} \rangle^{-1}$ (see Eq. 160) in different linear polymers of different molecular masses measured at 90 MHz as a function of the reciprocal temperature [33]. A characteristic temperature T_{char} shows up for $M_w \geq M_c$ in the form of a relatively sharp bend in the temperature dependence. This bend disappears when approaching the critical molecular weight from above. For $M_w < M_c$ an additional molecular weight dependence reveals itself as a consequence of the chain length dependent free-volume effects. **a** Polyisobutylene. Symbols for the data points have been plotted for three different molecular weights only, while the data for all other molecular weights are represented by *polygonal lines* for the sake of clearness. From the *top* to the *bottom*, the lines refer to data for M_w=1,600; 5,800; 15,800; 34,000; 55,000; 80,600; 122,000; 182,000; 393,000; 610,000; 830,000; 1,100,000. The critical molecular

The transverse relaxation rate to be discussed in the following will be identified with this average expression, that is, an effective transverse relaxation time is defined as $T_2 = \left\langle T_{2j}^{-1} \right\rangle^{-1}$.

4.1.5.1
Characteristic Temperature

Experimental data of the effective transverse relaxation time are plotted in Fig. 17 as a function of the reciprocal temperature for different polymers. The molecular weight dependence for M_w<M_c is due to free-volume effects. We will come back to this sort of molecular weight effect later in the review.

In contrast to short polymer chains, the molecular weight dependence occurring for M_w>M_c at temperatures above a certain characteristic value T_{char} given in Table 4 is of a completely different nature. The temperature dependence of T_2 for high molecular weights in a sense reflects the hierarchy of segment and chain dynamics in terms of the three components A, B, and C. That is, more and more of the spectral density of dipolar fluctuations is shifted into the motional averaging regime when the temperature is increased (compare the Anderson/Weiss formula in the spectral density variant given in Eq. 40). Figure 18 shows a schematic representation of the low-frequency section of the spectral density that predominantly determines the transverse relaxation rate according to Eq. 40. As a consequence, the spectral densities specific for components A, B, and C come into play only one by one in a more or less combined way as suggested by Eq. 156 upon temperature variation.

At very low temperatures, none of the components is fast enough to be subject to motional averaging. The "rigid-lattice limit" begins to apply (possibly with the exception of methyl group rotation or other fast side group motions if appropriate). Increasing the temperature, the fastest chain motion component, namely component A, is the first one causing motional averaging to some degree. The transverse relaxation time is hence getting longer with increasing temperature. The next slower process is component B, still of no or very little molecular weight dependence. The characteristic

weight is about M_c≈15,000 [35]. The characteristic temperature is T_{char}=333 K. **b** Atactic polystyrene. The data refer to M_w=9,200 (<M_c≈31,000) and M_w=340,000 (>>M_c). The characteristic temperature is T_{char}=448 K. **c** Polyisoprene. Symbols for the data points have been plotted for three different molecular weights only, while the data for all other molecular weights are represented by *polygonal lines* for the sake of clearness. From the *top* to the *bottom*, the lines refer to data for M_w=720; 1,000; 3,280; 3,450; 5,100; 8,150; 10,800; 19,400; 34,000; 40,500; 58,000; 67,000; 130,000; 205,000; 280,000; 790,000. The critical molecular weight is about M_c≈19,000 [35]. The characteristic temperature is T_{char}=285 K

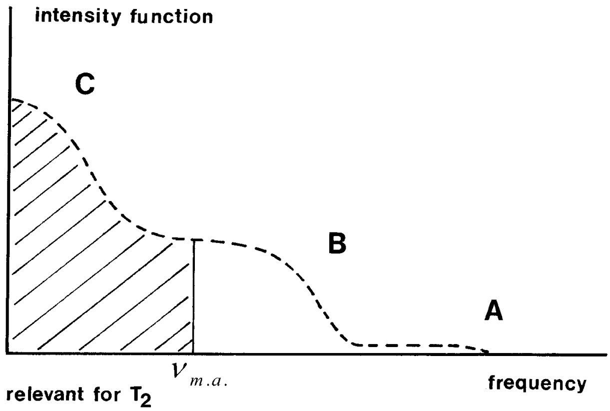
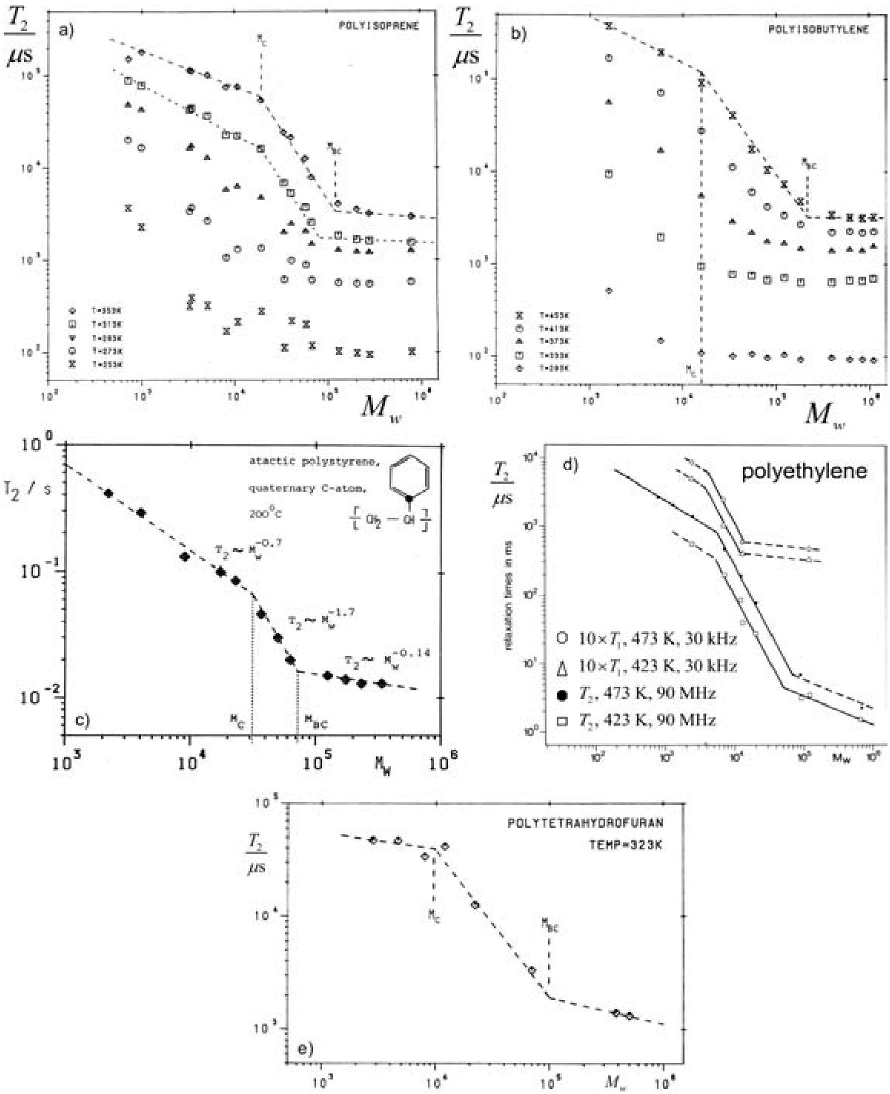


Fig. 18. Schematic representation of the spectral density (or intensity function) for spin couplings in the frame of the three-component analysis of molecular motions in polymers. The dipolar broadening region represented as the *hatched section* at low fluctuation rates is predominantly responsible for the transverse relaxation rate (compare Eq. 40 and Ref. [2]). Variation of the temperature shifts the components across the fluctuation rate defined by the motional-averaging condition, so that the influence of the individual components changes one by one. Variation of the molecular weight or the polymer concentration likewise shifts the molecular weight or concentration dependent components across the motional averaging fluctuation rate $\nu_{m.a}$

temperature T_{char} finally marks the onset of motional narrowing for component C with its strong dependence on the chain length. That is, for $T > T_{char}$ all three components give rise to motional averaging, whereas for $T < T_{char}$ only component A and, with gradually decreasing efficiency, component B contribute to the motion-based averaging of dipolar coupling. A semiempirical three-component formalism using the same evaluation procedure for T_2 and describing this sort of phenomenon in great detail has been reported [33, 35]. Note that the same sort of characteristic temperature was observed with spin–lattice relaxation data in the rotating frame, $T_{1\rho}$, at 1 kHz for polyisoprene. The characteristic temperatures found for a series of different polymer species are listed in Table 4.

Fig. 19a–e. Molecular weight dependence of the effective proton and ^{13}C transverse relaxation times $T_2 = \langle 1/T_{2,I} \rangle^{-1}$ (see Eq. 160) in different linear polymers at different temperatures. The data were measured at 2.1 T [35, 138, 139]. The classical critical molecular weight M_c and a value characteristic for the three-component nature of molecular motions in polymers, M_{BC}, reveal themselves as bends in the slopes of the power laws approaching the dependences on the molecular weight. Values of the characteristic molecular weights are listed in Table 4. For $M_w < M_c$ an additional molecular weight dependence



may matter as a consequence of the chain length dependent free-volume effects). **a** Polyisoprene. From the *top* to the *bottom*, these proton data refer to the temperatures 353, 313, 293, 273, 253 K. **b** Polyisobutylene. From the *top* to the *bottom*, these proton data refer to the temperatures 453, 413, 373, 333, 293 K. **c** Atactic polystyrene. These data refer to the ^{13}C line of the quaternary carbon (see *inset*) and were measured at 473 K. Note that the double-bend behavior found with ^{13}C NMR [139] was not observed with proton NMR where only a single bend showed up [138]. That is, M_{BC} is concealed by ring flips in the latter case. **d** Polyethylene. These proton data refer to 423 and 473 K. For comparison, low-frequency T_1 data are also plotted to demonstrate that spin–lattice relaxation at frequencies in a range of those relevant for T_2 is also subject to characteristic molecular weights. **e** Polytetrahydrofuran. These proton data refer to 323 K

Table 4. Characteristic temperatures and molecular weights evaluated from ^{1}H and ^{13}C (of quaternary carbons) relaxation data of linear polymers [33, 35, 139, 140]. PE, polyethylene; PIP, polyisoprene; PIB, polyisobutylene; PS, atactic polystyrene; PTHF, polytetrahydrofuran; PDMS, polydimethylsiloxane. Literature data for critical molecular weights determined with rheology are also listed for comparison and completeness

	T_{char}/K	M_c		M_{BC}	
PE	–	4,000	(T_1, ^{1}H, 30 kHz)	52,000	(423 K, T_2, ^{1}H, 90 MHz)
		5,600	(T_2, ^{1}H, 90 MHz)	70,000	(473 K, T_2, ^{1}H, 90 MHz)
		3,800	(rheology [52])		
PIP	285 K (T_2, ^{1}H, 90 MHz)	19,000	(T_2, ^{1}H, 90 MHz)	119,000	(353 K, T_2, ^{1}H, 90 MHz)
		20,000	(T_2, ^{13}C, 23 MHz)	80,000	(313 K, T_2, ^{1}H, 90 MHz)
		10,000	(cis, rheology [52])	70,000	(293 K, T_2, ^{1}H, 90 MHz)
				140,000	(T_2, 353 K, ^{13}C, 23 MHz)
PIB	333 K (T_2, ^{1}H, 90 MHz)	15,000	(T_2, ^{1}H, 90 MHz)	215,000	(453 K, T_2, ^{1}H, 90 MHz)
		15,200	(rheology [52])	133,000	(413 K, T_2, ^{1}H, 90 MHz)
				80,000	(373 K, T_2, ^{1}H, 90 MHz)
PS	448 K (T_2, ^{1}H, 90 MHz)	31,000	(T_2, ^{13}C, 23 MHz)	71,000	(T_2, 473 K, ^{13}C, 23 MHz)
		31,200	(rheology [52])		
PTHF	–	10,000	(T_2, ^{1}H, 90 MHz)	100,000	(323 K, T_2, ^{1}H, 90 MHz)
PDMS	–	20,000	(T_2, ^{1}H, 90 MHz)	$\geq$200,000	(353 K, T_2, ^{1}H, 90 MHz)
		24,000	(rheology [52,147])		
PEO	–	3,600	(rheology [52])	–	

4.1.5.2
Characteristic Molecular Weights

Figure 19 shows typical data for the effective transverse relaxation time $T_2=\langle T^{-1}{}_{2,j}\rangle^{-1}$ as a function of the molecular weight. Analogous to the characteristic temperature discussed above, the molecular weight dependences show two sharp bends defining molecular weights again characteristic for the hierarchical role of the three components of chain dynamics.

At very high molecular weights, it is only the molecular weight independent component A and to some extent component B that contribute to motional averaging. In terms of the schematic representation of the spectral density in Fig. 18, the combined fluctuation intensity of these components is the only one that acts above the motional averaging frequency. Reducing the molecular weight means faster component C fluctuations. This manifests itself at the characteristic molecular weight M_{BC} below which the molecular weight dependent component C gradually increases T_2 with decreasing molecular weight. M_{BC} separates the regimes where component C is subject to motional averaging ($M_w<M_{BC}$) and where not ($M_w>M_{BC}$).

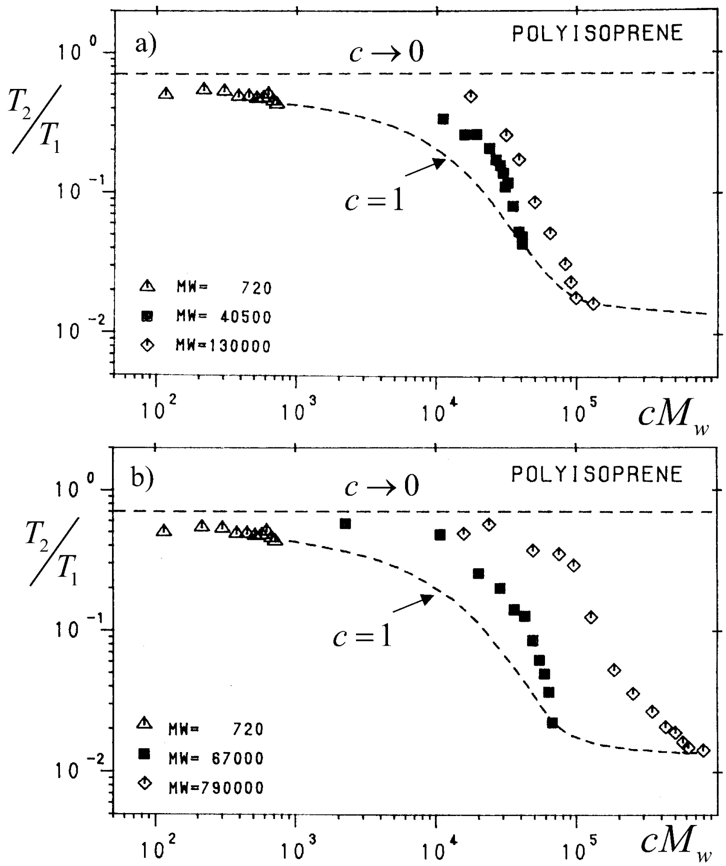


Fig. 20. Combined molecular weight and concentration dependence of T_2/T_1 in solutions of polyisoprene in CCl_4 [34]. The data refer to five different molecular weights (M_w=720; 40,500; 67,000; 130,000; 790,000). All data have been measured at 90 MHz proton resonance and 313 K. The concentration c is given as the volume fraction of the polymer. The *lower broken line* refers to pure melts (c =1; compare Fig. 19), the *upper straight line* to dilute solutions (c<<1)

Before complete motional averaging is reached at very low molecular weights, another phenomenon interferes, namely the crossover from entangled to Rouse-like chain dynamics at the critical molecular weight M_c well-known from rheology [52]. At this second characteristic molecular weight, component B and component C adopt a different character as discussed above in context with the model considerations. For M_w<M_c chain dynamics tends to be governed by Rouse-like dynamics, whereas entangled chain dynamics shows up for M_w>M_c.

A semiempirical three-component formalism using the same evaluation procedure for T_2 and describing the molecular weight dependence with both characteristic breaks can be found in Ref. [35]. Note that the same sort of phenomenon also occurs with ^{13}C transverse relaxation [139] and spin–lattice relaxation in the kHz frequency regime [138]. Table 4 summarizes the characteristic molecular weights observed for different polymer species.

Note that the molecular weight dependence of the NMR relaxation times in polymer melts is generally much weaker than that of the viscosity in the same materials [52]. Even below M_{BC} the exponent of power laws for the molecular weight dependence of NMR relaxation times is roughly only half as large. This statement applies to molecular weights both below and above M_c. The reason is that the zero-shear viscosity is dominated by component C of chain dynamics alone, whereas the NMR relaxation times are affected by components B and A as well. In the frame of the three-component concept it was concluded that the molecular weight dependence of the NMR relaxation times for $M_w<M_{BC}$ are governed by an effective correlation time, τ_c^{eff} ,formed as the geometrical average of the time constants of component B, τ_l, and component C,τ_t [35, 138, 139]. That is

$$\tau_c^{eff} = \sqrt{\tau_l \tau_t}. \tag{161}$$

Since τ_l does not or only weakly depend on the chain length, the molecular weight dependence of the effective correlation time for entangled dynamics results in $\tau_c^{eff} \propto M_w^{1.5\ldots2.0}$ [35, 139, 144], whereas the terminal relaxation time of rheology scales as $\tau_1 \propto M_w^{3.4}$ [52]. The effect of the crossover from unentangled to entangled dynamics on spin–lattice relaxation will be discussed later (see Fig. 31b).

4.1.5.3
Crossover from the Melt to Dilute Solutions

Swelling and dissolving polymers in low-molecular solvents changes chain dynamics. This refers to components B and C in particular. Actually, the critical molecular weight that indicates the onset of entangled behavior changes with the polymer concentration according to [52]

$$M_c^{sol} c = const, \tag{162}$$

where c is the polymer concentration and M_c^{sol} is the critical molecular weight of the solution. Decreasing the polymer concentration means increasing the critical molecular weight effective in the solution. Expressed the other way round, the crossover from entangled to unentangled dynamics of a polymer of a given molecular weight $M_w>M_c(melt)$ can be passed through by reducing the polymer concentration. This is visualized in Fig. 20 for polyisoprene of various molecular weights dissolved in CCl_4 [34]. In order to remove any effects by modified local motions (in particular due to the monomeric friction coefficient) and by the free volume that is likely affected by the solvent, the ratio T_2/T_1 is plotted. For c=1, the data refer to unswollen melts. For $c<<1$ a plateau is reached independent of the chain length. This plateau corresponds to the limit of dilute solutions.

The T_2/T_1 data demonstrate the entanglement effect which gradually increases with the molecular weight and the polymer concentration. The data are dominated by local motions (component A) for low concentrations and/or for low molecular weights. On the other hand, the entanglement effect reveals itself for high molecular weights and high polymer concentrations by the correspondingly modified components B and C. This phenomenon can again be described by the semiempirical three-component formalism mentioned before [34]. The dilution effect on spin–lattice relaxation will be discussed later in the review (see Fig. 31d).

4.2 Chain-end Dynamics ("Whip Model")

Segments located in blocks near the chain ends are expected to be subject to molecular motions different from those in the central block provided that the molecular mass of the chain is above the critical value, so that entangled dynamics is established in principle. The dynamics of segments in the chain-end blocks should generally be faster than those in the central block. Actually, in a ^{13}C transverse relaxation study of a polystyrene–polyisoprene–polystyrene three-block copolymer with a monomer ratio of 10:1600:10, this dynamical heterogeneity could be demonstrated qualitatively by selectively measuring ^{13}C signals specific for the polystyrene blocks and for the polyisoprene block [141].

The same sort of three-block dynamics is also expected for homopolymers where different transverse relaxation components on these grounds show up, provided that the molecular weight distribution is practically monodisperse [36]. This suggests experimental investigations in a rather direct way.

Two different scenarios for chain-end dynamics have been suggested. Doi [142] introduced the so-called contour length fluctuation (CLF) as a modification of the tube/reptation model. Due to the stochastic nature of chain modes in the tube, the chain ends are fluctuating back and forth a length proportional to the square root of the chain length and, hence, are subject to tube constraints to a much lower degree than the central part of the chain. The chain-end blocks are therefore expected to have a molecular mass obeying

$$M_{eb} \propto M_w^{1/2} \quad (\text{“CLF model”}). \tag{163}$$

On the other hand, the mobile chain-end blocks may be identified as chain sections not subject to entanglements so that they move in a whip-like fashion. In view of the critical molecular mass M_c indicating the minimum

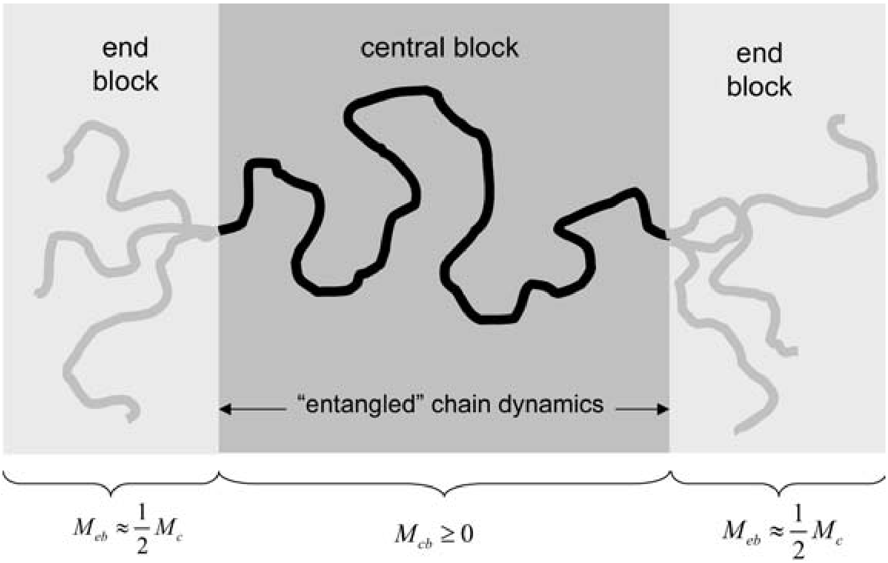


Fig. 21. Schematic representation of the three blocks that can be distinguished in a homopolymer melt with respect to segment dynamics ("whip model") [36]. The two chain-end blocks of a molecular mass $M_{eb}\approx 0.5\ M_c$ each are not expected to be subject to entangled dynamics, whereas the central block of molecular mass $M_{cb} \geq 0$ is. Below the critical molecular mass, $M_w \leq M_c$, no central block exists according to this definition, so that entangled behavior does not arise at all

chain length for entangled dynamics, the chain-end blocks are expected to have a molecular mass independent of the total chain length:

$$M_{eb} \approx \frac{1}{2} M_c = const \quad (\text{“whip model”}). \tag{164}$$

Figure 21 shows an illustration of this sort of three-block distinction.

The transverse relaxation functions can be analyzed for two components related to the central and chain-end blocks according to

$$S(t) \propto g(t) + q \exp\left(-t/T_2^l\right), \tag{165}$$

where T^l_2 is the transverse relaxation time of the longest component which is attributed to the two chain-end blocks. The quantity $q=2M_{eb}/M_w$ is the "amplitude" of the chain-end block component relative to the signal of the whole chain. The first term on the right-hand side, $g(t)$, comprises all faster components including potentially Gauss-like contributions with very long chains and at low temperatures. This fast-decaying part of the transverse relaxation curve is supposed to represent the central chain block. Experimentally an analysis of this sort will be the more reliable the faster $g(t)$ decays relative to the exponential function in Eq. 165. That is, the experiments described in the following refer mainly to $M_w >> M_c$ (compare the molecular weight dependences given in Fig. 19).

For the CLF model, one expects

$$q \propto \frac{\sqrt{M_w}}{M_w} = M_w^{-1/2}. \tag{166}$$

This molecular weight dependence is contrasted by the constant chain-end blocks assumed in the whip model:

$$q = \frac{2M_{eb}}{M_w} = \frac{M_c}{M_w} \propto M_w^{-1}. \tag{167}$$

The objective is now to determine q from experimental data as a function of the molecular mass of the whole chain in order to differentiate the two model predictions.

Figure 22 shows typical transverse relaxation curves. For $M_w<M_c$ the decay is monoexponential (apart from a slight influence of translational diffusion relative to the inhomogeneities of the magnet). The chain-end blocks reveal themselves only above the critical molecular weight M_c as visualized in Fig. 22 by the transverse relaxation curve for $M_w>>M_c$.

Fitting the second term on the right-hand side of Eq. 165 to the long-time tail of experimental transverse relaxation curves permits one to determine

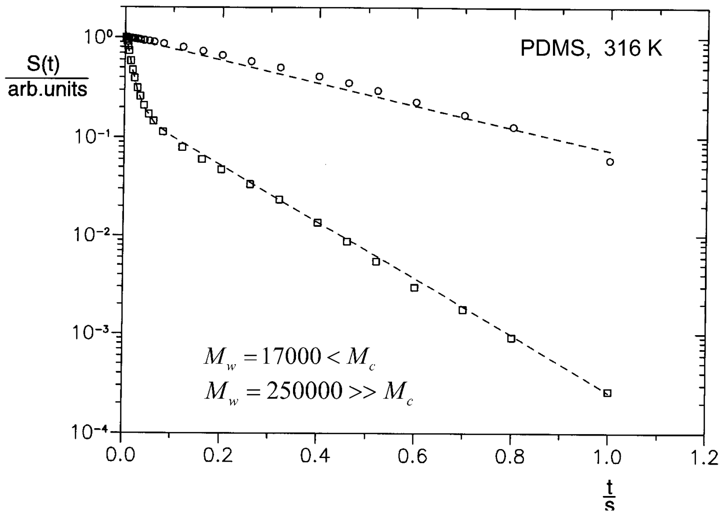


Fig. 22. Typical transverse relaxation curves for polydimethylsiloxane (PDMS) above and below the critical molecular weight M_c=24,000 [52, 147]. The data were recorded with the Hahn echo at 316 K [36]. The slowly decaying exponential tail indicative for the fast chain-end block dynamics reveals itself only above M_c. The *broken lines* have been calculated with the aid of Eq. 165 and a formalism described in Ref. [125]

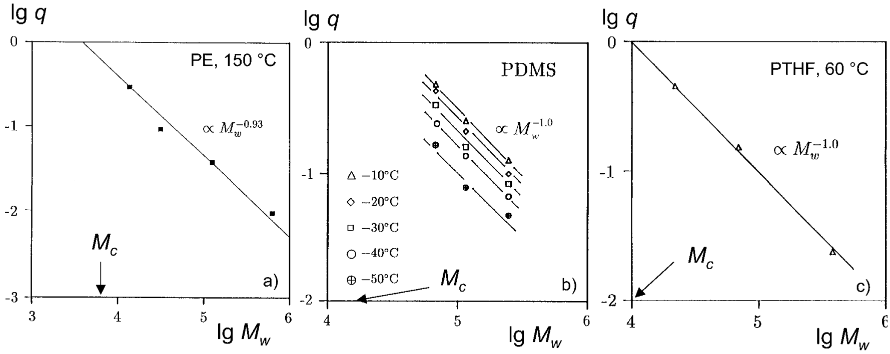


Fig. 23a–c. Relative amplitude of the slowly decaying tail of the transverse relaxation curve measured in **a** linear polyethylene (PE), **b** polydimethylsiloxane (PDMS), and **c** polytetrahydrofuran (PTHF) as a function of the molecular mass [36]. The data have been recorded with standard Hahn echo and Carr/Purcell/Meiboom/Gill pulse sequences. The values of the critical molecular weight are indicated on the abscissa axes. Extrapolating the data for $M_w \rightarrow M_c$ suggests $q \rightarrow 1$ as predicted by the whip model (Eq. 167)

the relative amplitude q. Figure 23 shows results for a series of different polymers as a function of the molecular mass. The data contradict the CLF model represented by Eq. 166, whereas they are largely in accordance with Eq. 167 predicted for the whip model. Moreover, extrapolating the data to q=1 suggests abscissa values very close to the critical molecular weight as suggested by Eq. 167. This, and the fact that the chain-end block relaxation time T_2^l of PDMS turned out to be essentially independent of the molecular weight in contrast to the central block relaxation function $g(t)$ [36, 143], strongly support the whip model further.

Figure 24 shows a schematic illustration of the scenario that chain-end dynamics is underlying. Any chain fold that happens to be formed near a chain end consists of a more mobile section (drawn in light gray) that can diffuse back and forth on a curvilinear path along the chain contour between the chain end and the tip of the fold. If this chain section moves towards the chain end, the mass of the moving entity is *increasing* by "unrolling" the fold (the position of which is assumed to be static for simplicity). If the chain section moves towards the fold tip, its mass is *decreasing* by "rolling up" the fold. The friction obstructing this sort of curvilinear displacement is proportional to the length ξ of the moving entity, so that the curvilinear diffusion coefficient of the mobile chain section will be proportional to ξ^{-1}. That is, displacements directed towards the fold tip are favored since ξ is reduced, whereas displacements towards the chain end are slowed down because of the increase of ξ associated with this direction of motion. This mechanism results in a motion resembling that of the cord of a cracking whip. A formalism describing this peculiar whip-like motion due to ter-

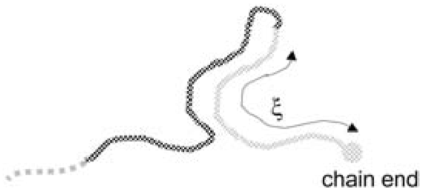


Fig. 24. Illustration of the whip model of chain-end dynamics. A chain fold that happens to be near a chain end consists of a relatively immobile strand towards the middle of the chain (*left*; drawn in *dark gray*) and a chain section on the right (drawn in *light gray*) that can quickly migrate back and forth. Its curvilinear length ξ between the chain end and the tip of the fold is decreased when it is displaced towards the tip of the fold and is increased when it moves in the chain-end direction. As a consequence, the curvilinear diffusion coefficient is increased or decreased, respectively, depending on the displacement direction. Therefore, motions towards the tip of the fold are favored, and the fold tends to unroll not unlike the cord of a cracking whip

minal chain folds close to chain ends has been described [36]. Recent ^{13}C spin–lattice relaxation measurements in the rotating frame corroborate the different dynamics of chain-end blocks [129].

4.3 Free-Volume and Void Effects

The whip effect on chain ends may be considered to be the molecular origin of the free volume which is known to be connected with chain ends [145, 146]. Since the relative number of segments near chain ends decreases with increasing molecular weight, free-volume effects show up predominantly at low molecular weights and in particular below the critical value M_c. The consequence is an additional chain length dependence of segment motions particularly for $M_w<M_c$. That is, the molecular weight dependence intrinsic to chain modes is superimposed by that due to free-volume variations. This effect is accounted for in the empirical Williams/Landel/Ferry (WLF) formalism that can be used to correct data for free-volume effects [52, 147–150].

The free volume and its distribution in heterogeneous samples can be directly probed by the enhancement of spin–lattice relaxation by molecular oxygen dissolved in the sample. Molecular oxygen is electron paramagnetic and, hence, an efficient relaxation agent. The local spin–lattice relaxation rate therefore monitors the local oxygen concentration. Exposing the sample to an oxygen atmosphere at an elevated pressure prompts oxygen molecules to diffuse into areas of reduced density. The concentration reached in equilibrium is a matter of the available free volume. A corresponding formalism can be found in Ref. [151]. In order to avoid leveling of the local relaxation rates by flip-flop spin diffusion, a perdeuterated semicrystalline polyethyl-

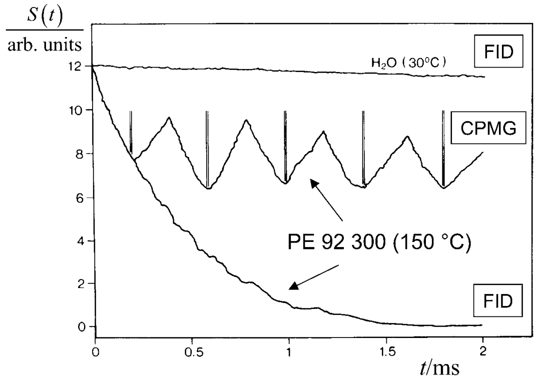


Fig. 25. Free-induction decay (FID) after a 90° pulse at t=0 and a series of Carr/Purcell/Meiboom/Gill (CPMG) spin echoes in linear polyethylene (M_w=92,300) at 150 °C. The 180° pulses of the CPMG sequence are indicated by *vertical lines*. The homogeneity of the external magnetic flux density is demonstrated by an FID of water at 30 °C. The data demonstrate the existence of strong internal field gradients arising from voids [152]

ene sample leaving only a few isolated undeuterated ethylene groups was examined [128]. It was shown that all free volume accessible by oxygen must be attributed to the amorphous part of the sample.

As a consequence of the existence of free volume related to chain ends, the density in polymer melts is expected to be heterogeneous even if the polymer chain length is practically monodisperse. In NMR, density inhomogeneities reveal themselves via the magnetic susceptibility distribution in the form of magnetic field gradients. The corresponding (inhomogeneous) broadening or the corresponding shortening of the free-induction decays in polyethylene were shown to reach two orders of magnitude [152]. Figure 25 demonstrates the effect of such internal field gradients on the free-induction decay (FID). The decay time to $1/e$ of the initial signal intensity is plotted in Fig. 26 as a function of the molecular weight. Interestingly a characteristic molecular weight similar to M_{BC} for the proper (effective) transverse relaxation time (see Fig. 19) shows up in the form of a crossover from a power law decay to a plateau at high molecular weights. This is an indication that component C acts as a very efficient motional narrowing mechanism as discussed above.

4.4 Evidence for Rouse Dynamics ($M<M_c$)

Field-cycling NMR relaxometry is an excellent monitor of chain modes. As illustrated in Fig. 5, it specifically and predominantly probes component B. The other components, A and C, matter only at low temperatures/high frequencies and high temperatures/low molecular weights, respectively. That is,

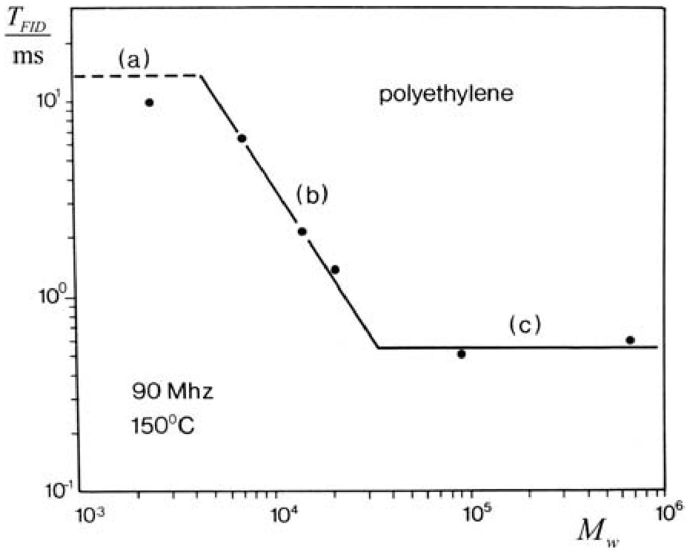


Fig. 26. Decay times T_{FID} to $1/e$ of the initial value of the free-induction signals in melts of linear polyethylene at 150 °C as a function of molecular mass. The *horizontal broken line* (a) indicates the instrumental homogeneity limitation determined with a water sample at 30 °C (see Fig. 25). The motional averaging region (b) is characterized by a strong molecular mass dependence equivalent to that of the proper (effective) transverse relaxation time (see Fig. 19d). It is therefore attributed to the combined influence of components B and C. The plateau (c) reached at high molecular masses reflects the unaveraged effect of internal field inhomogeneities corresponding to a full width at half maximum of the resonance line of about 7 ppm. This value can be estimated on the basis of the magnetic susceptibility of polyethylene relative to that of empty voids [152]

field-cycling NMR relaxometry is particularly suitable to examine component B and to directly test predictions of the model theories.

Rouse dynamics is expected to apply to molecular weights below the critical value where entanglement effects are not yet effective. Experimental data sets for the proton spin–lattice relaxation dispersion [47, 49, 153] are shown in Fig. 27 in comparison to the theoretical frequency dependence predicted by Eq. 64. Very interestingly, the values for the segment fluctuation time τ_s fitted to the T_1 data coincide with those derived from the T_1 minima (see Fig. 14) corrected for the temperature of the field-cycling measurements. That is, the two independent determination methods lead to consistent results.

The logarithmic Rouse formula given by Eq. 64 is valid for $\omega \ll \tau_s^{-1}$. Deviations of the theoretical curves from the experimental data points are therefore only expected at frequencies close to the "minimum condition" $\omega\tau_s \approx 1$. where simultaneously component A starts to become perceptible (compare Fig. 16). This explains why the PDMS data fit better to the model than poly-

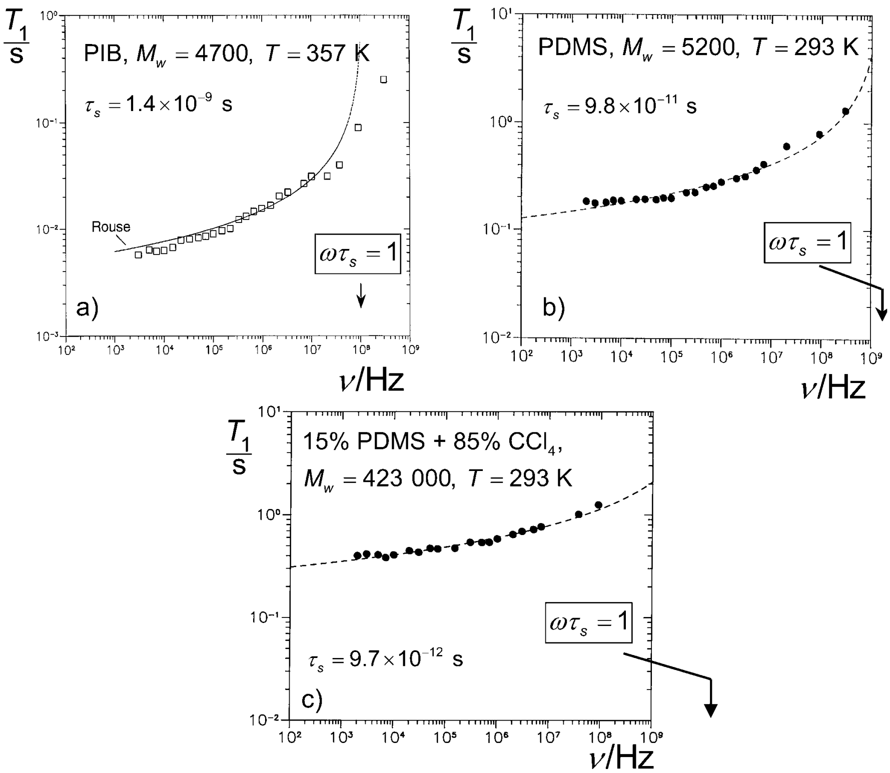


Fig. 27a–c. Proton spin–lattice relaxation dispersion under conditions where Rouse dynamics is expected to apply. The theoretical curves have been calculated with the aid of Eq. 64. The validity of this model is restricted to $\omega<<\tau_s^{-1}$. The positions on the frequency axes where the condition $\omega\tau_s=1$ applies are indicated by *arrows* for the segment fluctuation time τ_s fitted to the experimental data. The τ_s values are in accord with those derived from the T_1 minimum data (see Fig. 14) where applicable. **a** Polyisobutylene (M_w=4,700 < M_c≈15,000), melt at 357 K [49]. **b** Polydimethylsiloxane (M_w=5,200 < M_c≈20,000), melt at 293 K [47, 125, 153]. **c** Solution of 15% polydimethylsiloxane (M_w=423,000) in CCl_4 at 293 K [47, 125, 153]. The critical molecular weight of solutions is modified according to Eq. 162

isobutylene (PIB) with a segment fluctuation time one order of magnitude longer than that of PDMS.

Entanglement effects can be reduced by dissolving the polymer even if its molecular weight is well above the critical value in the melt. This is demonstrated in Fig. 27c for PDMS dissolved in CCl_4. The data can again be well described by the Rouse model.

The conclusion from these findings is that the Rouse model is perfectly corroborated by NMR relaxometry experiments provided that entanglement effects are excluded or not effective on the time/frequency scale of the experiment. On the other hand, Rouse dynamics was also predicted for entangled dynamics in limit $(\mathrm{I})_{\mathrm{DE}}$ of the tube/reptation model and for the short-time limits of the renormalized Rouse models (see Tables 1, 2 and 3). However, no such behavior was ever found for polymer melts on the basis of NMR relaxometry. The conclusion is that the short-time/high-frequency dynamics of entangled-polymer melts is dominated by component A (including side-group motions) rather than by component B in the Rouse limit. For polymer melts, there is no evidence for an intermittent Rouse regime under entanglement conditions. On the other hand, entangled-polymer solutions where the critical molecular weight is shifted to correspondingly higher values (see Eq. 162) obviously do show such an intermittent Rouse limit as demonstrated in Figs. 27c and 31d.

4.5
The Three Regimes of Spin–Lattice Relaxation Dispersion in Entangled Polymer Melts, Solutions, and Networks ($M<M_c$)

Figure 5 schematically shows the experimental window probed with field-cycling NMR relaxometry in combination with conventional pulse techniques for the determination of spin–lattice relaxation times. In the following, we specifically refer to molecular weights and temperatures large enough so that the relaxation data recorded exclusively refer to component B. Very large molecular weights also exclude any perceptible influence of the whip effect and of free volume connected with chain-end blocks. That is, segments can be assumed to form a dynamically homogeneous phase on the time scale of spin–lattice relaxation experiments (an exception is reported below in context with mesogenic order). Under such conditions, field-cycling NMR relaxometry [2, 18–21] turned out to be an unmatched tool for the elucidation of dynamics in the chain-mode regime.

Figures 28, 29, 30, 31, 32, 33, and 34 show a series of typical spin–lattice relaxation dispersion curves. The technique has been applied to melts, solutions, and networks of numerous polymer species. As experimental parameters, the temperature, the molecular weight, the concentration, and the cross-link density were varied. For control and comparison, the studies are partly supplemented by rotating-frame spin–lattice relaxation data and, of course, by high-field data of the ordinary spin–lattice relaxation time. Furthermore, the deuteron spin–lattice relaxation was employed for identifying the role different spin interactions are playing for relaxation dispersion.

Remarkably, a series of distinct and apparently universal NMR relaxation dispersion regimes can be identified from the data in Figs. 28 to 34 for the diverse polymer species. We consider the time/frequency window $\tau_t << (t,\omega^{-1}) << \tau_s$ (see Fig. 5). The terminal chain relaxation time, τ_t, is indicated by a crossover to a low-frequency T_1 plateau (see Fig. 16); the seg-

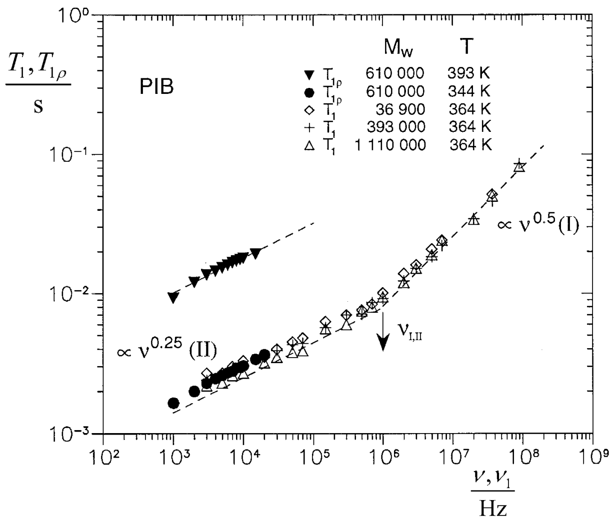


Fig. 28. Proton spin–lattice relaxation times in the laboratory system (T_1) and in the rotating frame ($T_{1\rho}$) of polyisobutylene (PIB) melts as a function of the frequency (ν, Larmor frequency in the laboratory frame; $\nu_1=\gamma B_1/(2\pi)$, rotating-frame nutation frequency) [125]. The data refer to the molecular weight independent chain-mode regimes I (high-mode-number limit) and II (low-mode-number limit) [49]. The *arrow* indicates the crossover frequency $\nu_{I,II}$ between regions I and II

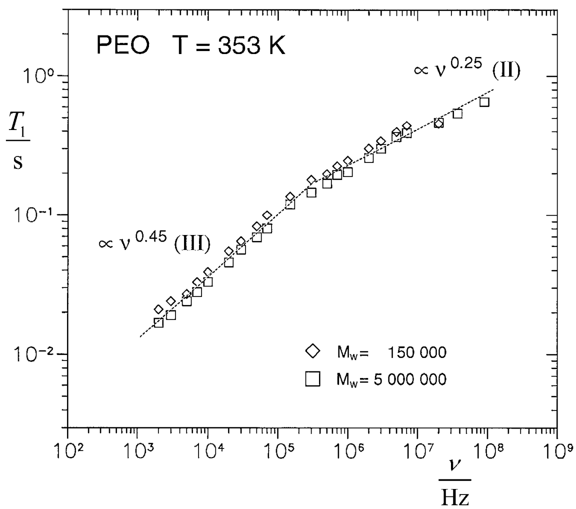


Fig. 29. Proton spin–lattice relaxation time of polyethyleneoxide (PEO) melts as a function of the frequency [49]. The data refer to the molecular weight independent chain-mode regime II (low-mode-number limit) and regime III influenced by intersegment dipolar interactions [154]

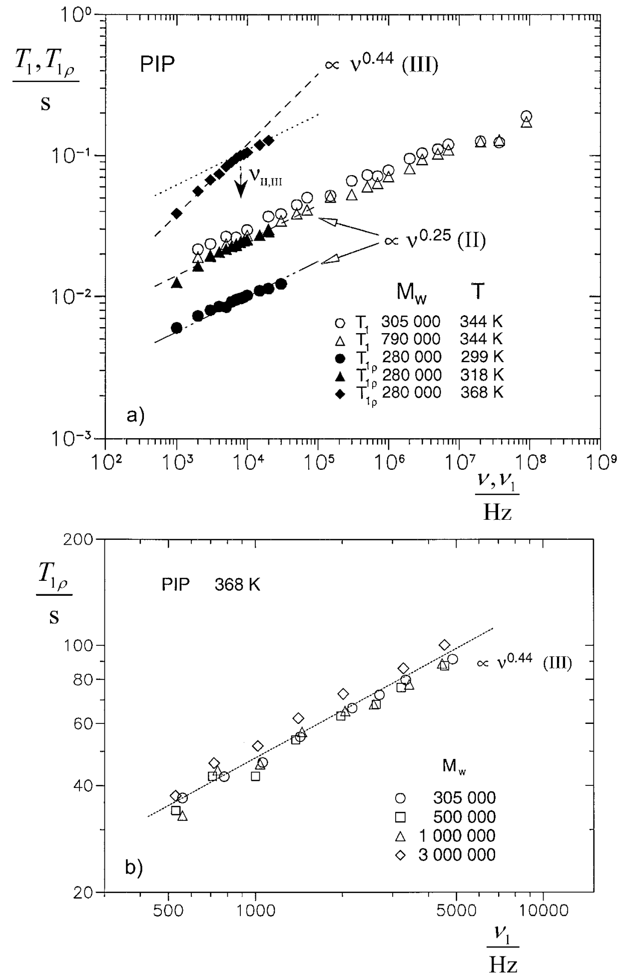


Fig. 30a, b. Proton spin–lattice relaxation times in the laboratory system (T_1) and in the rotating frame ($T_{1\rho}$) of polyisoprene (PIP) melts as a function of the frequency (ν, Larmor frequency in the laboratory frame; $\nu_1=\gamma B_1/(2\pi)$, rotating-frame nutation frequency) [125]. The data refer to the molecular weight independent chain-mode regime II (low-mode-number limit) and regime III influenced by intersegment dipolar interactions [154]. **a** Comparison of T_1 and $T_{1\rho}$ data. The *arrow* is to indicate the crossover frequency $\nu_{I,II}$ between regions I and II. **b** $T_{1\rho}$ data in detail

ment fluctuation time, τ_s, can be determined from T_1 minima (see Figs. 14 and 15). Under such conditions, there is clear experimental evidence for three distinct proton dispersion regimes for entangled polymers, $M_w >> M_c$:

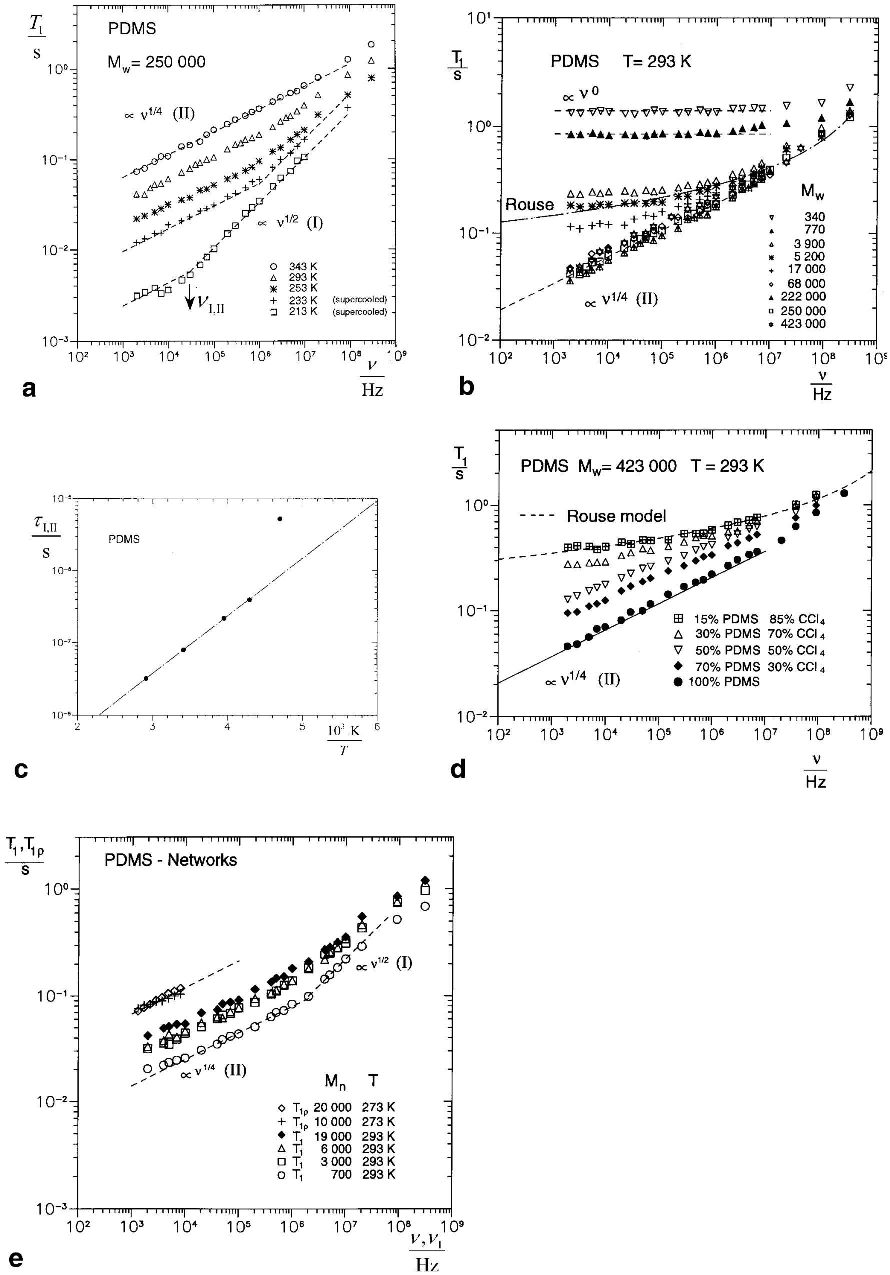


Fig. 31a–e. Proton spin-lattice relaxation times in the laboratory system (T_1) and in the rotating frame ($T_{1\rho}$) of polydimethylsiloxane (PDMS) melts and solutions as a function of the frequency (ν, Larmor frequency in the laboratory frame; $\nu_1=\gamma B_1/(2\pi)$, rotating-frame nutation frequency) [125]. The melt data for $M_w>M_c\approx 24{,}000$ refer to the molecular weight independent chain-mode regimes I (high-mode-number limit) and II (low-mode-

$$\uparrow^{\omega}\downarrow_{T}\begin{cases} T_1 \propto M_w^0 \omega^{0.5\pm0.05} & \text{(region I; “high mode number limit”)} \\ T_1 \propto M_w^0 \omega^{0.25\pm0.05} & \text{(region II; “low mode number limit”)} \\ T_1 \propto M_w^0 \omega^{0.45\pm0.05} & \text{(region III; “inter - segment interaction limit”)} \end{cases} \tag{168}$$

These power laws are in accordance with data for the frequency dependence of the spin–lattice relaxation time in the rotating frame, $T_{1\rho}$, in a range as far as accessible by this technique (see Figs. 28 and 30–32).

The three proton relaxation dispersion regions I (fast motions), II, and III (slow motions) appear in that sequence from high to low frequencies or from low to high temperatures (see the up and down arrows in Eq. 168. The frequency window of the field-cycling NMR relaxometry technique (Fig. 5) is often not broad enough to cover all three regions at once. Mobile polymers like polyisobutylene (Fig. 28), polydimethylsiloxane (Fig. 31), and polydiethylsiloxane in the isotropic phase (Fig. 32) are subject to regions I and II under the experimental conditions. Less mobile polymers like polyethyleneoxide (Fig. 29) and polyisoprene (Fig. 30) tend to reveal regions II and III in the instrumental frequency window.

Nevertheless it is possible to demonstrate that all three proton relaxation dispersion regions are intrinsic to entangled polymer dynamics. The temperature/frequency range accessible with NMR relaxometry in polybutadiene (Fig. 33) permits the recording of all three regions one after the other in the same sample. At low temperatures, regions I and II dominate; at elevated temperatures region III comes into play. The shift of the crossover time between regions I and II,

$$\tau_{I,II} = 1/\left(2\pi\nu_{I,II}\right), \tag{169}$$

number limit) [49]. **a** Melts of PDMS 250,000 at different temperatures. The crossover between regimes I and II at $\nu_{I,II}$ is shifted to higher frequencies with increasing temperature. **b** Melts of PDMS at 293 K for different molecular weights. For M_w<M_c the chain-mode regimes I and II characteristic for entangled dynamics are absent and are replaced by motions not subject to entanglements. The theoretical T_1 dispersion curve expected for the Rouse model is shown for comparison. **c** Temperature dependence of the “crossover time constant” $\tau_{I,II}=1/(2\pi\nu_{I,II})$ evaluated from the plot in Fig. 31a. The *line* represents the Arrhenius law $\tau_{I,II}=1.1\times10^{-10}$ s $\exp\{(15.8\ \text{kJ mol}^{-1})/RT\}$. The deviation of the data point at 213 K indicates the influence of the supercooled state at this temperature. **d** Solutions of PDMS 423,000 in CCl_4 at 293 K for different concentrations. For low concentrations the chain-mode regimes I and II characteristic for entangled dynamics fade more and more. The theoretical T_1 dispersion curve expected for the Rouse model is shown for comparison. **e** Cross-linked PDMS for different cross-link densities specified by the mesh molecular weight M_n (number average)

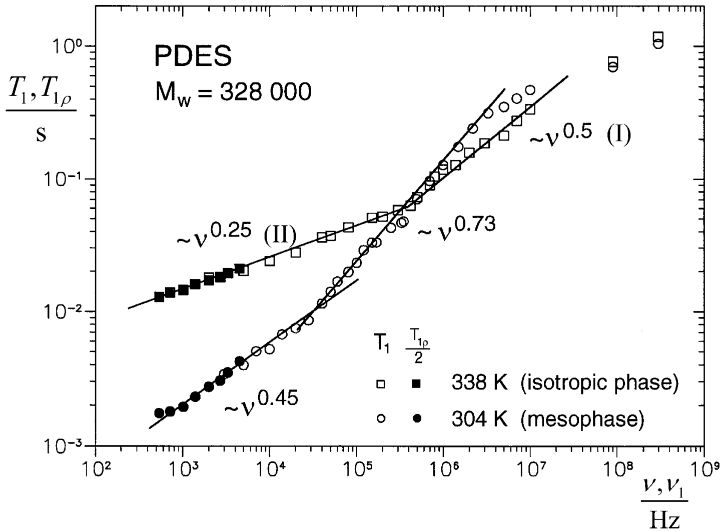


Fig. 32. Proton spin–lattice relaxation times in the laboratory system (T_1) and in the rotating frame ($T_{1\rho}$) of polydiethylsiloxane (PDES) melts in the isotropic and mesomorphic phases as a function of the frequency (ν, Larmor frequency in the laboratory frame; $\nu_1=\gamma B_1/(2\pi)$, rotating-frame nutation frequency) [155]. The data of the isotropic melt phase refer to the molecular weight independent chain-mode regimes I (high-mode-number limit) and II (low-mode-number limit). In the (ordered) mesomorphic phase the dispersion slopes are much steeper and the crossover frequency is shifted to a lower value. At very high frequencies, the influence of component A (including side chain motions) becomes gradually visible

for polydimethylsiloxane is plotted in Fig. 31c as a function of the reciprocal temperature. The Arrhenius-like behavior corroborates that the chain-mode regimes I and II are subject to thermal activation.

These three proton spin–lattice relaxation dispersion regions must not be identified with the Doi/Edwards limits $(\mathrm{I})_{\mathrm{DE}}$, $(\mathrm{II})_{\mathrm{DE}}$, and $(\mathrm{III})_{\mathrm{DE}}$ which are predicted for dynamic ranges with significantly different frequency and molecular weight dependences (see Table 1). No Rouse-like dynamics corresponding to limit $(\mathrm{I})_{\mathrm{DE}}$ can be identified in the experimental data sets shown for entangled polymer melts. In the frame of the tube/reptation model, there is no such thing as a frequency dependence like region II, $T_1 \propto M_w^0 \omega^{0.25\pm0.05}$. The square root molecular weight dependence predicted in that model for limit $(\mathrm{III})_{\mathrm{DE}}$ does not occur either. However, it will be shown that Doi/Edwards predictions can indeed be verified in perfect detail if chains are confined to artificial tubes prepared in a solid polymer matrix [95]. On the other hand, the high- and low-mode-number limits discussed in the frame of the renormalized Rouse models (Tables 2 and 3) provide a perfect explana-

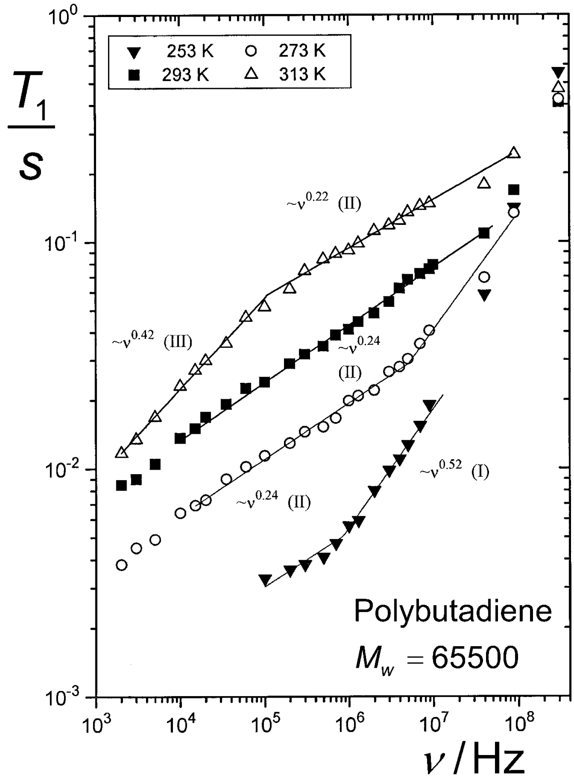


Fig. 33. Proton spin–lattice relaxation time of a melt of linear polybutadiene (M_w= 65,500) at different temperatures as a function of the frequency [154]. By varying the temperature, all three dispersion regimes show up in the experimental frequency window one by one (see Fig. 5)

tion of dispersion regions I and II, whereas region III can be shown to be due to intersegment dipolar interactions.

4.5.1
High- and Low-Mode-Number Limits (Dispersion Regions I and II)

The appearance of dispersion regions I and II in experiments is an exultant confirmation of the high- and low-mode-number, short-time limits predicted by the twice renormalized Rouse models (Tables 2 and 3). The exponents of the power laws suggested by the experimental data even match the theoretical predictions almost perfectly [47, 49]. Nevertheless, the good coincidence of the numerical values of these exponents is not considered to be the decisive finding backing up the renormalized Rouse models. The problem is that the theoretical exponents are slightly affected by the renormalization

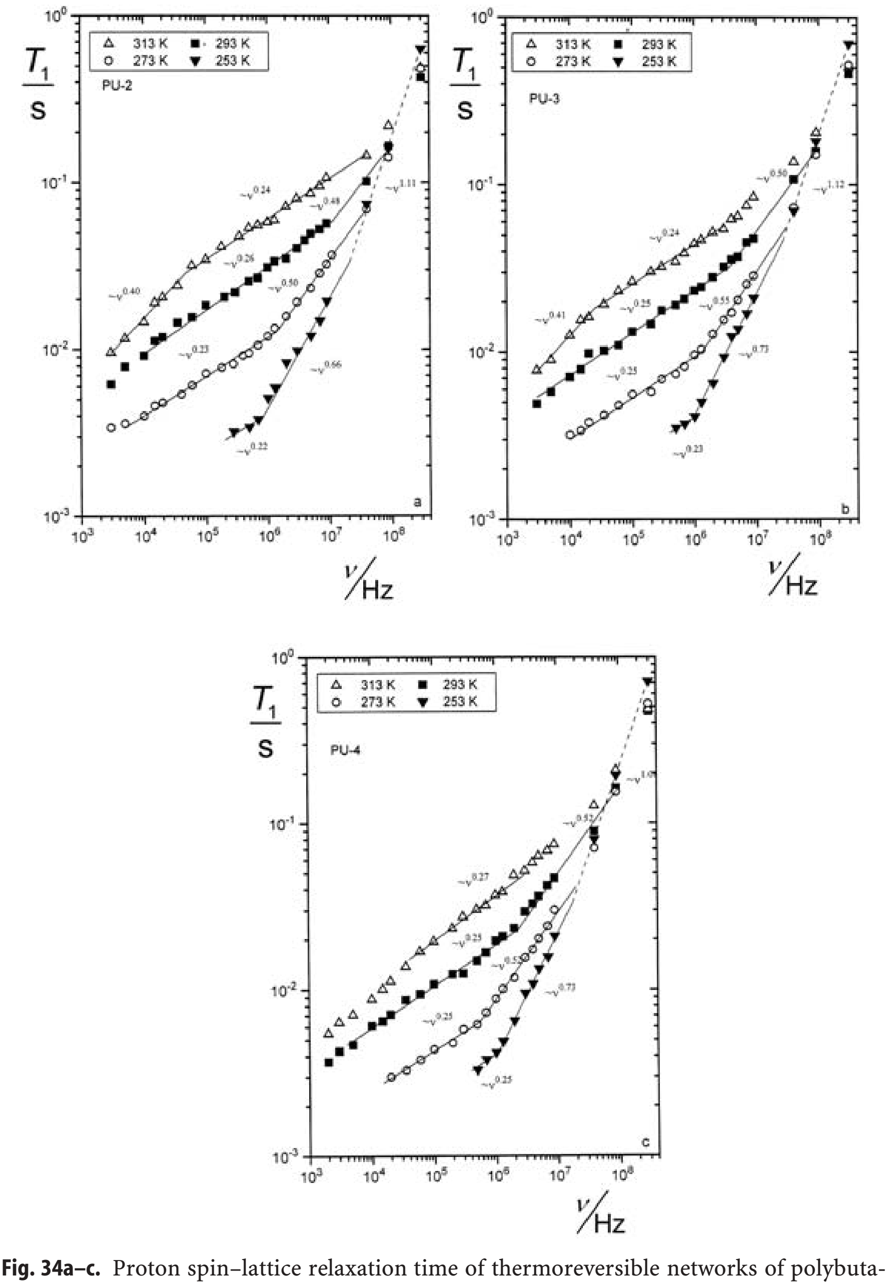


Fig. 34a–c. Proton spin–lattice relaxation time of thermoreversible networks of polybutadiene at different temperatures as a function of the frequency [131]. Linear PB (M_w=51,000) was cross-linked by addition of 4-phenyl-1,2,4-triazoline-3,5-dione (phenylurazole, PU). The crossover frequencies between regimes I, II, and III are shifted depending on the cross-link density. At the lowest temperatures the dispersion slopes tend to be steeper than in ordinary melts (see Eq. 168). **a** 19 phenylurazole groups per chain. **b** 28 phenylurazole groups per chain. **c** 37 phenylurazole groups per chain

ansatz that cannot be traced back to elementary principles and is subject to some ambiguity. The important result is rather that (i) two such limits become visible and that (ii) the exponents change almost exactly as predicted for the crossover between the high- and low-mode-number limits.

The distinction of two such limits with the right variation of the power law exponents appears to be a result reflecting the analytical structure of the generalized Langevin equation, which is thus proven to represent the essential features of entangled chain dynamics in the short-time limit, $t \ll \tau_t$. The generalized Langevin equation obviously contains crucial elements determining entangled-chain dynamics.

4.5.2
Intra- and InterSegment Spin Interactions (Dispersion Region III)

Intrasegment dipolar interaction fluctuates as a consequence of segment reorientations and conformational changes. If the interacting nuclear dipoles are residing on different segments or even different chains, variations of the internuclear vector tend to take much longer because they are the consequence of displacements of the dipole hosting segments by self-diffusion relative to each other. Any spin–lattice relaxation dispersion affected by intersegment dipolar interactions is therefore expected at very low frequencies. This can be tested by comparing proton with deuteron spin–lattice relaxation dispersion of the same polymer species. Deuteron relaxation is dominated by (intrasegment) quadrupole coupling with the local electric field gradient, whereas proton relaxation can be determined by intra- as well as intersegment dipolar interactions [2, 154].

Figure 35 shows such comparisons for polyethyleneoxide and polybutadiene melts. The crossover between dispersion regions II and III observed with proton NMR obviously disappears in the deuteron studies. Region III does not occur with deuterons. The different dispersion slope measured with deuteron NMR was also demonstrated with samples of lower molecular weights in Figs. 36 and 45b. That is, dispersion region III must not be considered as a limit specific for polymer theories. Rather it appears that is mainly an effect intrinsic to NMR relaxation by dipolar interaction.

A schematic representation of the situation one is dealing with in this context is shown in Fig. 37. We consider the representative segments # k on chain # α and # l on chain # β. The internuclear vector, $\vec{r}_{kl}(t)$, fluctuates because of self-diffusive displacements $\vec{R}_{rel}(t)$ of segment # l relative to segment # k. That is, the origin of the reference frame is considered to be fixed at segment # k.

The average correlation function for the intersegment dipolar interaction can be expressed as [154]

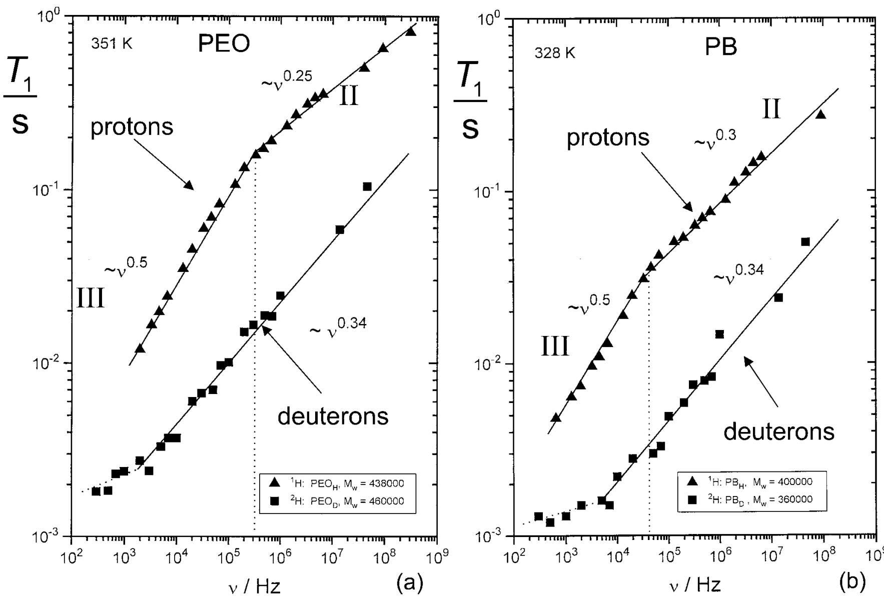


Fig. 35a, b. Proton and deuteron spin–lattice relaxation times of polyethyleneoxide (PEO) and polybutadiene (PB) melts as a function of the frequency [154]. In addition to intrasegment dipolar coupling, proton relaxation is also subject to intersegment dipolar couplings leading to the dispersion regime III specific for this sort of relaxation mechanism. Deuteron relaxation is predominantly due to quadrupole interaction which is of an intrasegment nature. A crossover from regime II to regime III therefore does not occur with deuteron resonance. **a** Polyethyleneoxide (for further deuteron relaxation data see Figs. 36 and 45b). **b** Polybutadiene

$$G^{(m)}_{\text{inter}}(t) = \rho_s \int g(r_0) G^{m}_{kl}(t) d^3 r_0 \quad (m = 1,2), \tag{170}$$

where

$$G^{(m)}_{kl}(t) = \left\langle \frac{Y_{2,m}(\vartheta_0, \varphi_0) Y_{2,-m}(\vartheta_t, \varphi_t)}{r_0^3 r_t^3} \right\rangle \Big/ G^{(m)}_{kl}(0) \tag{171}$$

(see Eq. 28), $g(r_0)$ is the radial segment pair correlation function, and ρ_s is the spin number density. The variables r_0 and r_t in Eqs. 170 and 171 stand for r_{kl} (0) and r_{kl} (t), respectively. That is, $r_0 \equiv r_{kl}$ (0) and $r_t \equiv r_{kl}$ (t). For the analytical treatment the radial segment pair correlation function may be approximated by (see Fig. 38)

$$g(r_0) \approx \begin{cases} 0 \text{ if } r_0 \leq \sigma \\ 1 \text{ otherwise} \end{cases} . \tag{172}$$

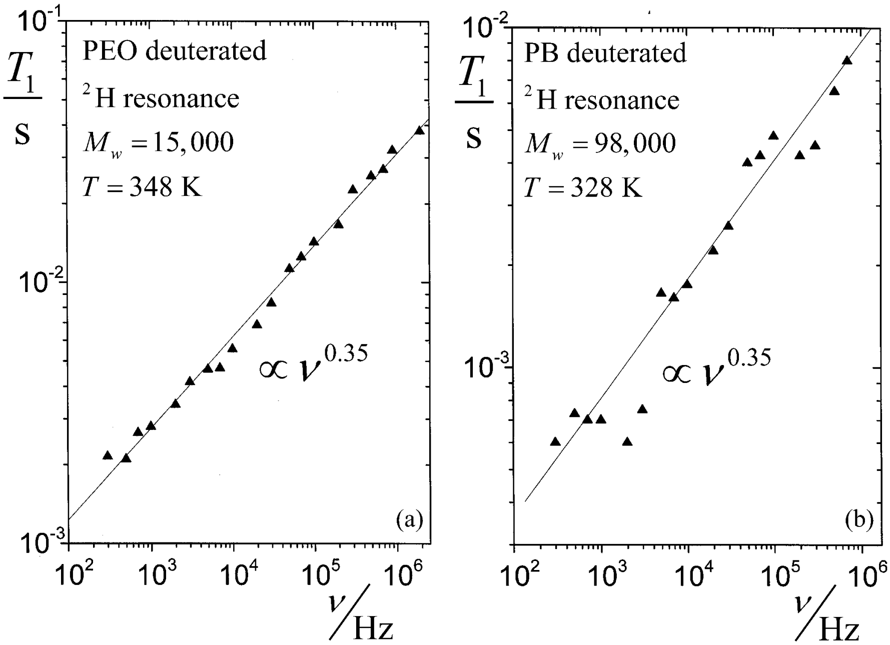


Fig. 36a, b. Deuteron spin–lattice relaxation times of deuterated polyethyleneoxide (PEO) (**a**) and polybutadiene (PB) (**b**) as a function of the frequency [156]

The correlation function in Eq. 171 can be approximated by the following consideration. For relative displacements $R(t)$ (see Fig. 37) much larger than the initial internuclear distance $r_0 \equiv r_{kl}(0)$, the distance and polar angle variation leads to total loss of any correlation to the initial distance vector, whereas the correlation is completely retained in the opposite limit. That is,

$$G_{kl}^{(m)}(t) \rightarrow 0 \text{ for } R_{rel}(t) >> r_0 \tag{173}$$

and

$$G_{kl}^{(m)}(t) \approx 1 \text{ for } R_{rel}(t) << r_0. \tag{174}$$

An ansatz accounting for these limits is

$$G_{kl}^{(m)}(t) \approx \left\langle \left| \frac{Y_{2,m}(0)}{r_0^3} \right|^2 \right\rangle P(t) = \frac{1}{r_0^6} P(t), \tag{175}$$

where $P(t)$ is the probability that segment # l is in a spherical volume $\propto r_0^3$ around its initial position $\vec{r}_0$ relative to segment # k. This probability is specified by the following limits:

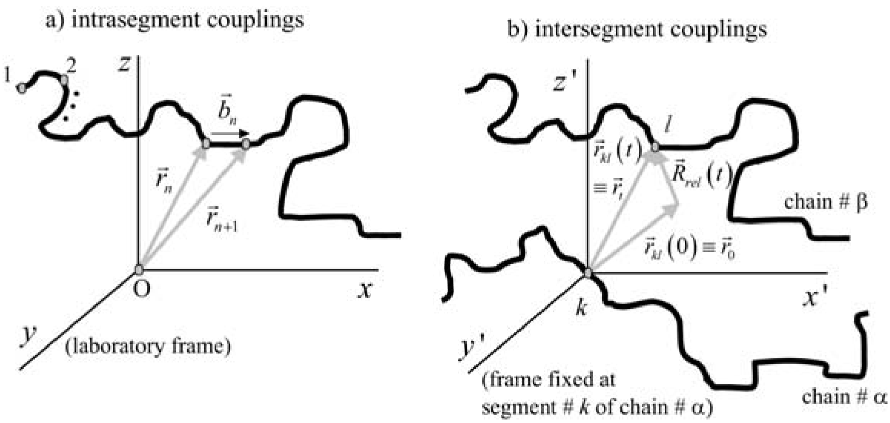


Fig. 37a, b. Intra- and intersegment spin couplings. **a** In the frame of the experiments referred to in this article, intrasegment spin couplings can be dominated by dipole–dipole interactions (^{1}H, ^{13}C), quadrupole couplings (^{2}H), and anisotropic chemical shift interactions (^{13}C). They fluctuate due to segment reorientations relative to the laboratory frame. For component B, the relevant segment orientation is represented by the chain tangent vector $\vec{b}_n$ at segment # n. **b** Intersegment interactions (segments on the same or different chains) are exclusively of a dipolar nature. Spin–lattice relaxation on these grounds originate from reorientation and length variation of the distance vector $\vec{r}_{kl}$ between segments (in the scheme denoted as # k and #l). These fluctuations are caused by translational displacements $\vec{R}_{rel}$ of one segment relative to the other. This is described best in a frame fixed on one of the segments of the interacting pair. Since self-diffusion is a relatively slow process, spin–lattice relaxation by intersegment interactions becomes relevant only at relatively low frequencies

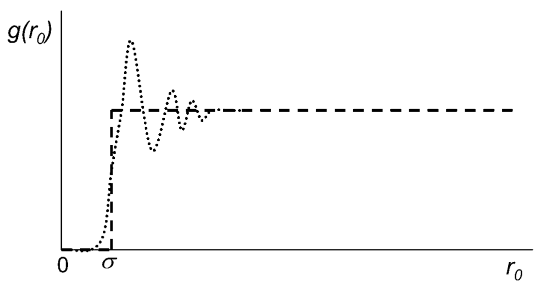


Fig. 38. Schematic representation of the radial segment pair correlation function. The "real" function characterized by a peak series of decaying amplitudes at the nearest coordination shells (*dotted line*). For the analytical treatment, this course is approximated by a step function (*broken line*). σ is the nearest-neighbour distance of segments

$$P(t) \to 1 \qquad \text{for } \left\langle R_{rel}^2(t) \right\rangle << r_0^2,$$

$$P(t) \to \frac{r_0^3}{\left\langle R_{rel}^2(t) \right\rangle^{3/2}} \quad \text{for} \quad \left\langle R_{rel}^2(t) \right\rangle >> r_0^2. \tag{176}$$

The expression

$$P(t) \approx \frac{r_0^3}{\left[r_0^2 + \left\langle R_{rel}^2(t) \right\rangle \right]^{3/2}} \tag{177}$$

complies to both limits given in Eq. 176 and will therefore be taken as an approximation. Inserting Eqs. 172, 175, and 177 in Eq. 170 gives

$$G_{\text{inter}}^{(m)}(t) \approx \rho_s \frac{\ln\left\{ 2\left\langle R_{lab}^2(t) \right\rangle / \sigma \right\}}{\left\langle R_{lab}^2(t) \right\rangle^{3/2}}, \tag{178}$$

where the mean squared displacement in the laboratory frame is related to the mean squared displacement in the frame fixed at segment # k by $\left\langle R_{lab}^2(t) \right\rangle = \frac{1}{2} \left\langle R_{rel}^2(t) \right\rangle$. The logarithmic term in Eq. 178 varies slowly with time so that we may approximate further

$$G_{\text{inter}}^{(m)}(t) \approx \frac{\rho_s}{\left\langle R_{lab}^2(t) \right\rangle^{3/2}}. \tag{179}$$

The time dependence of the mean squared segment displacement in the laboratory frame was derived on the relevant time scale as low-mode-number, short-time limits of the renormalized and twice renormalized Rouse models as $\left\langle R_{lab}^2(t) \right\rangle \propto t^{2/5}$ and $\left\langle R_{lab}^2(t) \right\rangle \propto t^{1/3}$ (limit $(\text{II})_{\text{TRR}}$), respectively (see Tables 2 and 3). Inserting these power laws in Eq. 179, one obtains after Fourier transformation according to Eqs. 29 and 30

$$\frac{1}{T_1} = \frac{1}{T_1^{\text{intra}}} + \frac{1}{T_1^{\text{inter}}}, \tag{180}$$

where ${T_1}^{\text{intra}}$ refers to the expressions given in Tables 2 and 3 for the low-mode-number, short-time limits, and ${T_1}^{\text{inter}}$ scales as

$$T_1^{\text{inter}} \propto \nu^{0.4\cdots0.5}. \tag{181}$$

That is, dispersion region III is explained by the dominance of this inter-segment contribution, whereas regions I and II are dominated by intrasegment spin interactions.

All three dispersion regions occurring in the frame of component B of chain dynamics can be described in a consistent way based on the same elementary theory. Region III very remarkably is determined by the effect of a certain relaxation mechanism in combination with segment self-diffusion properties as independent phenomena. The consistency of the interpretation of the three proton dispersion regions I, II, and III with the aid of the renormalized Rouse theory is striking. Quantitative evaluations of proton spin–lattice relaxation times on this basis can be found elsewhere [157].

The slope of the deuteron spin–lattice relaxation dispersion curves representing intrasegment interactions is characterized by the exponent 0.34 (Figs. 35, 36, and 42b). This is somewhat larger than the exponent 0.25 found in the average for the proton relaxation dispersion in region II (Eq. 168) interpreted above also to be due to intrasegment interactions alone. This discrepancy may be due to some influence of intersegment dipolar interactions on the proton relaxation dispersion already in region II.

Actually, taking the first nearest neighbor peak (first segment coordination shell) of the radial segment pair correlation function (see Fig. 38 and compare Ref. [69]) into account in a numerical evaluation permitted Herman recently to predict a corresponding effect [158]. One may conclude that dispersion region II of proton relaxation represents a superposition of intrasegment and short-range intersegment relaxation mechanisms, whereas the proton dispersion region III is dominated by long-range intersegment interactions. Here "short" and "long-range" are meant relative to the radius of the first segment coordination shell (Fig. 38).

4.5.3 Mesomorphic Phases

Linear polydiethylsiloxane (PDES) and its higher homologues form ordered mesophases in temperature intervals between the solid and the isotropic melt phases [159, 160]. This mesophase is to be juxtaposed to monomeric [161–166] or main-chain polymeric nematic liquid crystals [167]. The results to be described in the following demonstrate the sensitivity of NMR techniques to microstructural changes.

Figure 32 shows spin–lattice relaxation dispersion data for PDES (having $-CH_2CH_3$ as side groups) both in the isotropic and in the mesomorphic phases [155]. The dispersion of the isotropic melt is governed by the same empirical power laws for regions I and II as stated before (Eq. 168) and in particular as measured in PDMS melts (side groups: $-CH_3$).

Two power law regimes were also found in the mesophase, but with larger exponents and a crossover frequency shifted by a factor of about 10 (after correction for the different temperatures) to a lower value. In summary, the power laws observed in PDES in broad frequency ranges are (see Fig. 32 and Ref. [155])

$$T_1 \propto \begin{cases} M_w^0 \omega^{0.50\pm0.05} & \text{(region I “isotropic phase”)} \\ M_w^0 \omega^{0.73\pm0.05} & \text{(region I “mesophase”)} \\ M_w^0 \omega^{0.25\pm0.05} & \text{(region II “isotropic phase”)} \\ M_w^0 \omega^{0.45\pm0.05} & \text{(region II “mesophase”)} \end{cases} . \tag{182}$$

The slopes in region II for both the isotropic and the mesomorphic phases are in accordance with data for the spin–lattice relaxation time in the rotating frame, $T_{1\rho}$.

These results for the ordered mesophase are to be compared with the spin–lattice relaxation dispersion expected for nematic liquid crystals where a power law

$$T_1 \propto \nu^{0.5} \quad \text{(ODF)} \tag{183}$$

is predicted for order director fluctuations (ODF) at low frequencies [161–167]. With nematic compounds this law and the collective fluctuations in the ordered state it represents should show up irrespective of whether it is a monomeric or polymeric substance.

This is in contrast to the mesophase of PDES. The monomers of this compound do not show any order because they do not contain mesogenic groups. That is, the order observed in the PDES mesophase must have a thermodynamic background different from that of nematic phases.

It is concluded that the modified power laws for spin–lattice relaxation dispersion in the mesophase reflect a modified behavior of chain modes rather than collective fluctuations of ensembles of molecules in ordered domains. This conclusion is corroborated by the identical frequency dispersion of the spin–lattice relaxation times T_1 and $T_{1\rho}$ in the laboratory and in the rotating frames, respectively. If the order in the PDES mesophase would be of a nematic nature and the fluctuations causing dispersion region II consequently would be of the ODF type, the frequency dispersion of $T_{1\rho}$ should vanish while that of T_1 is retained [22].

In this context, a study of the dipolar correlation effect in the mesophase of PDES is of particular interest [168]. Using this technique, it was concluded that the sample consists of defect-enriched areas and ordered domains. The two dynamic states are connected with segments of different mobility, and they fluctuate temporally and spatially. The exchange times between the two mobility states range between 0.1 and 1.0 s in the mesomorphic temperature range.

The exchange process between the two environments is slow compared with the transverse relaxation rates so that the two signal components can clearly be distinguished. This is in contrast to spin–lattice relaxation where flip-flop spin diffusion as an additional exchange mechanism mediates averaging over all heterogeneities. Spin–lattice relaxation curves therefore decay mono-exponentially in a wide range [155], whereas the transverse magnetization is subject to a superposition of virtually two exponential functions [168]. As concerns the different power law regimes of spin–lattice relaxation given in Eq. 182, it is not yet known to which of the two segment phases they must be attributed or to what degree exchange averaging matters in this respect.

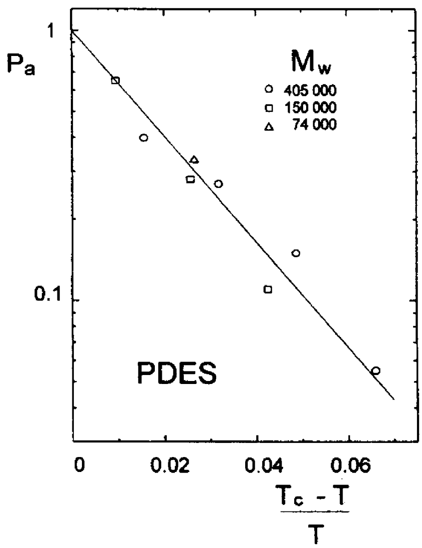


Fig. 39. Fraction of the more mobile segments in the mesophase of polydiethylsiloxane (PDES) as a function of the reduced temperature. The *solid line* represents Eq. 184

The dipolar correlation effect is based on stimulated-echo signals [2] and therefore contains elements both of transverse and longitudinal relaxation. Flip-flop spin diffusion as well as material transport may contribute to the exchange mechanism identified this way. An interesting result of the dipolar correlation effect study is the temperature dependence of the fraction of the more mobile segments which was shown to obey the empirical law (see Fig. 39) [168]

$$P_a = \exp\left\{45 \frac{T - T_c}{T}\right\}, \tag{184}$$

where T_c is the (molecular weight dependent) isotropization temperature. Approaching T_c from below increases the fraction of the more mobile segments exponentially as a function of the reduced temperature.

Figure 40 shows an attempt to visualize how the areas of different mobility could arise based on "collapsed" coils. The "gain" in chain mobility in the ordered mesophase is attributed to conformations containing many chain folds compensating the dynamic restrictions entailing the almost nematic-like order in this phase (compare the discussion in context with chain-end dynamics and the statistical physics treatment of chain folds in Ref. [124]).

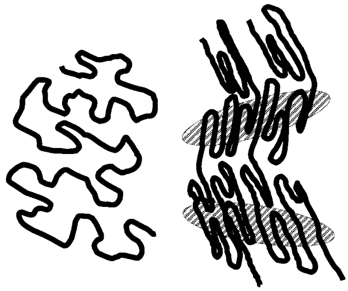


Fig. 40. Attempt to schematically represent the ordered mesophase of PDES juxtaposed to the isotropic phase. The “collapsed” coil structure assumed in the mesophase entails chain folds that provide additional motional degrees of freedom and is therefore thermodynamically stabilized in a certain temperature range (compare the statistical physics treatment in Ref. [124]). The *shaded ellipses* indicate somewhat closer packed, more ordered and less mobile areas in contrast to the more mobile, more distorted chain conformations in regions where many folds accumulate

4.5.4 Solutions

The features of entangled polymer dynamics fade upon dilution by a low-molecular solvent. In other words, the critical molecular weight characteristic for the crossover between Rouse-like and entangled dynamics is growing with the degree of dilution (see Eq. 162). NMR relaxation measurements turned out to be particularly suitable for studies of the dilution process. The crossover from the melt to the dilute solution was already demonstrated with the T_2/T_1 ratio in Fig. 20. It can also be visualized with the aid of spin–lattice relaxation dispersion of PDMS dissolved in CCl_4 as a proton-free solvent. Figure 31d shows corresponding data.

Compared to the region II dispersion occurring as one of the characteristics of melts of entangled polymers the spin–lattice relaxation dispersion becomes flatter upon dilution. With a PDMS content of 15% in CCl_4 the data can be described by the Rouse model as the incarnation of free-chain dynamics (see Fig. 31d). This may appear surprising since the critical molecular weight for this concentration according to Eq. 162 is below the weight-average molecular weight of the polymer examined, M_w=423,000, so that entangled dynamics should still apply.

On the other hand, all polymer dynamics models unspecifically predict some short-time/high-frequency limit where Rouse-like behavior should dominate (see Tables 1–3). In melts of entangled polymers such a regime could never be identified by spin–lattice relaxation dispersion because component B in the form of the high-mode-number limit (region I) appears to overlap with the local segment fluctuation regime (manifesting itself as component A). That is, there is no intermediate time or frequency window left in entangled melts where Rouse dynamics could develop in full.

Such a dynamic gap obviously arises in solutions if the polymer concentration is low enough. Component A representing local segment fluctuations becomes accelerated upon dilution, whereas the entanglement effect on the chain modes is shifted to correspondingly longer length and time scales. That is, a very important theoretical requirement is satisfied in this way. We will come back to the topological constraint problem later in the review.

4.5.5 Networks

Permanent or thermoreversible cross-links mediate the opposite effect on chain dynamics compared with dilution by a solvent: instead of releasing topological constraints by dilution, additional hindrances to chain modes are established by the network. With respect to NMR measurands relatively large cross-link densities are needed to affect chain modes visible in the experimental time/frequency window, as demonstrated with proton spin–lattice relaxation dispersion of polyethylene cross-linked by 10-Mrad irradiation with electron beams [123] and with styrene–butadiene rubbers [29]. However, there is a very strong effect on the dipolar correlation effect which probes much slower motions and can therefore be used favorably for the determination of the cross-link density [29, 176, 177].

In the plot shown in Fig. 31e, spin–lattice relaxation dispersion curves of permanently cross-linked PDMS are shown. With decreasing mesh length, the chain modes appear to be shifted to lower frequencies. This is indicated by lower values of the relaxation times while the dispersion slopes of regions I and II are retained. On the other hand, the effect on the crossover frequency is minor.

The fact that the dispersion regions I, II, and III of proton spin–lattice relaxation retain their qualitative appearance in the presence of (in this case thermoreversible) cross-links is demonstrated in Fig. 34. The influence of fluctuating cross-links on the dispersion curves can be explained by an effectively modified monomeric friction coefficient [131]. A result of particular interest is the shift of the crossover frequencies between dispersion regions I, II, and III. The shifts can be explained by the ratios

$$\frac{\nu_{I,II}}{\tilde{\nu}_{I,II}} = \frac{\nu_{II,III}}{\tilde{\nu}_{II,III}} = \frac{\tilde{\tau}_s}{\tau_s}, \tag{185}$$

where the quantities with tilde refer to networks. The friction effect on segment fluctuations equally slows down chain modes at the lowest frequencies.

4.6 Translational Segment Diffusion and Spin Diffusion

The NMR diffusometry techniques described in this article are frequently and suitably used for tests of the limits for the mean squared segment displacement predicted by polymer theories. The time and length scales probed by the techniques can be adjusted by the corresponding pulse intervals and the field gradient strength (the "wave number"), respectively. In this way, one gets access to the center-of-mass diffusion limit as well as to the anomalous-segment diffusion regime. The root mean squared displacements examined in the former case are larger than the coil dimension, whereas the latter limit is probed if the displacements are shorter than the coil extension.

4.6.1 Center-of-Mass Diffusion

NMR diffusometry is suitable for studies of the center-of-mass diffusion coefficient provided that the diffusion time can be chosen longer than the terminal relaxation time. In terms of the tube/reptation model, this is limit $(\mathrm{IV})_{\mathrm{DE}}$ defined by $t>>t_d$. Eq. 75 suggests $D\propto M_w^{-2.0}$ which appears to have been verified several times in experiment [8, 148–150, 169–171]. Figure 41 shows polydimethylsiloxane melt data as a typical example [8]. Data for other polymer melt examples reported in the literature refer to polystyrene and polyethylene at different temperatures [169, 148, 149], and polyethyleneoxide [11, 170].

It appears that all those data perfectly corroborate the validity of limit $(\mathrm{IV})_{\mathrm{DE}}$ of the tube/reptation model. However, there are four reasons why the molecular weight dependence observed in experiments may be modified by factors other than the tube constraints:

- The condition $t>>t_d$ is difficult to achieve with high molecular weights so that the anomalous segment displacement regime may gain an increasing influence from low to high molecular weights, as was clearly demonstrated at least with the two highest molecular weights in the plot shown in Fig. 41 [8]. The consequence is an apparently *weaker* molecular weight dependence. The crossover regime between the center-of-mass and the anomalous segment displacement regimes was extensively studied [172] for polystyrene solutions. It appears that the center-of-mass diffusion coefficient cannot be evaluated reliably by the field-gradient technique without considering the crossover to the anomalous diffusion regime at the same time.
- Free volume connected with chain-end blocks influences the molecular weight dependence of chain dynamics in the vicinity of M_c. An apparently *steeper* molecular weight dependence results on these grounds particularly below M_c. The failure of NMR techniques to directly render the molecular

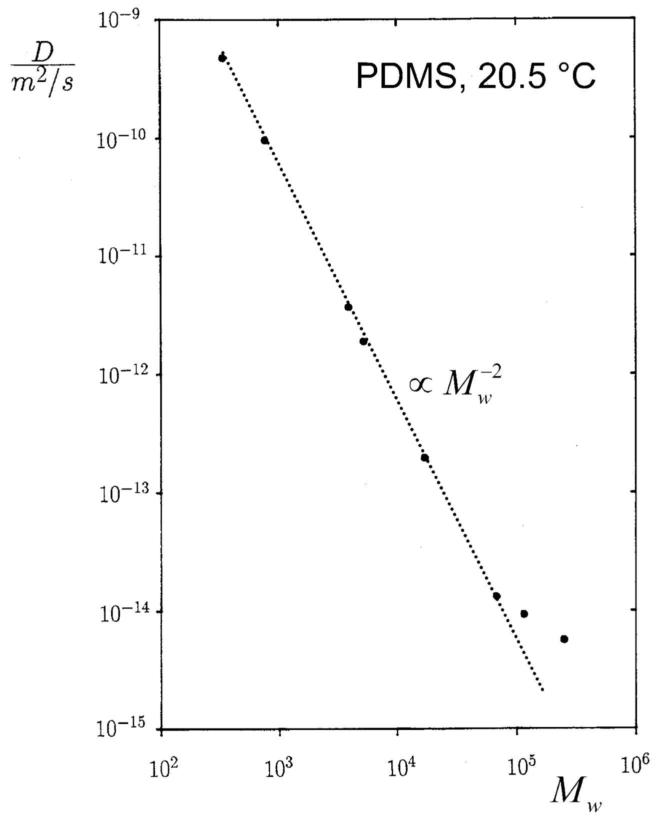


Fig. 41. Center-of-mass diffusion coefficient in polydimethylsiloxane (PDMS) melts at 20.5 °C [8] as a function of the molecular weight. This self-diffusion coefficient is relevant for root mean square displacements exceeding the random-coil dimension. The experimental condition is that the echo attenuation is recorded with small field-gradient strengths and correspondingly long diffusion times. The *straight line* represents the law $D \propto M_w^{-2}$ which seemingly coincides with the prediction by the tube/reptation model for limit $(IV)_{DE}$ (see Eq. 75). The deviation from the M_w^{-2} dependence above M_w=100,000 is due to the fact that in this case the diffusion time could not be chosen long enough to ensure center-of-mass diffusion. Actually we are here already in the anomalous segment diffusion regime. For M_w=250,000, for instance, the root mean square displacement reached along the gradient direction in a diffusion time of 30 ms is only 20 nm compared with a radius of gyration of 25 nm

weight dependence $D \propto M_w^{-1.0}$ predicted for the Rouse model (see Eq. 59) was explained in this way [148, 149].

- At very high molecular weights, when the diffusion coefficient due to flip-flop spin diffusion of the order 10^{-15} m²/s is getting comparable to the ordinary Brownian self-diffusion coefficient, a *flatter* chain length dependence of the effective diffusion coefficient is expected [10].

– If very broad, the molecular weight distribution may influence the (average) center-of-mass diffusion coefficient evaluated from echo attenuation curves [169, 175].

In view of the contour-length fluctuation mechanism suggested by Doi as an explanation of the fractional exponent of the power law for the zero-shear viscosity [142], $\eta \propto M_w{}^{3.4}$, a molecular weight dependence somewhat stronger than predicted by Eq. 75 was predicted also for the center-of-mass diffusion coefficient [169]. More recent studies mainly carried out with concentrated solutions even suggest power laws with universally stronger molecular weight dependences than predicted by the tube/reptation model [172–174]:

$$D \propto M_w^{-2.4\pm0.1}\phi^{-1.8\pm0.2},$$
$$\eta \propto M_w^{3.4\pm0.1}\phi^{3.8\pm0.2}, \tag{186}$$

where ϕ is the volume fraction of the polymer.

4.6.2 Anomalous Segment Diffusion

Because of the relatively short displacement time or length scales typically probed by NMR diffusometry, it is particularly well suited to detect anomalies in the segment displacement behavior expected on a time scale shorter than the terminal relaxation time, that is for root mean squared displacements shorter than the random-coil dimension. All models discussed above unanimously predict such anomalies (see Tables 1–3). Therefore, considering exponents of anomalous mean squared displacement laws alone does not provide decisive answers. In order to obtain a consistent and objective picture, it is rather crucial to make sure that (i) the absolute values of the mean squared segment displacement or the time-dependent diffusion coefficient are compatible with the theory, (ii) the dependence on other experimental parameters such as the molecular weight are correctly rendered, and (iii) the values of the limiting time constants are not at variance with those derived from other techniques.

For instance, the segmental friction coefficient or the mean squared end-to-end distance can be determined directly from viscoelasticity or neutron scattering data. Estimating segment diffusion with the aid of the tube/reptation model based on such data leads to unrealistic predictions. The predicted segment displacements tend to be much too large [9, 12, 178]. Figure 42 shows polyethyleneoxide data as an example. One identifies a time dependence $D(t) \propto t^{-0.5}$ at short diffusion times in contrast to the law $D(t) \propto t^{-3/4}$ predicted for $\tau_e << t << \tau_R$ as limit $(\mathrm{II})_{\mathrm{DE}}$ (see Table 1) apart from an inconsistent molecular weight dependence [12, 172]. The incompatibility becomes particularly obvious if the echo attenuation curves are evaluated in full

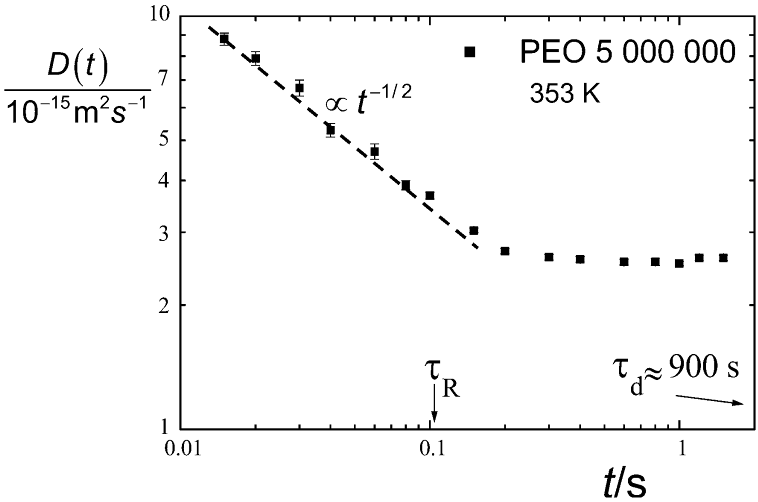


Fig. 42. Time-dependent diffusion coefficient measured in a polyethyleneoxide melt (M_w=5,000,000) at 353 K as a function of the diffusion time [12]. The data were evaluated according to Eq. 27. The Rouse relaxation time τ_R and the tube disengagement time τ_d predicted by the tube/reptation model based on the neutron scattering value of the ratio $\langle R^2_{ee}\rangle / M_w = 1.01\times10^{-20}$ m^{2}mol/g [179, 180] are indicated. The *broken line* represents the power law $D(t)\propto t^{-1/2}$ which, according to the tube/reptation model, should appear in the time interval *above* τ_R as limit $(\mathrm{III})_{\mathrm{DE}}$ and not below (see Table 1). The time range where this power law appears and the value for τ_R estimated on the basis of the tube/reptation model are not consistent with each other. The plateau of the experimental data for $t >> 0.1$ s is due to flip-flop spin diffusion which physically limits the detection of molecular displacements by NMR diffusometry

(compare Fig. 44) using the formalism specifically derived for the tube/reptation model (see Eqs. 76–79). In the frame of the experimental accuracy, it is not possible to describe the whole family of echo attenuation curves on the basis of the unmodified predictions in the full diffusion time range for all molecular weights at the same time [9, 12, 178], even if flip-flop spin diffusion (see the following section) is taken into account.

On the other hand, if a static "tube" really exists as anticipated in the tube/reptation model, the predictions for anomalous segment diffusion are adequate, of course, and can be verified in experiment. This will be demonstrated by considering linear polymers confined in nanopores.

4.6.3 Flip-Flop Spin Diffusion

Flip-flop spin diffusion is a phenomenon specific for NMR. It does not matter with other techniques. Spin diffusion based on Zeeman energy conserving flip-flop transitions of dipolar coupled spins, that is interchange of spin-up and spin-down states, is normally negligible compared to molecular self-diffusion in liquids. However, with viscous systems with little motional averaging of dipolar coupling and with the aid of very strong field gradients, NMR diffusometry is able to detect this immaterial transport mechanism competing with molecular diffusion. Spin echoes can be attenuated on this basis just as with any other incoherent displacement process [181].

Employing the strong fringe-field gradient of a superconducting magnet, it was possible to demonstrate the influence of flip-flop spin diffusion on the echo attenuation of polymer melts [10, 12]. The experiments were carried out by diluting the polymer under investigation isotopically. That is, undeuterated polymers were dissolved in a perdeuterated matrix of polymers of equivalent molecular weight. The reduction of intermolecular dipolar interactions diminishes the proton spin flip-flop rate as expected.

The effect is also evident in Fig. 42 referring to a polyethyleneoxide melt with an extremely large molecular weight, M_w=5,000,000. The time-dependent diffusion coefficient was evaluated in the low wave number approximation, Eq. 27, according to $\langle R^2(t)\rangle=6D(t)t$. The time-independent plateau appearing at times $t >> 0.1$ s was shown to be due to flip-flop spin diffusion [10, 12].

As a consequence, the proper evaluation of segment diffusion at long diffusion times turned out to be more complicated than often anticipated. However, even with flip-flop spin diffusion theoretically taken into account, it is not possible to fit the formulas predicted by the tube/reptation model to the experimental data with respect to both the time and molecular weight dependences in a consistent way and without assuming unrealistic parameters [12].

5
Chain Dynamics in Pores ("Artificial Tubes")

The motivation to study the dynamics of polymer chains confined in nanoporous materials with more or less rigid pore walls is twofold. Firstly, there may be important technological applications requiring knowledge on the dynamic behavior of polymers under such conditions [182]. The second reason making this field intriguing for polymer science is the possibility to study chain dynamics under topological model constraints in the form of artificial tubes. In the following we will focus on this latter point.

The tube introduced in the frame of the Doi/Edwards reptation model for the treatment of bulk systems of entangled polymers is a fictitious one. As outlined above, the laws predicted on this basis (see Table 1) can provide only a rather crude picture failing to account for numerous experimental findings quantitatively as well as qualitatively. It may therefore be helpful to study chain dynamics in tube-like pores of a physically real nature.

The diffusion and relaxation behavior of diverse oligomers and polymers has been studied in nanoporous silica glasses [183]. Both measuring techniques provide clear evidence for modified chain dynamics compared with the bulk. However, the problem with this sort of system is that interactions of the polymers with the pore walls play an important role and must be distinguished from the geometrical confinement effect. A corresponding analysis was possible by comparison of polymer data with data obtained with short oligomers of the same chemical species. In contrast to the polymers, the oligomers were assumed to be subject to adsorption but not to modifications of the chain modes by geometrical confinement. Subtraction of the oligomer relaxation rates from those measured with polymers in the pores suggests a spin–lattice relaxation dispersion reduced for the influence of geometrical confinements on the chain-mode distribution. Actually a power law dispersion reproducing that predicted for limit $(\mathrm{II})_{\mathrm{DE}}$ of the tube/reptation model (see Table 1) could be elucidated this way.

A more direct verification of tube/reptation features was possible with systems where the solid matrix and the mobile polymer chains confined to nanopores are of similar organic chemical composition. In the experiments referred to in the following, linear polymers were confined in a solid, that is strongly cross-linked, polymer environment. Under such conditions, the geometry effect can be expected to dominate whereas the wall adsorption phenomenon is of negligible influence.

This sort of system was prepared in the form of so-called semiinterpenetrating networks. Preparation details are described in Ref. [184]. The matrix

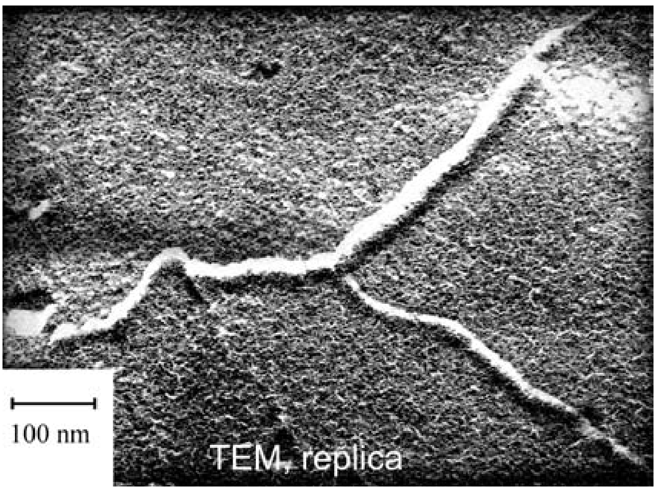


Fig. 43. Transmission electron micrograph of a replica of a freeze-fractured surface of polyethyleneoxide M_w=6,000 in PHEMA [11]. The pore channels have a width of about 10 nm

consisted of cross-linked polyhydroxyethyl methacrylate (PHEMA). Linear polyethyleneoxide (PEO) was incorporated in the 10-nm pores. The molecular weights of the PEO samples used in experiment were chosen large enough to ensure that the root mean squared random-coil diameter in bulk exceeds the pore diameter. Figure 43 shows an electron micrograph of pore channels in this material having a width of about 10 nm.

5.1 Diffusion in Pores

Segment diffusion in pores suggests itself as a typical model scenario representing the premisses of the tube/reptation model. NMR diffusometry is suitable to probe the time or length scales of the Doi/Edwards limits $(\mathrm{II})_{\mathrm{DE}}$, $(\mathrm{III})_{\mathrm{DE}}$ and beyond. Since the wall adsorption effect in the PHEMA system is expected to be negligible, one can therefore expect that the reptation features of the anomalous segment diffusion regime, especially with respect to limit $(\mathrm{III})_{\mathrm{DE}}$, are faithfully rendered by the experiments.

Figure 44 shows typical echo attenuation curves recorded with PEO in a solid PHEMA matrix [186]. The good coincidence of the theoretical curves calculated on the basis of Eqs. 76–79 for all accessible values of the experimental parameters k and t for different molecular weights [11] corroborates the validity of the tube/reptation model for linear polymers confined to arti-

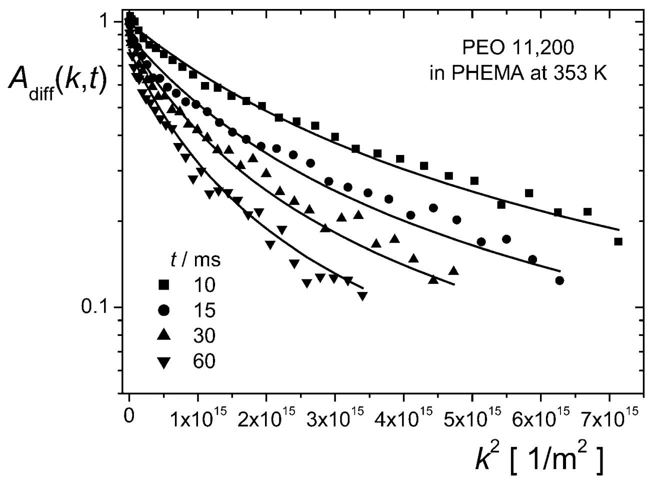


Fig. 44. Echo attenuation curves for polyethyleneoxide, M_w=11,200, confined to PHEMA pores at 80 °C as a function of the squared wave number, k^2, for different diffusion times. The *solid lines* represent a fit of Eqs. 76–79 to the experimental data. The tube diameter is found to be a=(9±1) nm as the only fitting parameter. Other parameter values such as $N=M_w/853$, $D_0=9.66\times10^{-10}$ m^2/s, and $b=8.38\times10^{-10}$ m were taken from the literature [180, 187]

ficial nanopores in a solid matrix. Very remarkably the tube diameter a (as the only free parameter) fitted to the diffusion data coincides with the pore diameter determined with other techniques such as electron microscopy, X-ray scattering or NMR spin diffusion [184].

5.2 Spin–Lattice Relaxation Dispersion in Pores

Field-gradient NMR diffusometry experiments tend to be largely dominated by the time and length scales of limit $(\mathrm{III})_{\mathrm{DE}}$. On the other hand, the much shorter time scale of limit $(\mathrm{II})_{\mathrm{DE}}$ can specifically be examined by NMR relaxometry, which under suitable conditions can indirectly probe translational diffusion of polymer segments. The laws predicted in Table 1 for anomalous segment diffusion on the one hand and spin–lattice relaxation on the other are based on the very same translational segment displacement mechanisms. They stipulate each other compellingly. Spin–lattice relaxation at low frequencies, as they can be covered with the aid of field-cycling NMR relaxometry, is therefore suitable to extend the range of field-gradient NMR diffusometry toward much shorter times.

The total proton frequency range that can be probed by NMR spin–lattice relaxation techniques is 10^3 Hz $< \nu <$ 10^9 Hz. For the present application, deuteron resonance selectively applied to perdeuterated polymers confined to the pores is superior to proton resonance which, unlike the situation in field-gradient experiments, is affected by signals from the matrix and flip-flop spin diffusion across the matrix. The deuteron frequency range is shifted by a factor of 0.15 to lower frequencies. This frequency window largely matches the time scale of chain modes of polymers with medium molecular masses.

The exceptional frequency and molecular weight dependence for the spin–lattice relaxation time expected for limit $(\mathrm{II})_{\mathrm{DE}}$,

$$T_1 \propto M^0 \omega^{3/4} \quad (\omega\tau_s \ll 1 \ll \omega\tau_e) \tag{187}$$

(see Table 1), is particularly indicative for reptational segment diffusion. Figure 45a shows the frequency dependence of the deuteron spin–lattice relaxation time of perdeuterated PEO of different molecular weights in a solid PHEMA matrix [95,185]. In the frequency regime, where the standard Bloch/Wangsness/Redfield relaxation theory is applicable, the law given in Eq. 187 is reproduced as

$$T_1 \propto M_w^{0\pm0.05} \nu^{0.75\pm0.02}. \tag{188}$$

The experimental deuteron frequency range in which this frequency dependence was observed is 5×10^5 Hz $< \nu <$ 6×10^7 Hz. De Gennes' prediction for limit $(\mathrm{II})_{\mathrm{DE}}$ has thus been verified for polymers confined in artificial

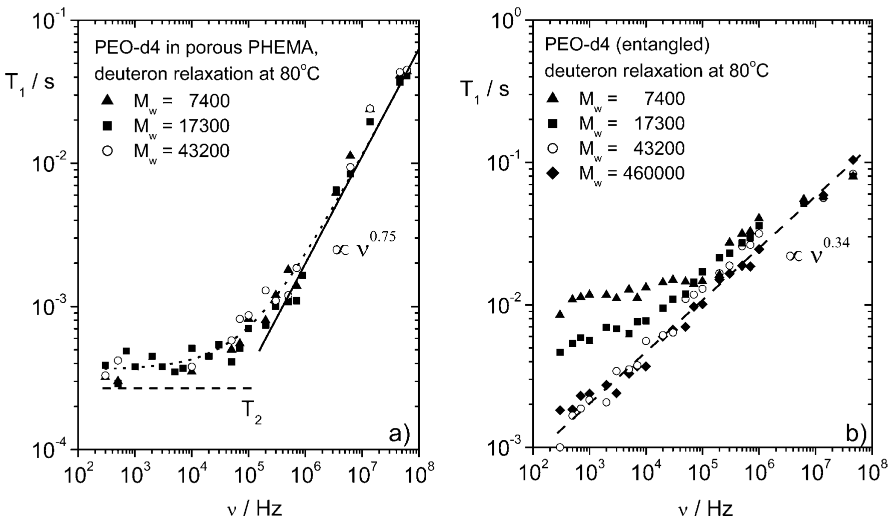


Fig. 45a, b. Frequency dependence of the deuteron spin–lattice relaxation time of perdeuterated PEO confined in 10-nm pores of solid PHEMA at 80 °C (**a**) and in bulk melts (**b**) [95, 185]. The dispersion of the confined polymers verifies the law $T_1 \propto M_w^0 \omega^{0.75}$at high frequencies as predicted for limit $(\mathrm{II})_{\mathrm{DE}}$ of the tube/reptation model (see Table 1). The low-frequency plateau observed with the confined polymers indicates that the correlation function implies components decaying more slowly than the magnetization relaxation curves, so that the Bloch/Wangsness/Redfield relaxation theory [2] is no longer valid in this regime. The plateau value corresponds to the transverse relaxation time, T_2, for deuterons extrapolated from the high-field value measured at 9.4 T

tubes in full accordance with the theory. This is in contrast to the bulk behavior of the very same polymers studied with deuteron resonance again. The data in Fig. 45b reproduce features observed with proton and deuteron resonance with bulk melts of other polymer species as discussed above (see Figs. 35 and 36). Polymer chains in bulk melts must obviously have additional degrees of freedom not existing under pore confinements.

5.3
Theoretical Crossover from Rouse to Reptation Dynamics

The theoretical background of the confinement effect in (artificial) tubes was recently examined in detail with the aid of an analytical theory as well as with Monte Carlo simulations [70]. The analytical treatment referred to a polymer chain confined to a harmonic radial tube potential. The computer simulation mimicked the dynamics of a modified Stockmayer chain in a tube with "hard" pore walls. In both treatments, the characteristic laws of the tube/reptation model were reproduced. Moreover, the crossover from reptation (tube diameter equal to a few Kuhn segment lengths) to Rouse dy-

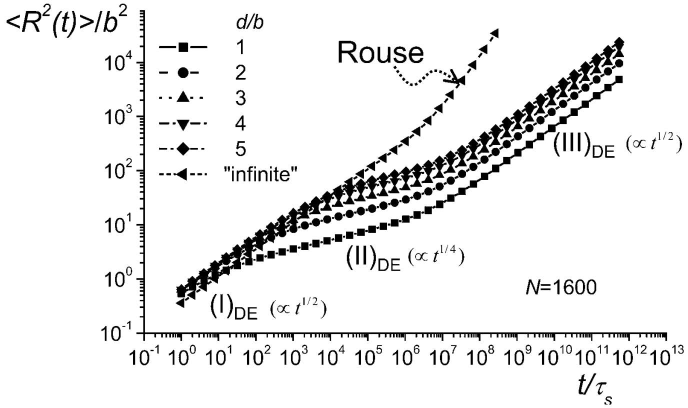


Fig. 46. Theoretical mean squared segment displacement of a chain confined in a randomly coiled tube versus time according to the harmonic radial potential theory [70]. The tube diameter d is given in multiples of the Kuhn segment length b. The crossover tendency to free, unconfined Rouse chain dynamics with increasing tube diameter is obvious. The mean squared displacement is given in units b^2, the diffusion time t in units of the segmental fluctuation time τ_s. The chain length was assumed to be N=1,600 Kuhn segments. The three "anomalous" Doi/Edwards limits (see Table 1) are reproduced with finite tube diameters

namics (tube diameter "infinite") was demonstrated by varying the tube diameter.

Figure 46 shows the time dependence of the mean squared segment displacement as predicted by the harmonic radial potential theory [70]. The three anomalous diffusion limits, $(\mathrm{I})_{\mathrm{DE}}$, $(\mathrm{II})_{\mathrm{DE}}$, and $(\mathrm{III})_{\mathrm{DE}}$, of the tube/reptation model are well reproduced. Note the extended width of the transition regimes between these limits, which should be kept in mind when discussing experimental data with respect to a crossover between different dynamic limits. Increasing the effective tube diameter is accompanied by the gradual transition to Rouse-like dynamics of an unconfined chain (where entanglement effects are not considered).

The spin–lattice relaxation dispersion was derived with the same sort of theory. Figure 47 shows data obtained under the same conditions as assumed for the mean squared displacement data in Fig. 46. Again the dispersion most specific for the tube/reptation model, namely $T_1 \propto \omega^{0.75}$ which is predicted for limit $(\mathrm{II})_{\mathrm{DE}}$, was perfectly reproduced at low frequencies.

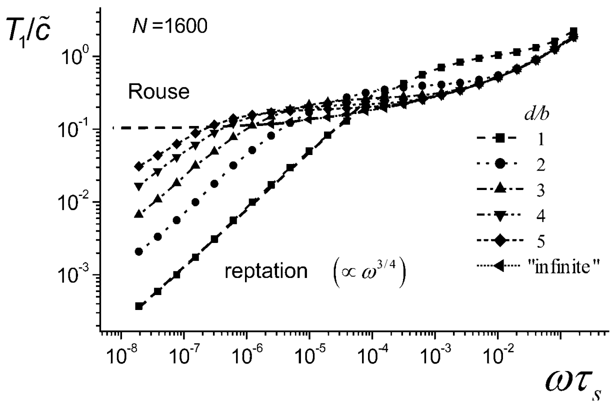


Fig. 47. Spin–lattice relaxation dispersion for a chain of N=1,600 Kuhn segments (of length b) confined to a randomly coiled tube with a harmonic radial potential with varying effective diameters d. The data were calculated with the aid of the harmonic radial potential theory [70]. $\tilde{c}$ is a constant. At low frequencies the curves visualize the crossover from Rouse dynamics depending on the effective tube diameter. The latter case is described by a T_1 dispersion proportional to $\omega^{3/4}$ characteristic for limit $(II)_{DE}$ of the tube/reptation model

6
Concluding Remarks

NMR may be considered as a particularly versatile class of investigation methods suitable for polymer dynamics studies. In particular, field-cycling NMR relaxometry employed in parallel with field-gradient NMR diffusometry is an unmatched combination of methods for the elucidation of features of chain dynamics. A great wealth of clearly documented phenomena for different measurands, experimental parameters, and different polymer species in different environments and under different thermodynamic conditions were discovered, and appear to be suitable for direct comparisons with theoretical model predictions. One can say that the experimental data available so far already render a closed picture from the empirical point of view.

Transverse and longitudinal NMR relaxometry of ^{1}H and ^{13}C nuclei was shown to be governed by three components of chain dynamics. With respect to tests of theoretical model concepts, the most important one is component B which reflects the influence of chain modes and can be probed with the field-cycling technique.

Field-gradient NMR diffusometry is suitable for recording translational displacement properties by self-diffusion. This in particular refers to the anomalous segment displacement regime. Center-of-mass diffusion is also accessible this way but must be handled with some care because of the experimental limitation of the maximum diffusion time. Furthermore the influence of flip-flop spin diffusion must be taken into account when examining long polymers.

Three basic model theories have been considered: the *Rouse model*, the *tube/reptation concept,* and the *renormalized Rouse formalism*. Depending on the sample system and the experimental conduct, characteristic features of all these theories have been shown. Some spectacular predictions of theories were verified, others were ruled out.

The frequency dependence of the spin–lattice relaxation time predicted by the *Rouse model* was verified with polymer melts for $M_w<M_c$ in accordance with the segmental fluctuation time determined from the T_1 minimum condition. Melts with $M_w>>M_c$ did not show any such behavior in the short-time limit although predicted so by all three model theories. On the other hand, dilution of the polymer by a low-molecular solvent increases M_c and diminishes the entanglement effect. Under such conditions, Rouse-like behavior was observed.

The *tube/reptation model* of entangled polymer dynamics suggests a number of peculiar anomalous-diffusion limits that were consistently and quantitatively verified in artificial tubes of a solid porous matrix with the aid of field-cycling NMR relaxometry and field-gradient NMR diffusometry. This clear corroboration of a model prediction is however in contrast to the dynamics concluded for bulk melts. In that case neither the prediected T_1 dispersion nor the anomalous segment diffusion behavior was found with respect to the frequency and molecular weight dependences. The assumption of static fictitious tubes as a replacement scheme for the entangling matrix effect is obviously too crude for bulk melts, and does not render the features of polymer dynamics correctly.

Based on the generalized Langevin equation, the *renormalized Rouse models* suggest dynamic high- and low-mode-number limits as an implicit structural feature of this equation of motion. This is a stand-alone prediction of paramount importance independent of any absolute values of power law exponents that arise and are measured in the formalism and in experiment, respectively. The two limits manifesting themselves as power law spin–lattice relaxation dispersions were clearly identified in bulk melts of entangled polymers of diverse chemical species.

Apart from these two relaxation dispersion regimes, which are mainly based on intrasegment spin interactions (and possibly nearest neighbor intersegment contributions), a third dispersion region specific for dipolar-coupled nuclei exists at low frequencies. This was shown to be due to (long-distance) dipolar interactions being relevant for protons but not for deuterons.

A most important finding in general and for establishing of model theories is the fact that segment dynamics is affected by microstructural details already at relatively short times. The local fluctuations of a Kuhn segment typically occur on a time scale shorter than 10^{-10} s and on a length scale of up to 1 nm. It was shown that factors such as chain entanglements, cross-linking or mesomorphic order quite amazingly, and contrary to most model expectations, have an influence on segment dynamics already rather close to these time and length scales. This finding became evident from the relaxometry studies reported and was also observed in the literature [129]. On the other hand, chain modes with low mode numbers and even global, molecular weight dependent motions corresponding to chain relaxation times beyond the Rouse relaxation time do show up at the predicted time and length scales.

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